

SmartDV IP Solutions

GravityXR SDIO/eMMC Host IIP Databook

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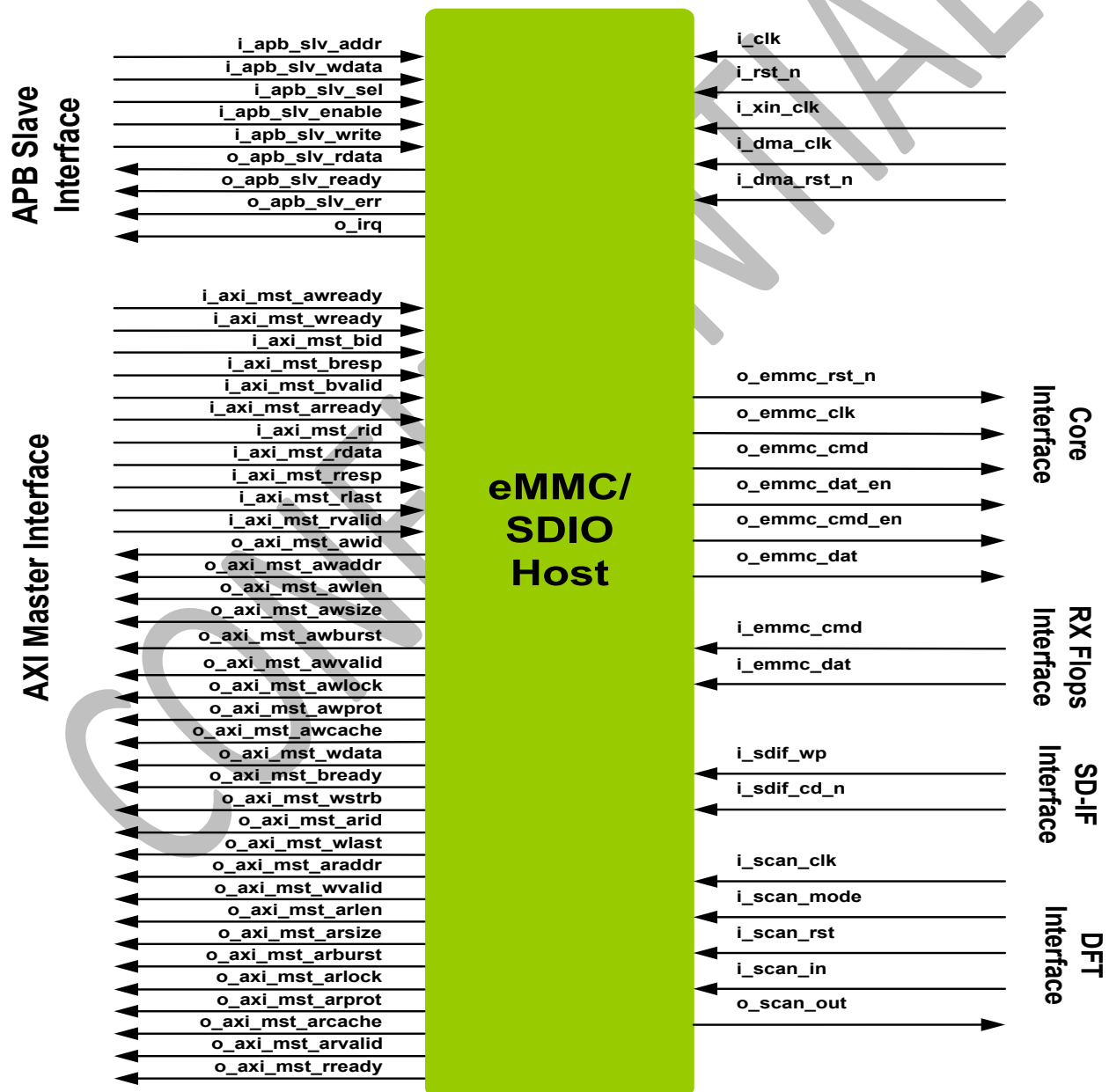
1 Revision History

Revision	Date	Changes
1.0	Mar-01-2025	Initial version
1.1	Mar-06-2025	Updated the Interface chapter and Clock Diagram chapter
1.2	Mar-18-2025	Added the C Driver Code chapter Updated the Interface chapter, Software Interface Chapter and Test plan Chapter Updated the Microarchitecture diagram and Module descriptions
1.3	Mar-25-2025	Updated the Clock Diagram chapter Added the IO Pads in Interface Chapter

2 Introduction

This user's guide describes the following SmartDV SDIO/eMMC Host core. This guide gives an overview of the SDIO/eMMC Host core, gives a section on getting started with SmartDV's SDIO/eMMC Host core and explains how to use it in a given design. The sections on the test environment are also included in this guide.

SmartDV SDIO/eMMC Host core is fully tested to meet the requirements of the JESD84-B45 Specification.



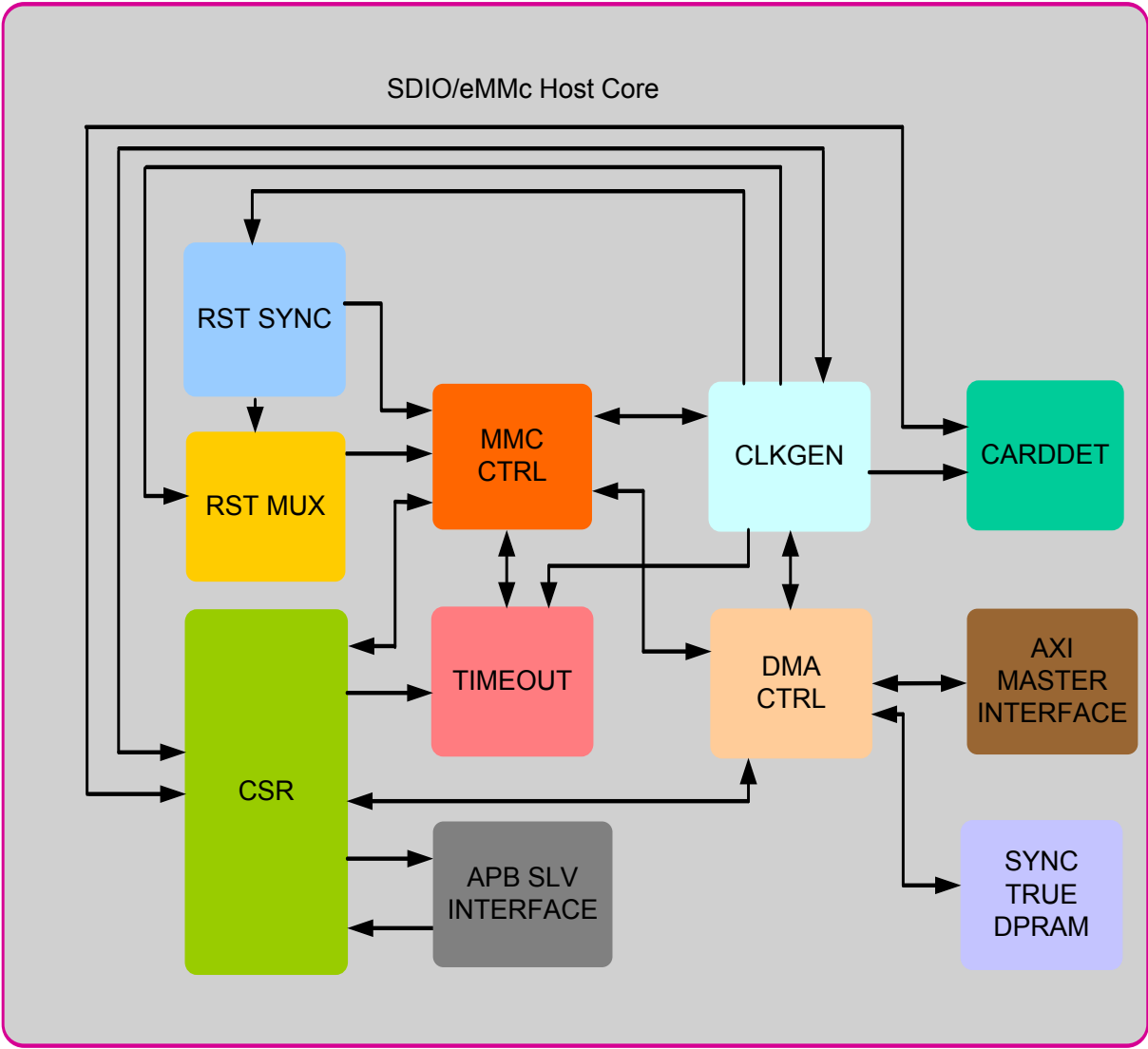
3 Feature Summary

- Compliant with Part A2 SD Host Controller Specification Ver4.0
- Compliant with SDIO Physical Specification version 4.0
- Compliant with Part E1 SDIO specification 3.00
- Compliant with JESD84-B45 Specification and earlier versions.
- Supports different data bus width modes: 1-bit, 4-bit, 8-bit.
- Supports Single byte, Single block, Multi block (finite and infinite) transfers.
- Supports CRC7 checking for command and CRC16 for Data integrity.
- Supports all commands/response types.
- Supports stream transfer operations.
- Supports boot operation mode with simple boot sequence method.
- Supports alternative boot operation mode.
- Supports high speed boot operation.
- Supports HS200 Mode.
- Supports single data rate & dual data rate.
- Supports SDMA and ADMA2 modes.
- Supports SD I/O card, Multi Media Card.
- Supports SDR12, SDR25, DDR50, SDR 50 and SDR104 modes.
- Supports suspend and resume.
- Supports read wait.
- Supports card detection
- Supports Synchronous/Asynchronous Interrupt.
- Fully synthesizable.
- Static synchronous design.
- Scan test ready.
- Simple interface allows easy connection to microprocessor/microcontroller devices.

4 Architecture

4.1 Block Diagram

The block diagram for SDIO/eMMC Host controller IIP is shown below.



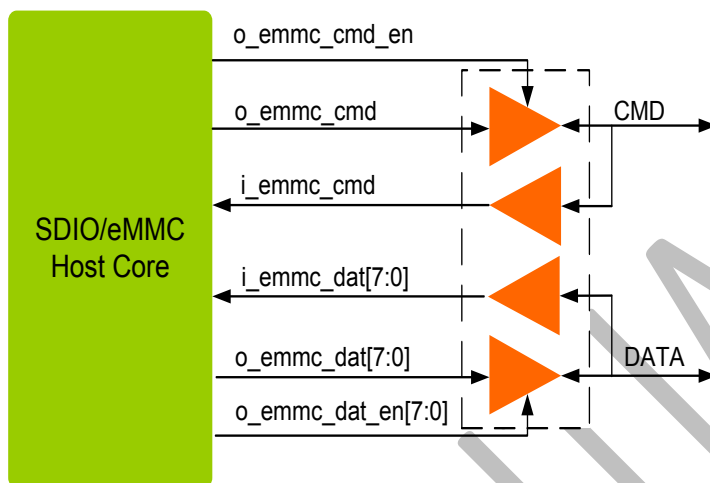
5 Interface

5.1 Core interface

PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
i_clk	INPUT	1		Interface for System Clock.
i_xin_clk	INPUT	1		Interface for Crystal Clock for MMC (200 MHz).
i_rst_n	INPUT	1	i_clk	Interface for Reset Synchronous to i_clk
i_dma_clk	INPUT	1		Interface for dma master Clock.
i_dma_rst_n	INPUT	1	i_dma_clk	Interface for Reset Synchronous to i_dma_clk.
o_emmc_clk	OUTPUT	1		Interface for eMMC clock
o_emmc_rst_n	OUTPUT	1	i_clk	Interface for Hardware reset indication. It is used for user programmable purpose. It will get updated according to VENDOR_SPECIFIC register offset (0x0F8).
o_emmc_cmd_en	OUTPUT	1	o_emmc_clk	Interface for command enable.
o_emmc_cmd	OUTPUT	1	o_emmc_clk	Interface for command.
o_emmc_dat_en	OUTPUT	8	o_emmc_clk	Interface for data enable.
o_emmc_dat	OUTPUT	8	o_emmc_clk	Interface for data.

5.2 IO Pads

GPIO IO pads can be used for connecting the eMMC/SDIO Host Core.



5.3 AXI Master Interface

PORTS	INPUT/OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
o_axi_mst_awid	OUTPUT	AXI_MST_ID_WIDTH	i_dma_clk	AXI Master - AWID - Write Address ID Tag to AXI Slave
o_axi_mst_awaddr	OUTPUT	32	i_dma_clk	AXI Master - AWADDR - Write Address to AXI Slave
o_axi_mst_awlen	OUTPUT	AXI_MST_BURST_LENGTH	i_dma_clk	AXI Master - AWLEN - Write Transfer Length to AXI Slave
o_axi_mst_awsz	OUTPUT	3	i_dma_clk	AXI Master - AWSIZE - Write Transfer Size to AXI Slave
o_axi_mst_awburst	OUTPUT	2	i_dma_clk	AXI Master - AWBURST - Write Transfer Burst Type to AXI Slave
o_axi_mst_awvalid	OUTPUT	1	i_dma_clk	AXI Master - AWVALID - Write Address/Control Valid Indication to AXI Slave

PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
i_axi_mst_awready	INPUT	1	i_dma_clk	AXI Master - AWREADY- Write Address/Control Ready Indication from AXI Slave
o_axi_mst_awlock	OUTPUT	2	i_dma_clk	AXI Master - AWLOCK - Type of access
o_axi_mst_awprot	OUTPUT	3	i_dma_clk	AXI Master - AWPROT - Protection type
o_axi_mst_awcache	OUTPUT	4	i_dma_clk	AXI Master - AWCACHE- Cache type
o_axi_mst_wdata	OUTPUT	32	i_dma_clk	AXI Master - WDATA - Write Data to AXI Slave
o_axi_mst_wstrb	OUTPUT	4	i_dma_clk	AXI Master - WSTRB - Write Data Strobe to AXI Slave
o_axi_mst_wlast	OUTPUT	1	i_dma_clk	AXI Master - WLAST - Write Data Last Indication to AXI Slave
o_axi_mst_wvalid	OUTPUT	1	i_dma_clk	AXI Master - WVALID - Write Data Valid Indication to AXI Slave
i_axi_mst_wready	INPUT	1	i_dma_clk	AXI Master - WREADY - Write Data Ready Indication from AXI Slave
i_axi_mst_bid	INPUT	AXI_MS T_ID_WI DTH	i_dma_clk	AXI Master - ABID - Write Response ID from AXI Slave
i_axi_mst_bresp	INPUT	2	i_dma_clk	AXI Master - ABRESP - Write Response from AXI Slave
i_axi_mst_bvalid	INPUT	1	i_dma_clk	AXI Master - ABVALID- Write Reaponse Valid Indication from AXI Slave
o_axi_mst_bready	OUTPUT	1	i_dma_clk	AXI Master - ABREADY- Write Response Ready Indication to AXI Slave
o_axi_mst_arid	OUTPUT	AXI_MS T_ID_WI DTH	i_dma_clk	AXI Master - ARID - Read Address ID Tag to AXI Slave

PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
o_axi_mst_araddr	OUTPUT	32	i_dma_clk	AXI Master - ARADDR - Read Address to AXI Slave
o_axi_mst_arlen	OUTPUT	AXI_MST_BURST_LENGTH	i_dma_clk	AXI Master - ARLEN - Read Transfer Length to AXI Slave
o_axi_mst_arsize	OUTPUT	3	i_dma_clk	AXI Master - ARSIZE - Read Transfer Size to AXI Slave
o_axi_mst_arburst	OUTPUT	2	i_dma_clk	AXI Master - ARBURST - Read Transfer Burst Type to AXI Slave
o_axi_mst_arlock	OUTPUT	2	i_dma_clk	AXI Master - ARLOCK - Type of access
o_axi_mst_arprot	OUTPUT	3	i_dma_clk	AXI Master - ARPROT - Protection type
o_axi_mst_arcache	OUTPUT	4	i_dma_clk	AXI Master - ARCACHE - Cache type
o_axi_mst_arvalid	OUTPUT	1	i_dma_clk	AXI Master - ARVALID - Read Address/Control Valid Indication to AXI Slave
i_axi_mst_arready	INPUT	1	i_dma_clk	AXI Master - ARREADY - Read Address/Control Ready Indication from AXI Slave
i_axi_mst_rid	INPUT	AXI_MST_ID_WIDTH	i_dma_clk	AXI Master - RID - Read Data ID Tag length from AXI Slave
i_axi_mst_rdata	INPUT	32	i_dma_clk	AXI Master - RDATA - Read Data from AXI Slave
i_axi_mst_rresp	INPUT	2	i_dma_clk	AXI Master - RRESP - Read Response from AXI Slave
i_axi_mst_rlast	INPUT	1	i_dma_clk	AXI Master - RLAST - Read Data Last from AXI Slave
i_axi_mst_rvalid	INPUT	1	i_dma_clk	AXI Master - RVALID - Read Data/Response Valid Indication from AXI Slave

PORTS	INPUT/OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
o_axi_mst_ready	OUTPUT	1	i_dma_clk	AXI Master - RREADY - Read Data/Response Ready Indication to AXI Slave

5.4 RX Flops Interface

PORTS	INPUT/OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
i_emmc_cmd	INPUT	1	o_emmc_clk	Interface for RxFlops Intf CMD In.(Domain emmc_clk)
i_emmc_dat	INPUT	[7:0]	o_emmc_clk	Interface for RxFlops Intf DAT In.(Domain emmc_clk)

5.5 Test/Scan Mode Interface

PORTS	INPUT/OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
i_scan_clk	INPUT	1		Interface for Scan Clock 1
i_scan_mode	INPUT	1	i_clk	Interface for Scan Mode Indication
i_scan_in	INPUT	1	i_clk	Scan - Scan serial data in
o_scan_out	OUTPUT	1	i_clk	Scan - Scan serial data out
i_scan_rst	INPUT	1	i_clk	Scan – Scan reset

5.6 SD IF Interface

PORTS	INPUT/OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
i_sdif_wp	INPUT	1	o_emmc_clk	Interface for SD Intf Write Protect (Active High)
i_sdif_cd_n	INPUT	1	o_emmc_clk	Interface for SD Intf Card Detect (Active Low)

5.7 APB Slave Interface

PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
i_apb_slv_addr	INPUT	[15:0]	i_clk	APB Slave Bus - External interface address input
i_apb_slv_wdata	INPUT	[31:0]	i_clk	APB Slave Bus - External interface write data input
o_apb_slv_rdata	OUTPUT	[31:0]	i_clk	APB Slave Bus - External interface Read data output
i_apb_slv_sel	INPUT	1	i_clk	APB Slave Bus - External interface transfer type input
i_apb_slv_enable	INPUT	1	i_clk	APB Slave Bus - External interface enable control input
i_apb_slv_write	INPUT	1	i_clk	APB Slave Bus - External interface read/write control input
o_apb_slv_ready	OUTPUT	1	i_clk	APB Slave Bus - External interface Ready output
o_apb_slv_err	OUTPUT	1	i_clk	APB Slave Bus - External interface Slave error output
o_irq	OUTPUT	1	i_clk	IRQ SOC - interrupt output

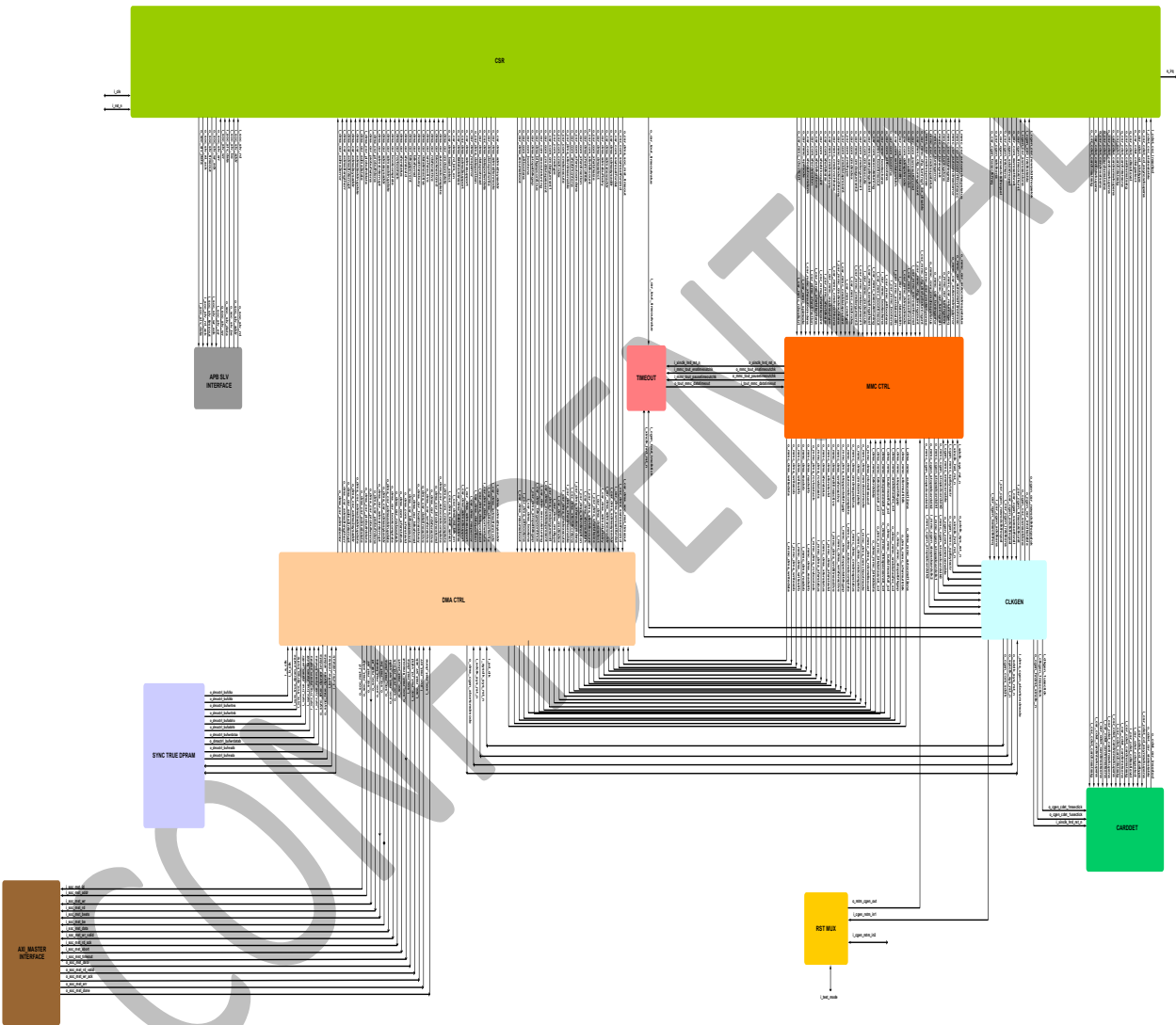
5.8 RTL Parameter

RTL Parameter	Default value	Description
P_CLK_COUNT_VALUE	10'h100	Parameter for Clock divider counter default value for divider value 10'h100. This Parameter (P_CLK_COUNT_VALUE) is used for unready state delay, when u_clkgen.i_csr_sdclkfreqsel [9:0] is switched to 0. The u_clkgen.i_csr_sdclkfreqsel [9:0] is switched to 0, and then the "o_emmc_clk" will be same of "i_xin_clk". To obtain that clock, we have provided delay state. The "o_emmc_clk" will follow the frequency of "i_xin_clk" after that delay. Since you have told, you don't need more delay to switch "o_emmc_clk" to same frequency of "i_xin_clk", We have provided you the parameter to control that delay. In Default it will be 10'h100. So, the "o_emmc_clk" will follow the "i_xin_clk" frequency after more delay time. If you need less number of delay you can change that parameter value to 1 or less value, and the delay state for "o_emmc_clk" to follow "i_xin_clk" will be reduced.
P_MSTAXI_DW	32	Parameter for Master Data Width.
P_BUFFER_SIZE	256	Parameter for Buffer Size. Minimum supported value is 256. For Buffer size 256, need to enable the RD_BLK_EN CSR register(offset – 0X314) to process second block of read data, once DMA transaction of first block data gets completed in Multi block read transaction.
P_MAXOUTSTANDING_REQ	8	Parameter for Maximum Number of Outstanding Requests
P_SWRSTTIMER	12'hFFF	Parameter for Number of System Clocks for SwReset_For_CMD/DAT
P_CARDDETMSCNT	12'h3FF	Parameter for Num of Micro/Milli Seconds for Card Detect (Removable).
P_CARDDETMSCNT_EMBEDDED	12'h002	Parameter for Num of Micro/Milli Seconds for Card Detect (Embedded)

RTL Parameter	Default value	Description
P_CARDDDET_TIMEUNIT	1'b0	Parameter for 1'b0 for milli sec tick and 1'b1 for micro sec tick.
P_NCR_VALUE	64	Parameter for NCR Value for Command Response Timeout. (Maximum 64)
P_NCRC_VALUE	16	Parameter for NCRC Value for CRC Response Timeout in Non-HS400 Mode.
AXI_MST_ID_WIDTH	2	Parameter for WID/AWID/RID/ARID Width in AXI Master bus
AXI_MST_BURST_LENGTH	8	Parameter for Burst Length AXI Master bus
FW	32	This parameter is used to define width of buffer/SRAM write/read data
FP	8	This parameter is used to define width of buffer/SRAM write/read address

6 Micro Architecture

6.1 Block Diagram



6.2 Module Description

6.2.1 MMC Control Module

6.2.1.1 Functionality

This module has command generation state machine. It monitors the command request from the CSR registers or the transfer done indication from the module DMA control and generates the command sequence. Once the command is transmitted, it waits for response state machine to receive response and after that the interrupts/flags are set for the software to read the status. Once the command is completed, the data state machine is started to send/receive data if the data has to be transferred. In case of read, the data reception starts with end of command transmission and also this module includes transmit data state machine. The receive CRC state machine starts receiving CRC from the DAT0 Line. If the CRC status is not received within 15 clocks, then a timeout indication is generated

6.2.1.2 Port Descriptions

S.No	PORTS	INPUT/OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
1	i_xin_clk	INPUT	1		XIN CLOCK - Interface for Crystal Clock for SD.
2	i_sd_clk	INPUT	1		SDCLOCK - Interface for SD Clock (Generated from xin_clk).
3	i_rclk	INPUT	1		RCLOCK - Interface for SD Clock (Receive Side from Card).
4	i_clk	INPUT	1		MMC CLOCK - Interface for SD Clock (Receive Side from Card).
5	i_xinclk_hrd_rst_n	INPUT	1	i_xin_clk	HARD RESET - Interface for Hard Reset Sync to xin_clk.
6	i_sdclk_sys_rst_n	INPUT	1	i_sd_clk	SDCLOCK RESET - Interface for System Reset Synchronous to sd_clk.
7	i_rclk_sys_rst_n	INPUT	1	i_rclk	RCLOCK RESET - Interface for System Reset Synchronous to rclk.

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
8	i_rst_n	INPUT	1	i_clk	SYSTEM CLOCK - Interface for System Reset Synchronous to emmc_clk.
9	i_csr_mmc_cfg_dis1 p5xtuningcnt	INPUT	1	i_clk	CFG - Interface for CoreCFG: Disable 1.5x Iterations from Tuning Count.
10	i_csr_mmc_cfg_tunin gcount	INPUT	6	i_clk	CFG - Interface for CoreCFG: Tuning Count (4/8/16/32). Number of Rx Taps Implemented.
11	o_mmc_core_txdat6e napos	OUTPUT	1	i_sd_clk	MMCCTRL TO CORE - Interface for Transmit DATA6 Output Enable on +ve Edge of Clock.
12	i_core_mmc_rxflops_ cmd	INPUT	1	o_emmc_cl k	RXFLOPS - Interface for RxFloPS: CMD Line (Flopped on +ve Edge)
13	i_core_mmc_rxflops_ dat0	INPUT	1	o_emmc_cl k	RXFLOPS - Interface for RxFloPS: DAT0 Line (Flopped on +ve Edge)
14	i_core_mmc_rxflops_ dat1	INPUT	1	o_emmc_cl k	RXFLOPS - Interface for RxFloPS: DAT1 Line (Flopped on +ve Edge)
15	i_core_mmc_rxflops_ dat2	INPUT	1	o_emmc_cl k	RXFLOPS - Interface for RxFloPS: DAT2 Line (Flopped on +ve Edge)
16	i_core_mmc_rxflops_ dat3	INPUT	1	o_emmc_cl k	RXFLOPS - Interface for RxFloPS: DAT3 Line (Flopped on +ve Edge)
17	i_core_mmc_rxflops_ dat4	INPUT	1	o_emmc_cl k	RXFLOPS - Interface for RxFloPS: DAT4 Line (Flopped on +ve Edge)
18	i_core_mmc_rxflops_ dat5	INPUT	1	o_emmc_cl k	RXFLOPS - Interface for RxFloPS: DAT5 Line (Flopped on +ve Edge)
19	i_core_mmc_rxflops_ dat6	INPUT	1	o_emmc_cl k	RXFLOPS - Interface for RxFloPS: DAT6 Line (Flopped on +ve Edge)

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
20	i_core_mmc_rxflops_dat7	INPUT	1	o_emmc_clk	RXFLOPS - Interface for RxFlops: DAT7 Line (Flopped on +ve Edge)
21	i_core_mmc_rxflops_dat0n	INPUT	1	o_emmc_clk	RXFLOPS - Interface for RxFlops: DAT0 Line (Flopped on -ve Edge)
22	i_core_mmc_rxflops_dat1n	INPUT	1	o_emmc_clk	RXFLOPS - Interface for RxFlops: DAT1 Line (Flopped on -ve Edge)
23	i_core_mmc_rxflops_dat2n	INPUT	1	o_emmc_clk	RXFLOPS - Interface for RxFlops: DAT2 Line (Flopped on -ve Edge)
24	i_core_mmc_rxflops_dat3n	INPUT	1	o_emmc_clk	RXFLOPS - Interface for RxFlops: DAT3 Line (Flopped on -ve Edge)
25	i_core_mmc_rxflops_dat4n	INPUT	1	o_emmc_clk	RXFLOPS - Interface for RxFlops: DAT4 Line (Flopped on -ve Edge)
26	i_core_mmc_rxflops_dat5n	INPUT	1	o_emmc_clk	RXFLOPS - Interface for RxFlops: DAT5 Line (Flopped on -ve Edge)
27	i_core_mmc_rxflops_dat6n	INPUT	1	o_emmc_clk	RXFLOPS - Interface for RxFlops: DAT6 Line (Flopped on -ve Edge)
28	i_core_mmc_rxflops_dat7n	INPUT	1	o_emmc_clk	RXFLOPS - Interface for RxFlops: DAT7 Line (Flopped on -ve Edge)
29	i_csr_mmc_swresetf_orcmd	INPUT	1	i_clk	CSR TO MMCCTRL - Interface for CSR: SW Reset for CMD.
30	i_csr_mmc_swresetf_ordat	INPUT	1	i_clk	CSR TO MMCCTRL - Interface for CSR: SW Reset for DAT.
31	i_csr_mmc_ddrspdm_ode	INPUT	1	i_clk	CSR TO MMCCTRL - Interface for CSR: DDR50 Mode Indication.

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
32	i_csr_mmc_datawidth	INPUT	1	i_clk	CSR TO MMCCTRL - Interface for CSR: Data Width (0:1Bit, 1:4Bits).
33	i_csr_mmc_extdatawidth	INPUT	1	i_clk	CSR TO MMCCTRL - Interface for CSR: Extendata Data Width (8-bit Mode)
34	i_csr_mmc_bootena	INPUT	1	i_clk	CSR TO MMCCTRL - Interface for CSR: Boot Enable.
35	i_csr_mmc_altbootena	INPUT	1	i_clk	CSR TO MMCCTRL - Interface for CSR: Alt Boot Enable.
36	i_csr_mmc_bootackena	INPUT	1	i_clk	CSR TO MMCCTRL - Interface for CSR: Boot Mode ACK Enable
37	i_csr_mmc_boottimeout	INPUT	32	i_clk	CSR TO MMCCTRL - Interface for CSR: Boot Timeout Count
38	i_csr_mmc_executetuning	INPUT	1	i_clk	CSR TO MMCCTRL - Interface for CSR: Execute Tuning Indication
39	i_csr_mmc_samplingclkssel	INPUT	1	i_clk	CSR TO MMCCTRL - Interface for CSR: Sampling Clock Selection.
40	i_csr_mmc_cmddexecute	INPUT	1	i_clk	CSR TO MMCCTRL - Interface for CSR: Command Execute Ind (Edge Det on sd_clk).
41	i_csr_mmc_cmddargument1	INPUT	32	i_clk	CSR TO MMCCTRL - Interface for CSR: Command Argument#1.
42	i_csr_mmc_cmddargument2	INPUT	32	i_clk	CSR TO MMCCTRL - Interface for CSR: Command Argument#2.
43	i_csr_mmc_cmddindex	INPUT	6	i_clk	CSR TO MMCCTRL - Interface for CSR: Command Index.

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
44	i_csr_mmc_cmdtype	INPUT	2	i_clk	CSR TO MMCCTRL - Interface for CSR: Command Type.
45	i_csr_mmc_cmddata present	INPUT	1	i_clk	CSR TO MMCCTRL - Interface for CSR: Command Data Present Indica
46	i_csr_mmc_cmdinde xchkena	INPUT	1	i_clk	CSR TO MMCCTRL - Interface for CSR: Command Index Check Enable.
47	i_csr_mmc_cmdcrcc hkena	INPUT	1	i_clk	CSR TO MMCCTRL - Interface for CSR: Command CRC Check Enable.
48	i_csr_mmc_cmdrespt ype	INPUT	2	i_clk	CSR TO MMCCTRL - Interface for CSR: Command Response Type.
49	i_csr_mmc_xferblksiz e	INPUT	12	i_clk	CSR TO MMCCTRL - Interface for CSR: Transfer Block Size value.
50	i_csr_mmc_xferblkco unt	INPUT	32	i_clk	CSR TO MMCCTRL - Interface for CSR: Transfer Block Count.
51	i_csr_mmc_xferblkcn tena	INPUT	1	i_clk	CSR TO MMCCTRL - Interface for CSR: Transfer Block Count Enable.
52	i_csr_mmc_xferdirect ion	INPUT	1	i_clk	CSR TO MMCCTRL - Interface for CSR: Transfer Direction (0:Write, 1:Read)
53	i_csr_mmc_xferautoc mdena	INPUT	2	i_clk	CSR TO MMCCTRL - Interface for CSR: Transfer Auto Command Enable (00:Dis, 01:CMD12, 10:CMD23).
54	o_mmc_core_txdat7o utpos	OUTPUT	2	i_sd_clk	MMCCTRL TO CORE - Interface for Transmit DATA7 Output (Both Bits) on +ve Edge of Clock.

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
55	o_mmc_core_txdat7e napos	OUTPUT	1	i_sd_clk	MMCCTRL TO CORE - Interface for Transmit DATA7 Output Enable on +ve Edge of Clock.
56	i_csr_mmc_spimode	INPUT	1	i_clk	CSR TO MMCCTRL - Interface for CSR: SPI Mode.
57	i_csr_mmc_rdwaitctrl	INPUT	1	i_clk	CSR TO MMCCTRL - Interface for CSR: Read Wait Control.
58	i_cgen_mmc_emmce nable	INPUT	1	l_clk	CLKGEN TO MMC - Interface for CLK_GEN: SD Interface Enable Indication.
59	i_cgen_mmc_spibyte ctr	INPUT	3	i_clk	CLKGEN TO MMC - Interface for CLK_GEN: SPI Byte Counter.
60	i_tout_mmc_datatime out	INPUT	1	i_xin_clk	DATA TIMEOUT TO MMC - Interface for P_TIMEOUT: Data Timeout Indication.
61	i_dma_mmc_autocm d12ena	INPUT	1	i_dma_clk	DMACTRL TO MMCCTRL - Interface for DMA_CTRL: Auto CMD12 Enable after Transfer.
62	i_dma_mmc_xfermult iblk	INPUT	1	i_dma_clk	DMACTRL TO MMCCTRL - Interface for DMA_CTRL: Transfer Multi Block Indication.
63	i_dma_mmc_stopatbl kgap	INPUT	1	i_dma_clk	DMACTRL TO MMCCTRL - Interface for DMA_CTRL: Stop at Block Gap Indication.
64	i_dma_mmc_enddata xfer	INPUT	1	i_dma_clk	DMACTRL TO MMCCTRL - Interface for DMA_CTRL: End Data Xfer Indication.
65	i_dma_mmc_bufalmo stfull	INPUT	1	i_dma_clk	DMACTRL TO MMCCTRL - Interface for DMA_CTRL: Buffer is getting Almost Full.
66	i_dma_mmc_blkspac eavail	INPUT	1	i_dma_clk	DMACTRL TO MMCCTRL - Interface for DMA_CTRL: Block Space Available Indication (Card Read).

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
67	i_dma_mmc_blkdata avail	INPUT	1	i_dma_clk	DMACTRL TO MMCCTRL - Interface for DMA_CTRL: Block Data Available Indication (Card Write)
68	i_dma_mmc_blkrddata	INPUT	64	i_dma_clk	DMACTRL TO MMCCTRL - Interface for DMA_CTRL: Block Read data from Buffer (Side B).
69	o_mmc_dma_rdcmdissued	OUTPUT	1	i_sd_clk	MMCCTRL TO DMACTRL - Interface for Read Command Issued Indication.
70	o_mmc_dma_mmcbootmode	OUTPUT	1	i_sd_clk	MMCCTRL TO DMACTRL - Interface for MMC Boot Mode Indication.
71	o_mmc_dma_rcvbootack	OUTPUT	1	o_emmc_clk	MMCCTRL TO DMACTRL - Interface for Receive Boot Ack Indication.
72	o_mmc_dma_cmdcomplete	OUTPUT	1	i_sd_clk	MMCCTRL TO DMACTRL - Interface for Command Complete Indication.
73	o_mmc_com_cmdrespstatus	OUTPUT	4	i_sd_clk	MMCCTRL TO COMMON - Interface for Command Response Status. {EB, IDX, CRC, TOUT}.
74	o_mmc_csr_cmdresponse	OUTPUT	128	o_emmc_clk	MMCCTRL TO CSR - Interface for Received Response.
75	o_mmc_dma_autocmdcomplete	OUTPUT	1	i_sd_clk	MMCCTRL TO DMACTRL - Interface for Auto Command Complete Indication.
76	o_mmc_csr_autocmdnexterror	OUTPUT	1	I_clk	MMCCTRL TO CSR - Interface for Auto Command Next Error.
77	o_mmc_csr_autocmdrespstatus	OUTPUT	4	i_sd_clk	MMCCTRL TO CSR - Interface for Auto Command Response Status {EB, IDX, CRC, TOUT}

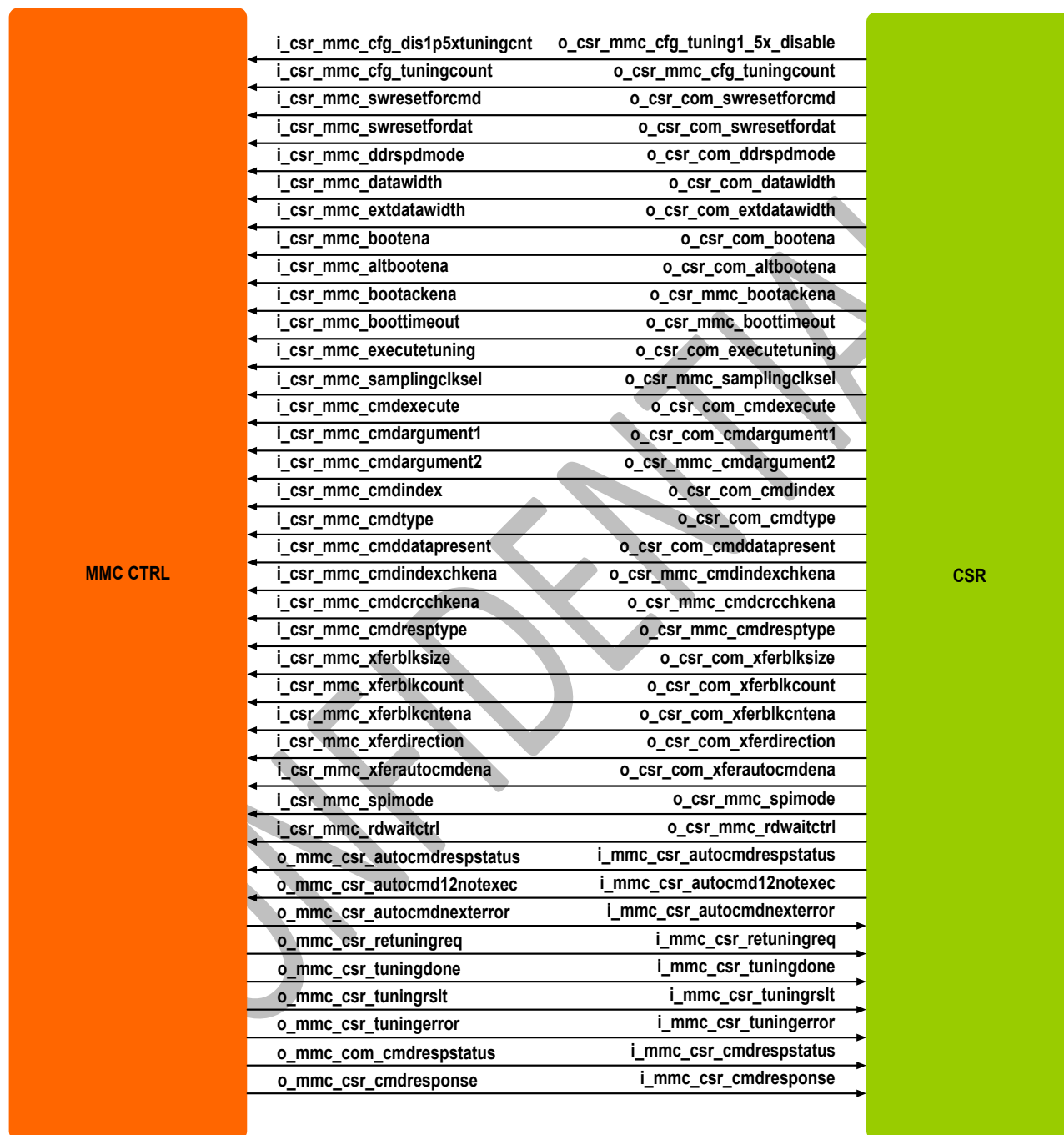
S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
78	o_mmc_csr_autocmd 12notexec	OUTPUT	1	i_sd_clk	MMCCTRL TO CSR - Interface for Auto Command (CMD12) Not Executed Indication.
79	o_mmc_tout_enatime outchk	OUTPUT	1	i_sd_clk	MMCCTRL TO TIMEOUT - Interface for Enable Data Timeout Check.
80	o_mmc_tout_pauseti meoutchk	OUTPUT	1	i_sd_clk	MMCCTRL TO TIMEOUT - Interface for Pause Data Timeout Check.
81	o_mmc_cgen_stopsd cardclk0	OUTPUT	1	i_sd_clk	MMCCTRL TO CLOCKGEN - Interface for Stop eMMC Clock Indication (Transmit Stream Mode, Empty BlkBuffer)
82	o_mmc_cgen_stopsd cardclk1	OUTPUT	1	i_sd_clk	MMCCTRL TO CLOCKGEN - Interface for Stop eMMC Clock Indication (Receive Normal Data Operation)
83	o_mmc_cgen_stopsd cardclk2	OUTPUT	1	i_sd_clk	MMCCTRL TO CLOCKGEN - Interface for Stop eMMC Clock Indication (Receive Stop at Block Gap).
84	o_mmc_cgen_stopsd cardclk3	OUTPUT	1	i_sd_clk	MMCCTRL TO CLOCKGEN - Interface for Stop eMMC Clock Indication (Receive Stream Mode, Full BlkBuffer).
85	o_mmc_dma_stoppe datblkgap	OUTPUT	1	i_sd_clk	MMCCTRL TO DMACTRL - Interface for Read Stopped at Block Gap Indication.
86	o_mmc_dma_wrxfera ctive	OUTPUT	1	i_sd_clk	MMCCTRL TO DMACTRL - Interface for Write Transfer Active Indication.
87	o_mmc_dma_wrdtlin eactive	OUTPUT	1	i_sd_clk	MMCCTRL TO DMACTRL - Interface for Write Data Line Active Indication.
88	o_mmc_dma_xfersts vld	OUTPUT	1	i_sd_clk	MMCCTRL TO DMACTRL - Interface for Xfer Status Valid.

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
89	o_mmc_dma_xferstat us	OUTPUT	3	i_sd_clk	MMCCTRL TO DMACTRL - Interface for Xfer Status {ENDBIT Err, CRC Err, TIMEOUT Err}
90	o_mmc_dma_rcvdata sts	OUTPUT	1	i_sd_clk	MMCCTRL TO DMACTRL - Interface for Receive Data Status
91	o_mmc_csr_retuning req	OUTPUT	1	i_clk	MMCCTRL TO CSR - Interface for Retuning Request.
92	o_mmc_csr_tuningdone	OUTPUT	1	i_sd_clk	MMCCTRL TO CSR - Interface for Tuning Done Indication.
93	o_mmc_csr_tuningresult	OUTPUT	1	i_sd_clk	MMCCTRL TO CSR - Interface for Tuning Result (1:Pass, 0:Fail) Indication.
94	o_mmc_csr_tuningerror	OUTPUT	1	i_sd_clk	MMCCTRL TO CSR - Interface for Tuning Error Indication.
95	o_mmc_core_rclksrc	OUTPUT	2	i_xin_clk	MMCCTRL - Interface for Receive Clock Source (00:Loopedback, 01:SDR, 10:DDR, 11:Tuned Clock)
96	o_mmc_core_rclktap	OUTPUT	5	i_xin_clk	MMCCTRL - Interface for Receive Clock Tap for Tuning.
97	o_mmc_cgen_rclkgate	OUTPUT	1	i_xin_clk	MMCCTRL TO CLOCKGEN - Interface for Receive Clock Gate.
98	o_mmc_dma_readstrobe	OUTPUT	1	i_sd_clk	MMCCTRL TO DMACTRL - Interface for Block Buffer Read Strobe.
99	o_mmc_dma_readendofblock	OUTPUT	1	i_sd_clk	MMCCTRL TO DMACTRL - Interface for Block Buffer Read End of Block Indication
100	o_mmc_dma_writestrobe	OUTPUT	1	i_sd_clk	MMCCTRL TO DMACTRL - Interface for Write Strobe Indication to Block Buffer (sd_clk)

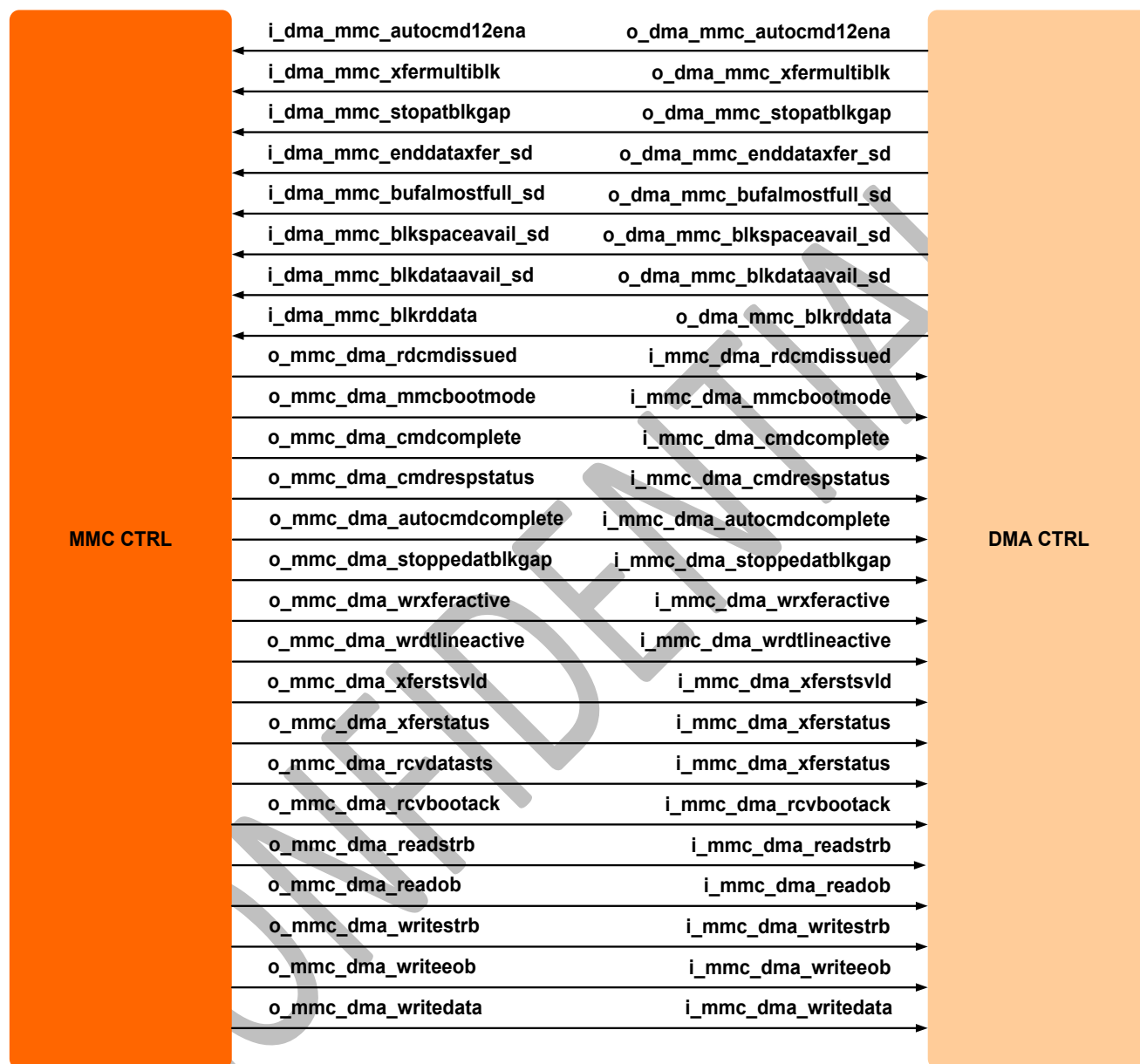
S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
101	o_mmc_dma_writeeob	OUTPUT	1	i_sd_clk	MMCCTRL TO DMACTRL - Interface for Write End of Block Indication to Block Buffer (sd_clk)
102	o_mmc_dma_writedata	OUTPUT	64	i_sd_clk	MMCCTRL TO DMACTRL - Interface for Write Data to Block Buffer (sd_clk)
103	o_mmc_core_txcmdoutpos	OUTPUT	1	o_emmc_clk	MMCCTRL TO CORE - Interface for Transmit CMD Output on +ve Edge of Clock.
104	o_mmc_core_txcmdenapos	OUTPUT	1	i_sd_clk	MMCCTRL TO CORE - Interface for Transmit CMD Output Enable on +ve Edge of Clock.
105	o_mmc_core_txdat0outpos	OUTPUT	2	i_sd_clk	MMCCTRL TO CORE - Interface for Transmit DATA0 Output (Both Bits) on +ve Edge of Clock.
106	o_mmc_core_txdat0enapos	OUTPUT	1	i_sd_clk	MMCCTRL TO CORE - Interface for Transmit DATA0 Output Enable on +ve Edge of Clock.
107	o_mmc_core_txdat1outpos	OUTPUT	2	i_sd_clk	MMCCTRL TO CORE - Interface for Transmit DATA1 Output (Both Bits) on +ve Edge of Clock.
108	o_mmc_core_txdat1enapos	OUTPUT	1	i_sd_clk	MMCCTRL TO CORE - Interface for Transmit DATA1 Output Enable on +ve Edge of Clock.
109	o_mmc_core_txdat2outpos	OUTPUT	2	i_sd_clk	MMCCTRL TO CORE - Interface for Transmit DATA2 Output (Both Bits) on +ve Edge of Clock.
110	o_mmc_core_txdat2enapos	OUTPUT	1	i_sd_clk	MMCCTRL TO CORE - Interface for Transmit DATA2 Output Enable on +ve Edge of Clock.

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
111	o_mmc_core_txdat3o utpos	OUTPUT	2	i_sd_clk	MMCCTRL TO CORE - Interface for Transmit DATA3 Output (Both Bits) on +ve Edge of Clock.
112	o_mmc_core_txdat3e napos	OUTPUT	1	i_sd_clk	MMCCTRL TO CORE - Interface for Transmit DATA3 Output Enable on +ve Edge of Clock.
113	o_mmc_core_txdat4o utpos	OUTPUT	2	i_sd_clk	MMCCTRL TO CORE - Interface for Transmit DATA4 Output (Both Bits) on +ve Edge of Clock.
114	o_mmc_core_txdat4e napos	OUTPUT	1	i_sd_clk	MMCCTRL TO CORE - Interface for Transmit DATA4 Output Enable on +ve Edge of Clock.
115	o_mmc_core_txdat5o utpos	OUTPUT	2	i_sd_clk	MMCCTRL TO CORE - Interface for Transmit DATA5 Output (Both Bits) on +ve Edge of Clock.
116	o_mmc_core_txdat5e napos	OUTPUT	1	i_sd_clk	MMCCTRL TO CORE - Interface for Transmit DATA5 Output Enable on +ve Edge of Clock.
117	o_mmc_core_txdat6o utpos	OUTPUT	2	i_sd_clk	MMCCTRL TO CORE - Interface for Transmit DATA6 Output (Both Bits) on +ve Edge of Clock.

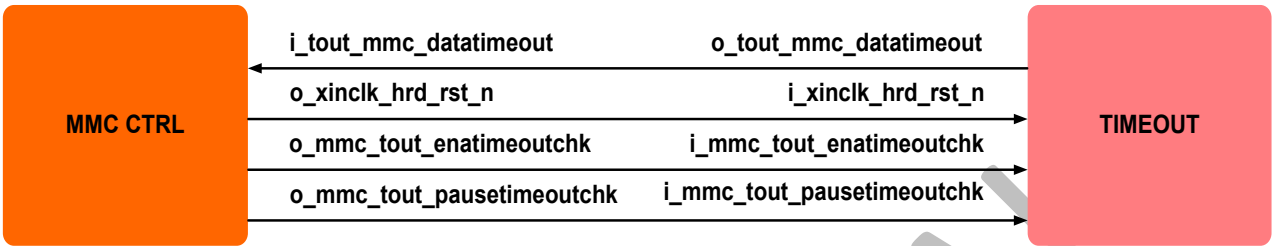
6.2.1.3 MMC and CSR Interface



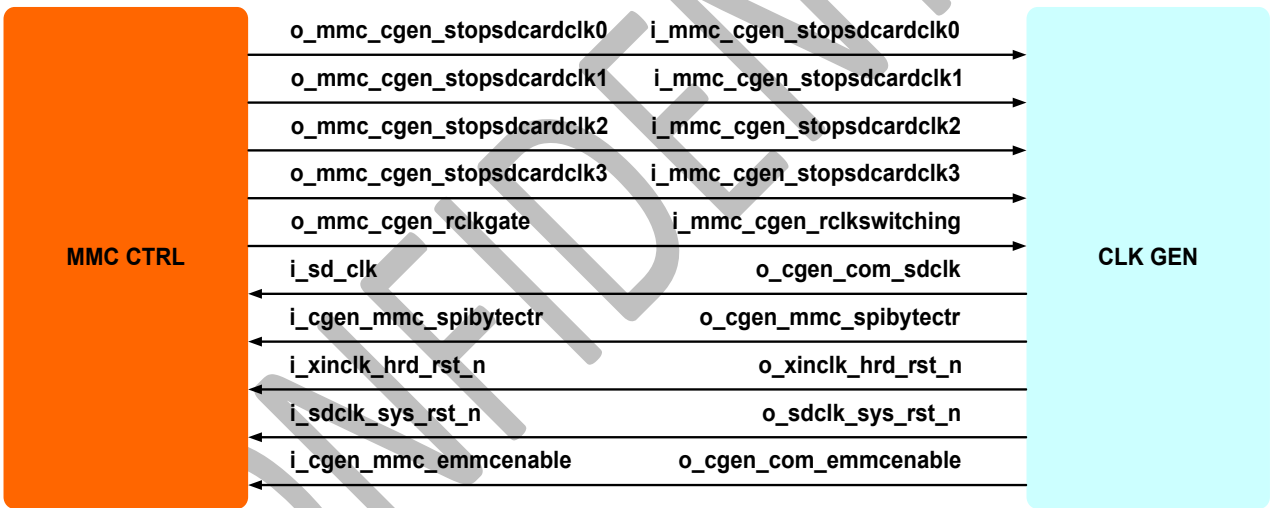
6.2.1.4 MMC and DMA Control Interface



6.2.1.5 MMC and Timeout Interface



6.2.1.6 MMC and CLKGEN Interface



6.2.2 DMA Control Module

6.2.2.1 Functionality

This module includes all SDMA and ADMA2 state machines. For the MMC Card write operation, the data from block buffer is transferred to MMC interface.

6.2.2.2 Port Descriptions

S.No	PORTS	INPUT/OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
1	i_clk	INPUT	1	i_dma_clk	DMACLK - Interface for Clock.
2	i_sd_clk	INPUT	1	i_sd_clk	SDCLK - Interface for SD Clock (Generated from xin_clk).
3	i_sysclk	INPUT	1	i_clk	SYSTEM CLK - Interface for System(Host Interface) Clock.
4	i_sysclk_sys_rst_n	INPUT	1	i_dma_clk	RESET - Interface for Reset Synchronous to clk.
5	i_sdclk_sys_rst_n	INPUT	1	i_sd_clk	SYSTEM RESET - Interface for System Reset Synchronous to sd_clk.
6	i_rst_n	INPUT	1	i_dma_clk	SYSTEM RESET - Interface for System Reset Synchronous to sd_clk.
7	i_core_dma_swrstdly_ena	INPUT	1	i_dma_clk	SOFTWARE RESET - Interface for Software reset Delaying Enable (For Async Abort and Error Recovery ECOs).
8	i_csr_dma_soc_mst_timeout	INPUT	16	i_clk	CSR to DMACTRL - SOC Master Timeout
9	i_csr_dma_swresetforcmd	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: SW Reset for CMD.
10	i_csr_dma_swresetfordat	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: SW Reset for DAT
11	i_csr_dma_sdmasysaddr	INPUT	64	i_clk	CSR to DMACTRL - Interface for CSR: SDMA System Address.

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
12	i_csr_dma_sdmanew addr	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: SDMA New Address Indication.
13	i_csr_dma_sdmabufb dry	INPUT	3	i_clk	CSR to DMACTRL - Interface for CSR: SDMA Buffer Boundary.
14	i_csr_dma_dmaselec t	INPUT	2	i_clk	CSR to DMACTRL - Interface for CSR: DMA Select Indication.
15	i_csr_dma_xferblksiz e	INPUT	12	i_clk	CSR to DMACTRL - Interface for CSR: Transfer Block Size value.
16	i_csr_dma_xferblkco unt	INPUT	32	i_clk	CSR to DMACTRL - Interface for CSR: Transfer Block Count value.
17	i_csr_dma_xfermultib lk	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: Transfer Multi Block Indication (0:Single Block).
18	i_csr_dma_xferdirecti on	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: Transfer Direction (0:Write, 1:Read)
19	i_csr_dma_xferautoc mdena	INPUT	2	i_clk	CSR to DMACTRL - Interface for CSR: Transfer Auto Command Enable (00:Dis, 01:CMD12, 10:CMD23).
20	i_csr_dma_xferblkcnt ena	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: Transfer Block Count Enable.
21	i_csr_dma_xferdmae nable	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: Transfer DMA Enable.
22	i_csr_dma_cmdexec ute	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: Command Execute Indication.
23	i_csr_dma_cmdindex	INPUT	6	i_clk	CSR to DMACTRL - Interface for CSR: Command Index.
24	i_csr_dma_cmdtype	INPUT	2	i_clk	CSR to DMACTRL - Interface for CSR: Command Type.

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
25	i_csr_dma_cmdargument1	INPUT	32	i_clk	CSR to DMACTRL - Interface for CSR: Command Argument #1.
26	i_csr_dma_cmddatapresent	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: Command Data Present Indication.
27	i_csr_dma_executetuning	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: Execute Tuning Indication.
28	i_csr_dma_blkgapcontinue	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: Continue after Block Gap Stop.
29	i_csr_dma_stopatblkgap	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: Stop at Block Gap.
30	i_csr_dma_rd_blk_en	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: read block en to process one after another block of data
31	i_csr_dma_beat_select	INPUT	2	i_clk	CSR to DMACTRL - Interface for CSR: AHB Transaction Beat select
32	i_csr_dma_bootenable	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: Boot Enable.
33	i_csr_dma_altbootenable	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: Alt Boot Enable.
34	i_csr_dma_resumeboot	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: Resume Boot Indication.
35	i_csr_dma_64bitaddress	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: Enable 64-bit Addressing
36	o_soc_mst_wr	OUTPUT	1	i_dma_clk	SOC Master - write input
37	o_soc_mst_rd	OUTPUT	1	i_dma_clk	SOC Master - read input
38	o_soc_mst_beats	OUTPUT	9	i_dma_clk	SOC Master - number of beats input

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
39	i_csr_dma_adma2sysaddr	INPUT	64	i_clk	CSR to DMACTRL - Interface for CSR: ADMA2 System Address (SDMA SysAddr in 64-bit Addressing).
40	i_csr_dma_adma2lenmode	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: ADMA2 Length Mode (0:16-bit Length, 1:26-bit Length)
41	i_csr_dma_dataportwrite	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: DataPort Write Indication.
42	i_csr_dma_dataportwdata	INPUT	32	i_clk	CSR to DMACTRL - Interface for CSR: DataPort Write Data.
43	i_csr_dma_dataportread	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: DataPort Read Indication.
44	i_csr_dma_dataportbenable	INPUT	4	i_clk	CSR to DMACTRL - Interface for CSR: DataPort Access ByteEnables.
45	i_csr_dma_resperror	INPUT	1	i_clk	CSR to DMACTRL - Interface for CSR: Response Error Indication.
46	i_csr_dma_sdmasupport	INPUT	1	i_clk	CSR to DMACTRL - Interface for SDMA Support (0:Not Supported, 1:Supported)
47	i_csr_dma_adma2support	INPUT	1	i_clk	CSR to DMACTRL - Interface for ADMA2 Support (0:Not Supported, 1:Supported)
48	o_soc_mst_wr_valid	OUTPUT	1	i_dma_clk	SOC Master - write data valid input
49	i_mmc_dma_rdcmdissued	INPUT	1	i_sd_clk	MMCCTRL to DMACTRL - Interface for SD_CTRL: Read Command Issued Indication (sd_clk domain signal)
50	i_mmc_dma_cmdcomplete	INPUT	1	i_sd_clk	MMCCTRL to DMACTRL - Interface for SD_CTRL: Command Complete Indication (sd_clk domain signal)

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
51	i_mmc_dma_cmdres pstatus	INPUT	4	i_sd_clk	MMCCTRL to DMACTRL - Interface for SD_CTRL: Command Response Status (sd_clk domain signal).
52	i_mmc_dma_autocm dcomplete	INPUT	1	i_sd_clk	MMCCTRL to DMACTRL - Interface for SD_CTRL: Auto Command Complete Indication (sd_clk domain signal)
53	i_mmc_dma_stopped atblkgap	INPUT	1	i_sd_clk	MMCCTRL to DMACTRL - Interface for SD_CTRL: Read Stopped at Block Gap Indication (sd_clk domain signal)
54	i_mmc_dma_wrxfera ctive	INPUT	1	i_sd_clk	MMCCTRL to DMACTRL - Interface for SD_CTRL: Write in Progress (sd_clk domain signal)
55	i_mmc_dma_wrtdline active	INPUT	1	i_sd_clk	MMCCTRL to DMACTRL - Interface for SD_CTRL: Write Data Line Active Indication (sd_clk domain signal)
56	i_mmc_dma_xferstsv ld	INPUT	1	i_sd_clk	MMCCTRL to DMACTRL - Interface for SD_CTRL: Read/Write Block Transfer Status Valid.
57	i_mmc_dma_xferstat us	INPUT	3	i_sd_clk	MMCCTRL to DMACTRL - Interface for SD_CTRL: Read/Write Block Transfer Status {ENDBIT Err, CRC Err, TIMEOUT Err}
58	i_mmc_dma_rcvdata sts	INPUT	1	i_sd_clk	MMCCTRL to DMACTRL - Interface for SD_CTRL: Receiving Data Status (sd_clk domain signal).
59	i_mmc_dma_readstr b	INPUT	1	i_sd_clk	MMCCTRL to DMACTRL - Interface for SD_CTRL: Read Strobe (sd_clk domain signal).

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
60	i_mmc_dma_readeo b	INPUT	1	i_sd_clk	MMCCTRL to DMACTRL - Interface for SD_CTRL: Read End of Block Indication (sd_clk domain signal).
61	i_mmc_dma_writestr b	INPUT	1	i_sd_clk	MMCCTRL to DMACTRL - Interface for SD_CTRL: Write Strobe (sd_clk domain signal).
62	i_mmc_dma_wroteo b	INPUT	1	i_sd_clk	MMCCTRL to DMACTRL - Interface for SD_CTRL: Write End of Block Indication (sd_clk domain signal).
63	i_mmc_dma_writedata	INPUT	64	i_sd_clk	MMCCTRL to DMACTRL - Interface for SD_CTRL: Write Data(sd_clk) domain bus.
64	i_mmc_dma_mmcbootmode	INPUT	1	i_sd_clk	MMCCTRL to DMACTRL - Interface for SD_CTRL: MMC Boot Mode Indication (sd_clk domain signal)
65	i_mmc_dma_rcvbootack	INPUT	1	o_emmc_clk	MMCCTRL to DMACTRL - Interface for SD_CTRL: Receive Boot Ack Indication.
66	o_soc_mst_rd_ack	OUTPUT	1	i_dma_clk	SOC Master - read ack input
67	o_soc_mst_abort	OUTPUT	1	i_dma_clk	SOC Master - abort transfer input
68	i_soc_mst_err	INPUT	1	i_dma_clk	SOC Master - error transfer output
69	i_core_dma_buffer_rddataa	INPUT	FW	i_a_clk	RAM to DMACTRL - Interface for SD_BUF: Read Data from Side A.
70	i_core_dma_buffer_rddatab	INPUT	FW	i_b_clk	RAM to DMACTRL - Interface for SD_BUF: Read Data from Side B
71	o_dma_core_bufclka	OUTPUT	1	i_dma_clk	DMACTRL to CORE - Interface for BufferA Clk.
72	o_dma_core_bufclkb	OUTPUT	1	i_dma_clk	DMACTRL to CORE - Interface for BufferB Clk.

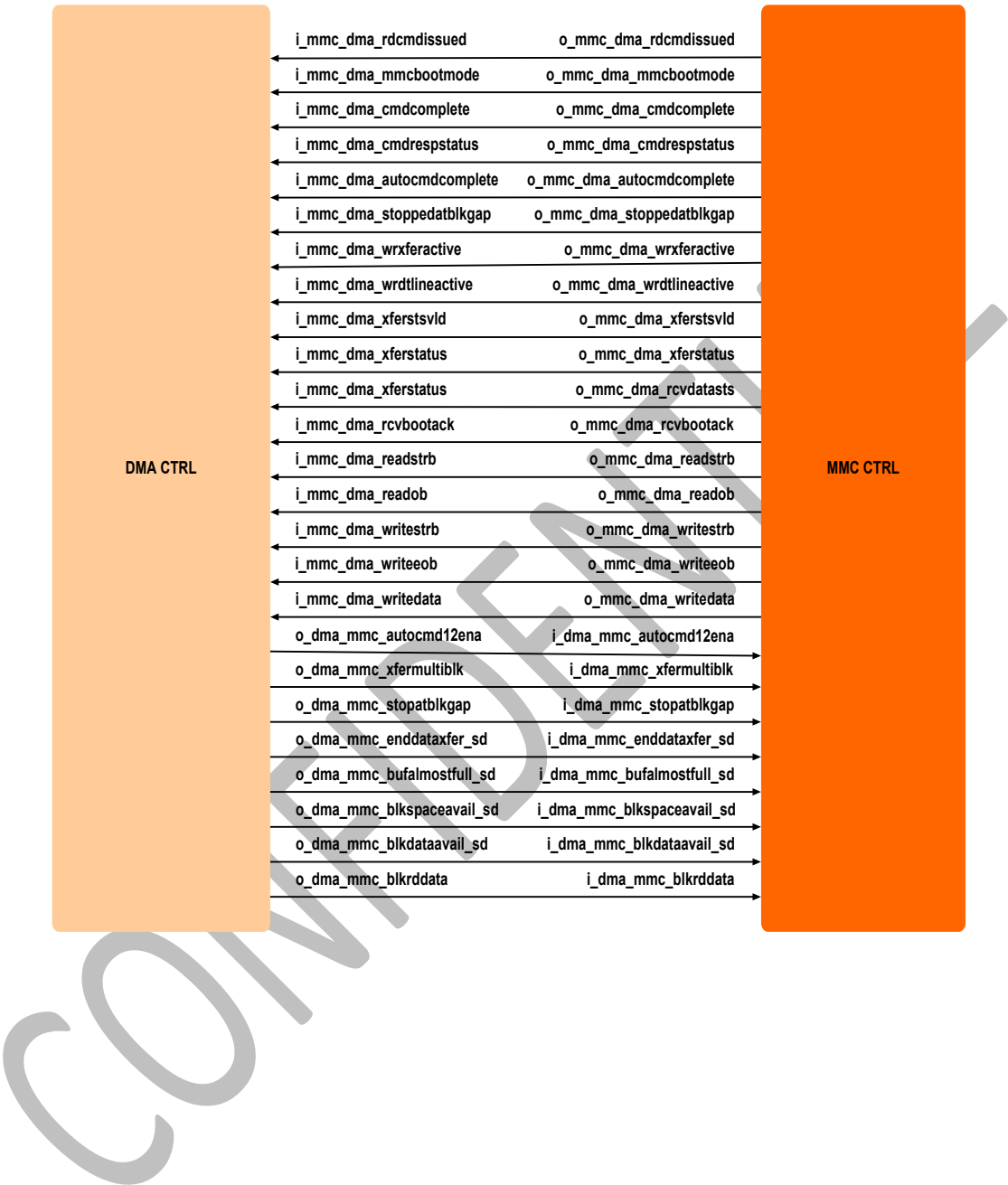
S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
73	o_dma_com_cmdcomplete	OUTPUT	1	i_dma_clk	DMACTRL to COM - Interface for Command Complete Indication
74	o_dma_csr_autocmdcomplete	OUTPUT	1	i_dma_clk	DMACTRL to CSR - Interface for Auto Command Complete Indication.
75	o_dma_csr_blocknextcmd	OUTPUT	1	i_dma_clk	DMACTRL to CSR - Interface for Block Next Command Indication.
76	o_dma_csr_rdxferactive	OUTPUT	1	i_dma_clk	DMACTRL to CSR - Interface for Read Transfer Active.
77	o_dma_csr_wrxferactive	OUTPUT	1	i_dma_clk	DMACTRL to CSR - Interface for Write Transfer Active.
78	o_dma_csr_data_line_active	OUTPUT	1	i_dma_clk	DMACTRL to CSR - Interface for Data Line Active Indication.
79	o_dma_csr_hosttransfer_busy	OUTPUT	1	i_dma_clk	DMACTRL to CSR - Interface for Host Xfer In Progress Indication
80	o_dma_csr_block_xfer_done	OUTPUT	1	i_dma_clk	DMACTRL to CSR - Interface for Block Transferred Indication (to/from Host memory)
81	o_dma_csr_block_gap_event	OUTPUT	1	i_dma_clk	DMACTRL to CSR - Interface for Block Gap Event Indication.
82	o_dma_com_xfer_status_valid	OUTPUT	1	i_dma_clk	DMACTRL to COM - Interface for Tx/Rx Transfer Status Valid Indication.
83	o_dma_com_xfer_status	OUTPUT	3	i_dma_clk	DMACTRL to COM - Interface for Tx/Rx Transfer Status Value.
84	o_dma_core_bufwrite_data	OUTPUT	FW	i_dma_clk	DMACTRL to CORE - Interface for BufferB WriteData.
85	o_dma_core_bufwrite_strobe	OUTPUT	1	i_dma_clk	DMACTRL to CORE - Interface for BufferB Write strobe.

S.No	PORTS	INPUT/OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
86	o_dma_core_bufreadb	OUTPUT	1	i_dma_clk	DMACTRL to CORE - Interface for BufferB Read Strobe.
87	o_dma_core_bufaddrb	OUTPUT	FW	i_dma_clk	DMACTRL to CORE - Interface for BufferB Address.
88	o_dma_core_bufwritea	OUTPUT	1	i_dma_clk	DMACTRL to CORE - Interface for BufferA Write Strobe.
89	o_dma_core_bufwrdataa	OUTPUT	FW	i_dma_clk	DMACTRL to CORE - Interface for BufferA Write Data.
90	i_soc_mst_cmd_ack	INPUT	1	i_dma_clk	SOC Master - write data ack output
91	i_soc_mst_cmd_full	INPUT	1	i_dma_clk	SOC Master - write data ack output
92	i_soc_mst_timeout_err	INPUT	1	i_dma_clk	SOC Master - error transfer output
93	i_soc_mst_done	INPUT	1	i_dma_clk	SOC Master - transfer done output
94	i_soc_mst_rd_valid	INPUT	1	i_dma_clk	SOC Master - read data valid output
95	o_dma_core_bufaddr_a	OUTPUT	FP	i_dma_clk	DMACTRL to CORE - Interface for BufferA Address.
96	o_dma_core_bufread_a	OUTPUT	1	i_dma_clk	DMACTRL to CORE - Interface for BufferA Read Strobe.
97	o_dma_csr_rcvbootack	OUTPUT	1	i_dma_clk	DMACTRL to CSR - Interface for Received Boot ACK Indication.
98	o_dma_csr_bootcomplete	OUTPUT	1	i_dma_clk	DMACTRL to CSR - Interface for Boot Complete Indication.
99	o_dma_cgen_abortcmdmode	OUTPUT	1	i_dma_clk	DMACTRL to CLKGEN - Interface for Abort Command Mode
100	o_dma_csr_sdmaaddressupdate	OUTPUT	1	i_dma_clk	DMACTRL to CSR - Interface for SDMA System Address Update Indication to CSR

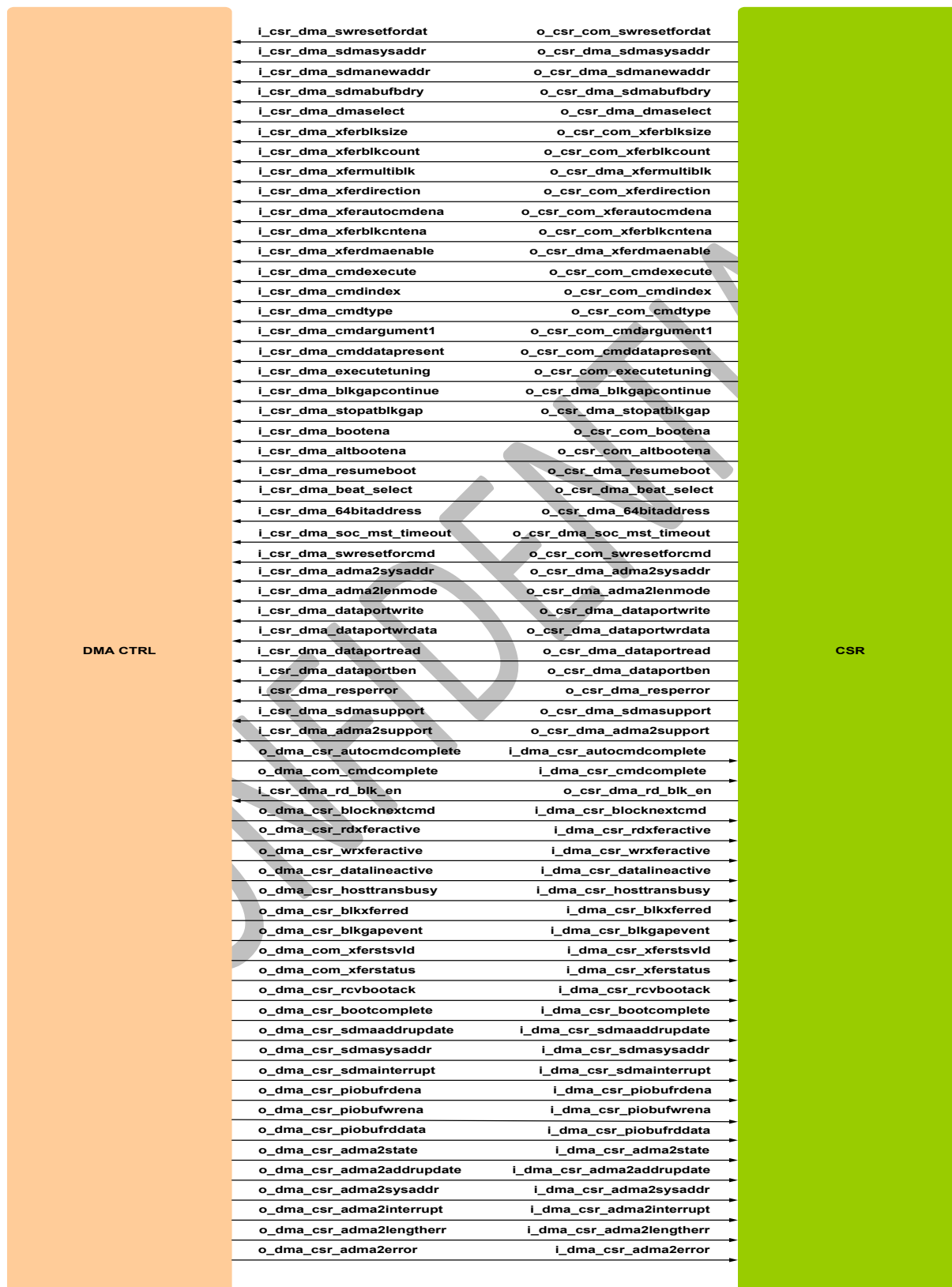
S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
101	o_dma_csr_sdmasys addr	OUTPUT	64	i_dma_clk	DMACTRL to CSR - Interface for SDMA System Address for Update.
102	o_dma_csr_sdmainte rrupt	OUTPUT	1	i_dma_clk	DMACTRL to CSR - Interface for SDMA Interrupt to CSR
103	o_dma_csr_piobufrd ena	OUTPUT	1	i_dma_clk	DMACTRL to CSR - Interface for PIO Buffer Read Enable to CSR.
104	o_dma_csr_piobufwr ena	OUTPUT	1	i_dma_clk	DMACTRL to CSR - Interface for PIO Buffer write Enable to CSR.
105	o_dma_csr_piobufrd data	OUTPUT	32	i_dma_clk	DMACTRL to CSR - Interface for PIO Buffer Read Data to CSR.
106	o_dma_csr_adma2st ate	OUTPUT	2	i_dma_clk	DMACTRL to CSR - Interface for ADMA2 Current State of ADMA2 States
107	o_dma_csr_adma2a ddrupdate	OUTPUT	1	i_dma_clk	DMACTRL to CSR - Interface for ADMA2 System Address Update Indication to CSR
108	o_dma_csr_adma2sy saddr	OUTPUT	64	i_dma_clk	DMACTRL to CSR - Interface for ADMA2 System Address for Update.
109	o_dma_csr_adma2int errupt	OUTPUT	1	i_dma_clk	DMACTRL to CSR - Interface for ADMA2 Interrupt to CSR.
110	o_dma_csr_adma2le ngtherr	OUTPUT	1	i_dma_clk	DMACTRL to CSR - Interface for ADMA2 Length Mismatch Error to CSR.
111	o_dma_csr_adma2er ror	OUTPUT	1	i_dma_clk	DMACTRL to CSR - Interface for ADMA2 Error to CSR.
112	o_soc_mst_be	OUTPUT	4	i_dma_clk	SOC Master - byte enable input
113	i_soc_mst_wr_ack	INPUT	1	i_dma_clk	SOC Master - write data ack output
114	o_soc_mst_id	OUTPUT	2	i_dma_clk	SOC Master - ID

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
115	o_dma_mmc_autocmd12ena	OUTPUT	1	i_dma_clk	DMACTRL to MMCCTRL - Interface for Auto CMD12 Enable after Transfer.
116	o_dma_mmc_xfermultiblk	OUTPUT	1	i_dma_clk	DMACTRL to MMCCTRL - Interface for Transfer Multi Block Indication.
117	o_dma_mmc_stopatblockgap	OUTPUT	1	i_dma_clk	DMACTRL to MMCCTRL - Interface for Stop at Block Gap Indication.
118	o_dma_mmc_enddataxfer_sd	OUTPUT	1	i_dma_clk	DMACTRL to MMCCTRL - Interface for End Data Transfer Indication. (In SD Mode)
119	o_dma_mmc_bufalmostfull_sd	OUTPUT	1	i_dma_clk	DMACTRL to MMCCTRL - Interface for Buffer is getting Almost Full (Card Read) (In SD Mode)
120	o_dma_mmc_blockspaceavail_sd	OUTPUT	1	i_dma_clk	DMACTRL to MMCCTRL - Interface for Block Space Available Indication (Card Read). (In SD Mode)
121	o_dma_mmc_blockdataavail_sd	OUTPUT	1	i_dma_clk	DMACTRL to MMCCTRL - Interface for Block Data Available Indication (Card Write) (In SD Mode)
122	o_dma_mmc_blockreaddata	OUTPUT	64	i_dma_clk	DMACTRL to MMCCTRL - Interface for Block Read data from Buffer
123	o_soc_mst_timeout	OUTPUT	16	i_dma_clk	SOC Master - address input
124	o_soc_mst_addr	OUTPUT	32	i_dma_clk	SOC Master - address input
125	o_soc_mst_data	OUTPUT	32	i_dma_clk	SOC Master - write data input
126	i_soc_mst_data	OUTPUT	32	i_dma_clk	SOC Master - read data output

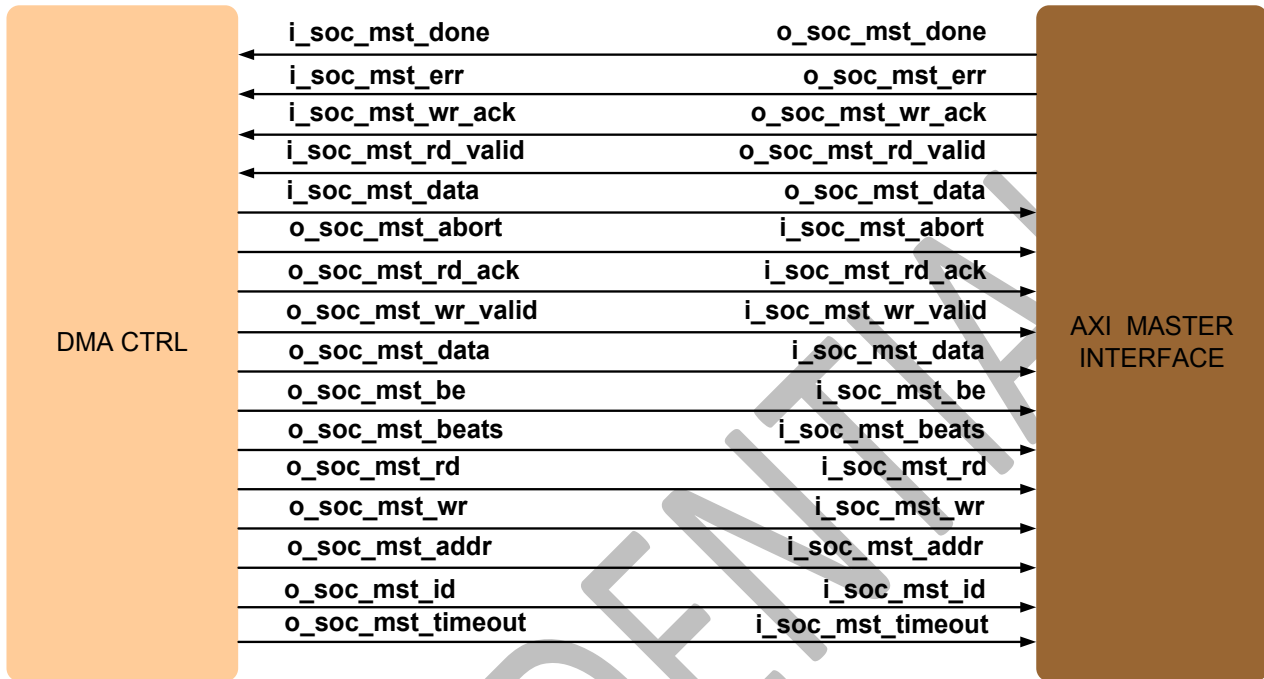
6.2.2.3 DMA Control and MMC Interface



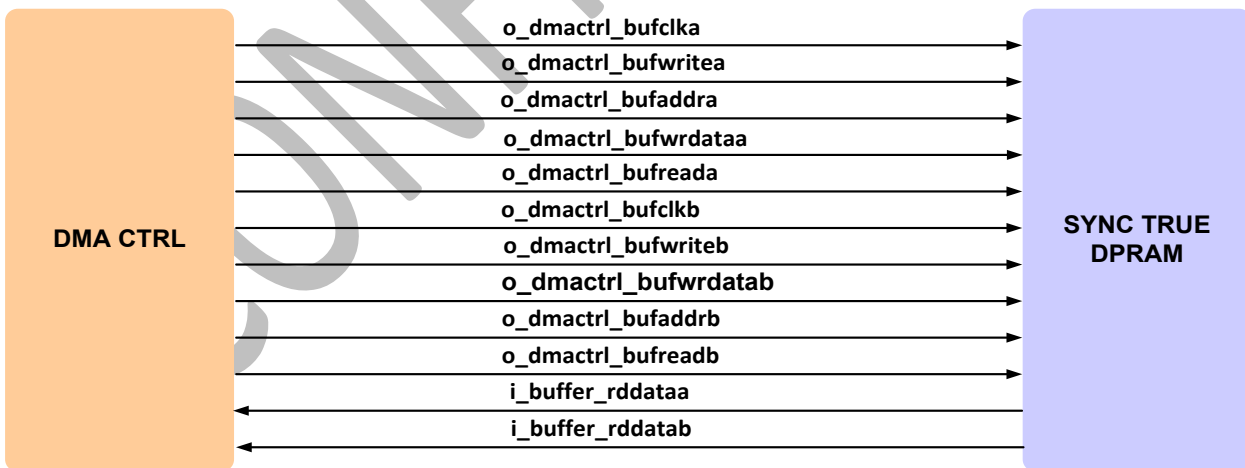
6.2.2.4 DMA Control and CSR Interface



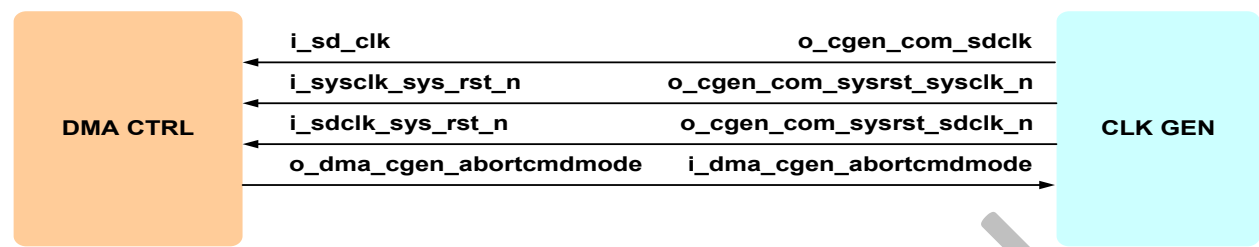
6.2.2.5 DMA Control and AXI Master Interface



6.2.2.6 DMA Control and Sync True Dpram Interface



6.2.2.7 DMA Control and CLKGEN Interface



6.2.3 Timeout Module

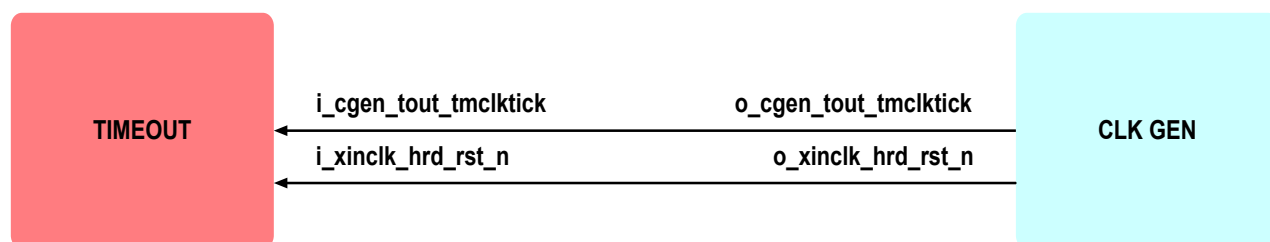
6.2.3.1 Functionality

This module contains data timeout generation. For data timeout, when the TXD control or RXD control enables the counting of timeout value, the counter start counting. When the counting is disabled, the counts are reset to zero.

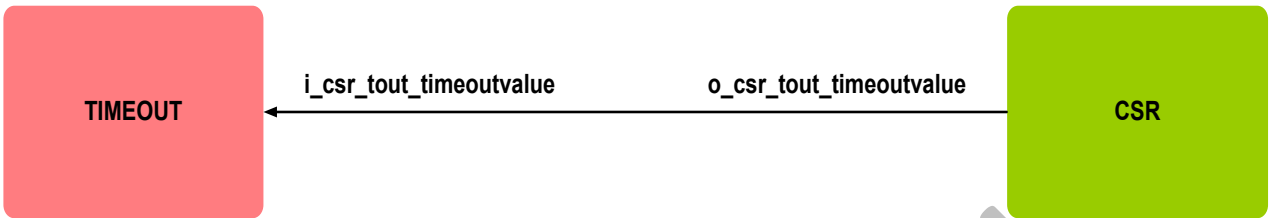
6.2.3.2 Port Descriptions

S.No	PORTS	INPUT/OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
1	i_xin_clk	INPUT	1		XIN CLOCK - Interface for Crystal Clock for SD.
2	i_xinclk_hrd_rst_n	INPUT	1	i_xin_clk	RESET - Interface for Hard Reset Sync to xin_clk.
3	i_csr_tout_timeoutvalue	INPUT	4	i_clk	CSR TO TIMEOUT - Interface for CSR: Data Time Out Value.
4	i_cgen_tout_tmclktick	INPUT	1	i_xin_clk	CLKGEN - Interface for CLK_GEN: TM_CLK Tick (1MHz or 1KHz).
5	i_mmc_tout_enatimeoutchk	INPUT	1	i_sd_clk	MMCCTRL TO TIMEOUT - Interface for SD_CTRL: Enable Data Timeout Check.
6	i_mmc_tout_pause_timeoutchk	INPUT	1	i_sd_clk	MMCCTRL TO TIMEOUT - Interface for SD_CTRL: Pause Data Timeout Check.
7	o_tout_mmc_data_timeout	OUTPUT	1	i_xin_clk	TIMEOUT - Interface for Data Timeout Indication

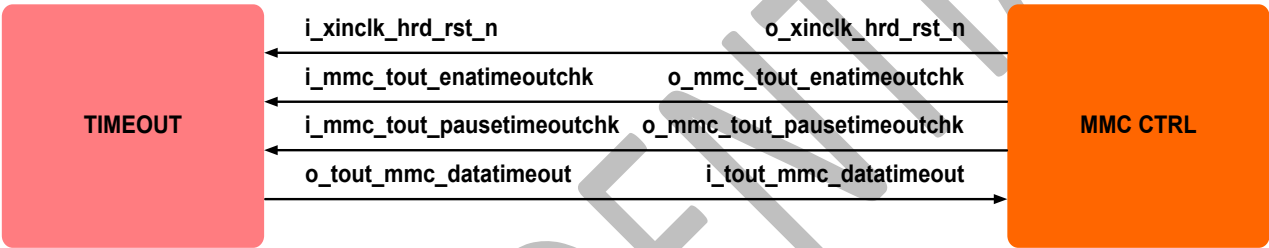
6.2.3.3 Timeout and CLKGEN Interface



6.2.3.4 Timeout and CSR Interface



6.2.3.5 Timeout and MMC Interface



6.2.4 CLKGEN Module

6.2.4.1 Functionality

This module has MMC clock generator logic. The clock divider value is loaded from the register CTRL2 (clock control). This value is from the MMC clock frequency select (either via driver programmed or selected preset value). The clock counter counts down from divider value down to 1. This is used to generate the MMC_CLK. The timer counts the MMC_CLK half period value. When the timer hits 1, then MMC_CLK is toggled. The clock counter counts down from divider value down to 1. This is used to generate the MMC_CLK. The timer counts the MMC_CLK half period value. When the timer hits 1, then MMC_CLK is toggled. The MMC_CARD clock is gated during blocks or under software control. Also the MMC Clock is gated when the RXD control indicates stop (between blocks and stop at block gap) and no abort command is being issued.

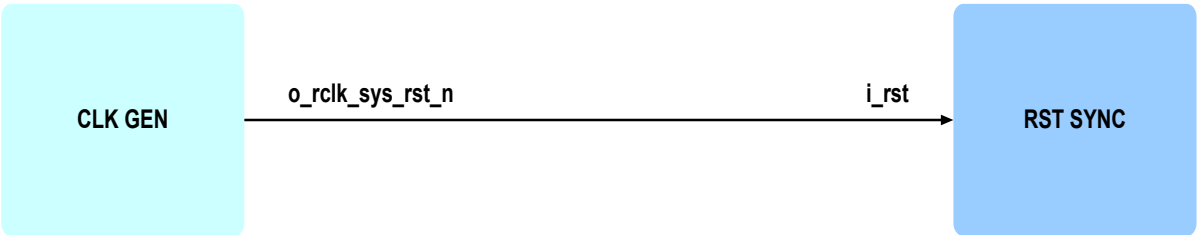
6.2.4.2 Port Descriptions

S.No	PORTS	INPUT/OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
1	i_xin_clk	INPUT	1		XIN CLK - Interface for Crystal Clock for SD (200 MHz).
2	i_clk	INPUT	1		SYSTEM CLK - Interface for System Clock (AHB/AXI/OCP)
3	i_rst_n	INPUT	1	i_clk	SYSTEM RESET - Interface for Hard Reset Synchronous to sys_clk.
4	i_scan_mode	INPUT	1	i_sd_clk	SCAN MODE - Interface for Scan Mode.
5	i_scan_clk	INPUT	1	i_clk	SCAN CLK -Interface for Scan Clk1.
6	i_csr_cgen_swresetf orall	INPUT	1		CSR to CLKGEN - Interface for CSR: SW Reset for ALL.
7	i_csr_cgen_newsdclk freqsel	INPUT	1	i_clk	CSR to CLKGEN - Interface for CSR: New SD Clock Frequency Select.
8	i_csr_cgen_sdclkfreq sel	INPUT	10	i_clk	CSR to CLKGEN - Interface for CSR: SD Clock Frequency Select.
9	i_csr_cgen_sdclkena	INPUT	1	i_clk	CSR to CLKGEN - Interface for CSR: SD Clock Enable.

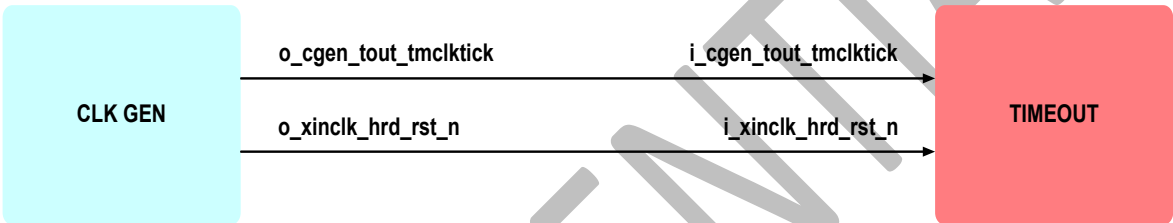
S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
10	i_csr_cgen_intclkena	INPUT	1	i_clk	CSR to CLKGEN - Interface for CSR: Internal Clock Enable.
11	i_csr_cgen_baseclkfreq	INPUT	8	i_clk	CSR to CLKGEN - Interface for Base clock freq
12	i_csr_cgen_timeoutclkunit	INPUT	1	i_clk	CSR to CLKGEN - Interface for timeout clk unit
13	i_dma_cgen_abortcmdmode	INPUT	1	i_dma_clk	DMACCTRL to CLKGEN - Interface for DMA_CTRL: Abort Command Mode.
14	i_mmc_cgen_stopsdcardclk0	INPUT	1	i_clk	MMCCTRL to CLKGEN - Interface for SD_CTRL: Stop eMMC Clock (Transmit Stream Mode, BlkBuffer Empty).
15	i_mmc_cgen_stopsdcardclk1	INPUT	1	i_sd_clk	MMCCTRL to CLKGEN - Interface for SD_CTRL: Stop eMMC Clock (Receive Normal Operation End of Block).
16	i_mmc_cgen_stopsdcardclk2	INPUT	1	i_sd_clk	MMCCTRL to CLKGEN - Interface for SD_CTRL: Stop eMMC Clock (Receive Stop at Block Gap).
17	i_mmc_cgen_stopsdcardclk3	INPUT	1	i_sd_clk	MMCCTRL to CLKGEN - Interface for SD_CTRL: Stop eMMC Clock (Receive Stream Mode, BlkBuffer Full)
18	i_mmc_cgen_rclkswitching	INPUT	1	i_sd_clk	MMCCTRL to CLKGEN - Interface for SD_CTRL: Receive Clock Switching (Xin_clk Domain Signal).
19	o_cgen_csr_intclkstable	OUTPUT	1	i_xin_clk	CLKGEN to CSR - Interface for Internal Clock Stable.
20	o_cgen_csr_swresetdone	OUTPUT	1	i_xin_clk	CLKGEN to CSR - Interface for Software Reset Done Indication.
21	o_cgen_csr_newsdclkfreqselok	OUTPUT	1	i_xin_clk	CLKGEN to CSR - Interface for New SD_CLK Freq Selection OK.

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
22	o_cgen_com_sdclk	OUTPUT	1	i_xin_clk	CLKGEN to COM - Interface for SD Clock used in the Host Ctrl.
23	o_cgen_core_sdcard clk	OUTPUT	1	o_emmc_ clk	CLKGEN to CORE - Interface for SD Clock sent to SDCARD and emmc_clk.
24	o_cgen_cdet_1msect ick	OUTPUT	1		CLKGEN to CARDDDET - Interface for 1 MilliSec Tick for Card Detect.
25	o_cgen_cdet_1usecti ck	OUTPUT	1	i_xin_clk	CLKGEN to CARDDDET - Interface for 1 MicroSec Tick for Card Detect.
26	o_cgen_mmc_spibyt ectr	OUTPUT	3	i_xin_clk	CLKGEN to MMCCTRL - Interface for CLK_GEN: SPI Byte Counter.
27	o_cgen_tout_tmclktic k	OUTPUT	1	i_clk	CLKGEN to TIMEOUT - Interface for TM_CLK Tick. (1MHz or 1KHz)
28	o_xinclk_hrd_rst_n	OUTPUT	1	i_xin_clk	CLKGEN to COM - Interface for Hard Reset Sync to xin_clk.
29	o_sysclk_sys_rst_n	OUTPUT	1	i_xin_clk	CLKGEN to COM - Interface for System Reset Sync to sys_clk.
30	o_sdclk_sys_rst_n	OUTPUT	1	i_sd_clk	CLKGEN to COM - Interface for System Reset Sync to sd_clk.
31	o_rclk_sys_rst_n	OUTPUT	1	o_emmc_ clk	CLKGEN to CORE - Interface for System Reset Sync to emmc_clk (sdcard_clk).
32	o_cgen_com_emmce nable	OUTPUT	1	i_clk	CLKGEN to COM - Interface for SD_IF Enable Indication.

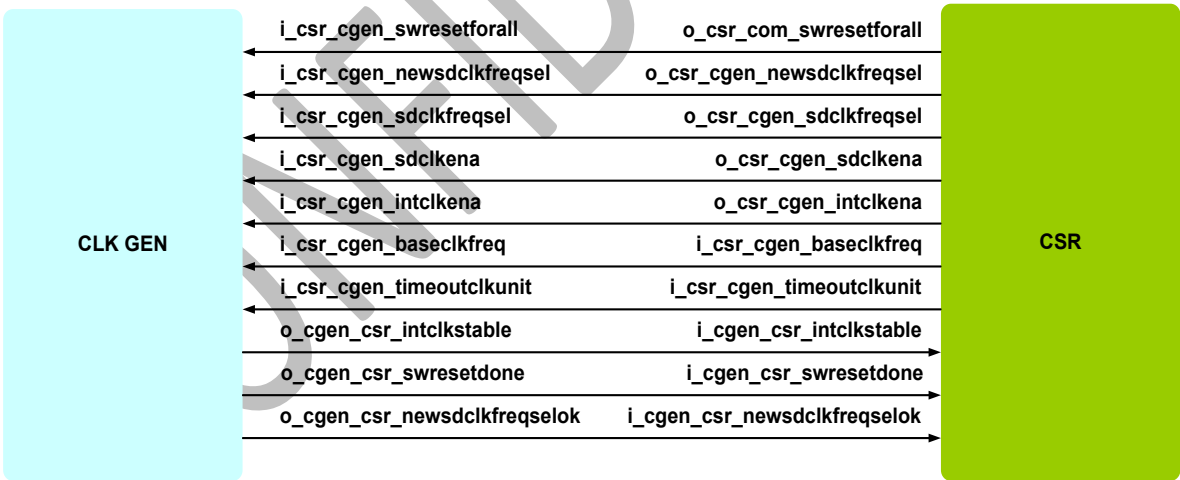
6.2.4.3 CLKGEN and RST SYNC Interface



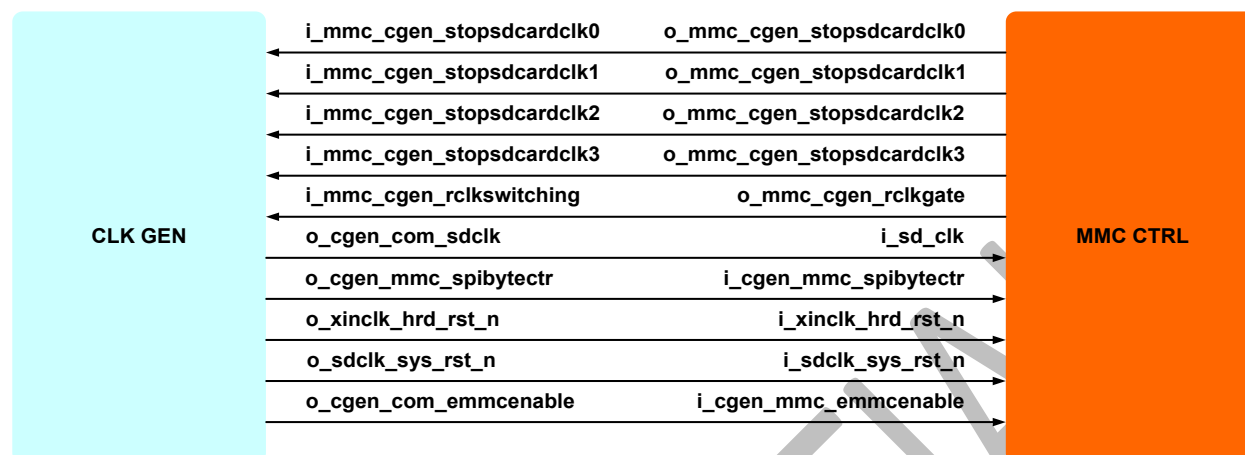
6.2.4.4 CLKGEN and Timeout Interface



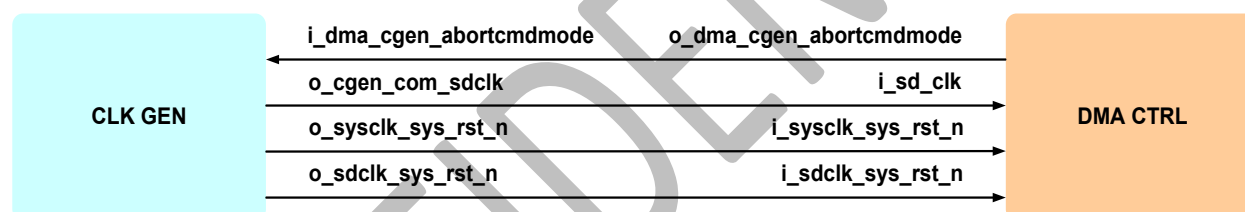
6.2.4.5 CLKGEN and CSR Interface



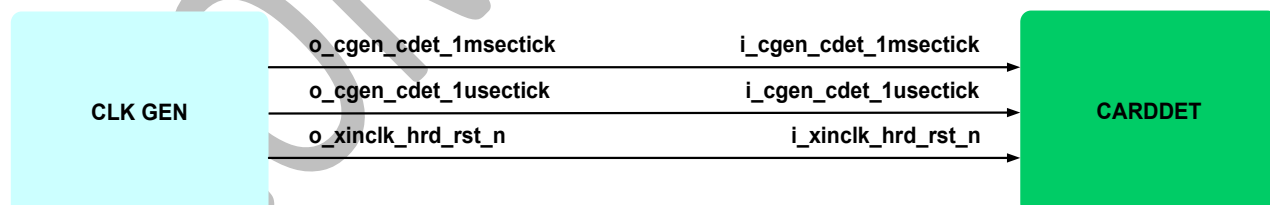
6.2.4.6 CLKGEN and MMC Interface



6.2.4.7 CLKGEN and DMA Interface



6.2.4.8 CLKGEN and CARDDDET Interface



6.2.5 CSR Module

6.2.5.1 Functionality

This module includes all the registers like block size, block count, address, argument ...etc and also it includes card interrupt logic. The contents of the registers are decoded and assigned to its respective output ports based on its functionality. The registers can get its data from both the internal and external system interface. Likewise, it can be retrieved by both the internal and external system interface.

➤ Write Interface

- ◆ When the registers address and data are given from external system interface based on register address, data will be written on corresponding register.
- ◆ In local interface, when register address matches with local interface address input along with local interface write enable, then the opcode will be written in respective register.

➤ Read Interface

- ◆ The contents of register will be retrieved, when the register address is provided.

6.2.5.2 Port Descriptions

S.No	PORTS	INPUT/OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
1	i_clk	INPUT	1		SYSTEM CLK - Interface for System Clock (AHB/AXI/OCP)
2	i_dma_clk	INPUT	1		DMA CLK - Interface for axi master Clock.
3	i_sd_clk	INPUT	1		SD CLK - Interface for eMMC Clock.
4	i_rst_n	INPUT	1	i_clk	SYSTEM RESET - Interface for Hard Reset Synchronous to sys_clk.
5	i_dma_rst_n	INPUT	1	i_dma_clk	DMA RESET - Interface for Reset Synchronous to i_dma_clk.
6	i_sdclk_sys_rst_n	INPUT	1	i_sd_clk	SD RESET - Interface for Reset Synchronous to emmc clk.
7	i_soc_slv_addr	INPUT	16	i_clk	SOC Bus - External interface address input

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
8	i_soc_slv_data	INPUT	32	i_clk	SOC Bus - External interface data input
9	o_soc_slv_data	OUTPUT	32	i_clk	SOC Bus - External interface data output
10	i_soc_slv_be	INPUT	4	i_clk	SOC Bus - External interface byte enable input
11	i_soc_slv_wr	INPUT	1	i_clk	SOC Bus - External interface write signal
12	i_soc_slv_rd	INPUT	1	i_clk	SOC Bus - External interface read signal
13	o_soc_slv_err	OUTPUT	1	i_clk	SOC Bus - External interface error output
14	o_soc_slv_wr_ack	OUTPUT	1	i_clk	SOC Bus - External interface write acknowledgement
15	o_soc_slv_rd_ack	OUTPUT	1	i_clk	SOC Bus - External interface read acknowledgement
16	o_soc_slv_timeout	OUTPUT	16	i_clk	SOC Bus - External interface timeout
17	o_irq	OUTPUT	1	i_clk	SOC Bus - External interface interrupt status output
18	i_core_csr_emmc_cmdin	INPUT	1	o_emmc_clk	EMMC CMD - Interface for SD_IF: Command Input.
19	i_core_csr_emmc_data0in	INPUT	1	o_emmc_clk	EMMC DATA - Interface for SD_IF: Data#0 Input.
20	i_core_csr_emmc_data1in	INPUT	1	o_emmc_clk	EMMC DATA - Interface for SD_IF: Data#1 Input.
21	i_core_csr_emmc_data2in	INPUT	1	o_emmc_clk	EMMC DATA - Interface for SD_IF: Data#2 Input.
22	i_core_csr_emmc_data3in	INPUT	1	o_emmc_clk	EMMC DATA - Interface for SD_IF: Data#3 Input.
23	i_core_csr_emmc_data4in	INPUT	1	o_emmc_clk	EMMC DATA - Interface for SD_IF: Data#4 Input.
24	i_core_csr_emmc_data5in	INPUT	1	o_emmc_clk	EMMC DATA - Interface for SD_IF: Data#5 Input.
25	i_core_csr_emmc_data6in	INPUT	1	o_emmc_clk	EMMC DATA - Interface for SD_IF: Data#6 Input.

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
26	i_core_csr_emmc_data7in	INPUT	1	o_emmc_clk	EMMC DATA - Interface for SD_IF: Data#7 Input.
27	i_dma_csr_sdmaininterrupt	INPUT	1	i_dma_clk	DMA to CSR - Interface for DMA_CTRL: SDMA Interrupt.
28	i_dma_csr_autocmdcomplete	INPUT	1	i_dma_clk	DMA to CSR - Interface for DMA_CTRL: Auto Command Complete Indication.
29	i_dma_csr_sdmaaddrupdate	INPUT	1	i_dma_clk	DMA to CSR - Interface for DMA_CTRL: SDMA System Address Update Indication.
30	i_dma_csr_xferstsvld	INPUT	1	i_dma_clk	DMA to CSR - Interface for DMA_CTRL: Tx/Rx Block Transfer Status Valid Indication.
31	i_dma_csr_wrxferactive	INPUT	1	i_dma_clk	DMA to CSR - Interface for DMA_CTRL: Write Transfer Active.
32	i_dma_csr_blocknextcmd	INPUT	1	i_dma_clk	DMA to CSR - Interface for DMA_CTRL: Block Next Command Indication.
33	i_dma_csr_xferstatus	INPUT	3	i_dma_clk	DMA to CSR - Interface for DMA_CTRL: Tx/Rx Block Transfer Status Value.
34	i_dma_csr_rdxferactive	INPUT	1	i_dma_clk	DMA to CSR - Interface for DMA_CTRL: Read Transfer Active.
35	i_dma_csr_cmdcomplete	INPUT	1	i_dma_clk	DMA to CSR - Interface for DMA_CTRL: Command Complete Indication
36	i_dma_csr_blkgapevent	INPUT	1	i_dma_clk	DMA to CSR - Interface for DMA_CTRL: Block Gap Event Indication (Pulse Synched).
37	i_dma_csr_blkxferred	INPUT	1	i_dma_clk	DMA to CSR - Interface for DMA_CTRL: Block Transferred Indication.

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
38	i_dma_csr_hosttrans busy	INPUT	1	i_dma_clk	DMA to CSR - Interface for DMA_CTRL: Host Transfer in Progress Indication.
39	i_dma_csr_data_line_active	INPUT	1	i_dma_clk	DMA to CSR - Interface for DMA_CTRL: Data Line Active Indication.
40	i_dma_csr_sdmasys addr	INPUT	64	i_dma_clk	DMA to CSR - Interface for DMA_CTRL: SDMA System Address for Update.
41	i_dma_csr_bootcomplete	INPUT	1	i_dma_clk	DMA to CSR - Interface for DMA_CTRL: Boot Complete Indication.
42	i_dma_csr_rcvbootack	INPUT	1	i_dma_clk	DMA to CSR - Interface for DMA_CTRL: Received Boot ACK Indication.
43	i_dma_csr_piobufwrenable	INPUT	1	i_dma_clk	DMA to CSR - Interface for DMA_CTRL: PIO Buffer write Enable.
44	i_dma_csr_piobufrdenable	INPUT	1	i_dma_clk	DMA to CSR - Interface for DMA_CTRL: PIO Buffer Read Enable.
45	i_dma_csr_piobufread data input	INPUT	32	i_dma_clk	DMA to CSR - Interface for DMA_CTRL: PIO Buffer Read Data.
46	o_csr_cdet_cardinsertenable	OUTPUT	1	i_clk	CSR to CARDDDET - Interface for Wakeup Event Enable on eMMC Insertion.
47	o_csr_cdet_cardremovalstatusenable	OUTPUT	1	i_clk	CSR to CARDDDET - Interface for Interrupt Status Enable on eMMC Removal.
48	o_csr_cdet_cardinsertstatusenable	OUTPUT	1	i_clk	CSR to CARDDDET - Interface for Interrupt Status Enable on eMMC Insertion.
49	o_csr_cdet_cardinterruptenable	OUTPUT	1	i_clk	CSR to CARDDDET - Interface for Wakeup Event Enable on eMMC Interrupt.

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
50	o_csr_cdet_cardinsst ssig	OUTPUT	1	i_clk	CSR to CARDDDET - Interface for eMMC Insertion Interrupt Status.
51	o_csr_mmc_rdwaitctr l	OUTPUT	1	i_clk	CSR to MMCCTRL - Interface for REG_SET: Read Wait Control.
52	i_sdif_wp	INPUT	1	i_clk	Interface for SD Intf - Write Protect (Active High)
53	i_sdif_cd_n	INPUT	1	i_clk	Interface for SD Intf - Card Detect (Active Low)
54	i_dma_csr_adma2sy saddr	INPUT	64	i_dma_cl k	DMA to CSR - Interface for DMA_CTRL: ADMA2 System Address for Update.
55	i_dma_csr_adma2ad drupdate	INPUT	1	i_dma_cl k	DMA to CSR - Interface for DMA_CTRL: ADMA2 System Address Update Indication.
56	i_dma_csr_adma2int errupt	INPUT	1	i_dma_cl k	DMA to CSR - Interface for DMA_CTRL: ADMA2 Interrupt.
57	i_dma_csr_adma2err or	INPUT	1	i_dma_cl k	DMA to CSR - Interface for DMA_CTRL: ADMA2 Error.
58	i_dma_csr_adma2len gtherr	INPUT	1	i_dma_cl k	DMA to CSR - Interface for DMA_CTRL: ADMA2 Length Mismatch Error.
59	o_csr_dma_sdmasup port	OUTPUT	1	i_clk	CSR to DMA - Interface for SDMA Support (0:Not Supported, 1:Supported)
60	o_csr_dma_adma2su pport	OUTPUT	1	i_clk	CSR to DMA - Interface for ADMA2 Support (0:Not Supported, 1:Supported)
61	o_csr_cdet_cardrem wkupena	OUTPUT	1	i_clk	CSR to CARDDDET - Interface for Wakeup Event Enable on eMMC Removal.
62	i_core_csr_hostif_ms terror	INPUT	1	i_clk	Interface for HOST_IF - Master Error (Received Error Response).
63	i_mmc_csr_autocmdr espstatus	INPUT	4	i_sd_clk	MMCCTRL to CSR - Interface for SD_CTRL: Auto Command Response status.

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
64	i_mmc_csr_autocmd 12notexec input	INPUT	1	i_sd_clk	MMCCTRL to CSR - Interface for SD_CTRL: AutoCMD12 Not Executed Indication.
65	i_mmc_csr_autocmd nexterror	INPUT	1	i_clk	MMCCTRL to CSR - Interface for SD_CTRL: Auto Command Next Error.
66	i_mmc_csr_retuningr eq	INPUT	1	i_clk	MMCCTRL to CSR - Interface for SD_CTRL: Retuning Request.
67	i_mmc_csr_tuningdo ne	INPUT	1	i_sd_clk	MMCCTRL to CSR - Interface for SD_CTRL: Tuning Done Indication.
68	i_mmc_csr_tuningrslt	INPUT	1	i_sd_clk	MMCCTRL to CSR - Interface for SD_CTRL: Tuning Result Indication.
69	i_mmc_csr_tuningerr or	INPUT	1	i_sd_clk	MMCCTRL to CSR - Interface for SD_CTRL: Tuning Error Indication.
70	i_mmc_csr_cmdresp status	INPUT	4	i_sd_clk	MMCCTRL to CSR - Interface for SD_CTRL: Command Response status {EB, IDX, CRC, TOUT}.
71	i_mmc_csr_cmdresp onse	INPUT	128	o_emmc_ clk	MMCCTRL to CSR - Interface for SD_CTRL: Command Response.
72	i_cdet_csr_inserted	INPUT	1	i_xin_clk	CARDDET to CSR - Interface for CARD_DET: Card Inserted Indication.
73	i_cdet_csr_statestabl e	INPUT	1	i_xin_clk	CARDDET to CSR - Interface for CARD_DET: Card State Stable Indication.
74	i_cgen_csr_newsdclk freqselok	INPUT	1	i_xin_clk	CLKGEN to CSR - Interface for CLK_GEN: New SD_CLK Freq Selection OK.
75	i_cgen_csr_intclkstab le	INPUT	1	i_xin_clk	CLKGEN to CSR - Interface for CLK_GEN: Internal Clock Stable.

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
76	i_cgen_csr_swresetdone	INPUT	1	i_xin_clk	CLKGEN to CSR - Interface for CLK_GEN: Software Reset Done Indication.
77	o_csr_cgen_baseclkfreq	OUTPUT	1	i_clk	CSR to CLKGEN - Interface for Base clock freq
78	o_csr_cgen_timeoutclkunit	OUTPUT	1	i_clk	CSR to CLKGEN - Interface for timeout clk unit
79	o_csr_dma_soc_mst_timeout	OUTPUT	16	i_clk	CSR to DMA - SoC Master Timeout value
80	o_csr_core_write_start_edge	OUTPUT	1	i_clk	EDGE Selection - Interface for write data start driving edge indicator.
81	o_csr_cmd_start_edge	OUTPUT	1	i_clk	EDGE Selection - Interface for cmd start edge driving indicator.
82	o_csr_core_rd_data_start_sampling_edge	OUTPUT	1	i_clk	EDGE Selection - Interface for read data start sampling edge indicator.
83	o_csr_core_resp_start_sampling_edge	OUTPUT	1	i_clk	EDGE Selection - Interface for response start sampling edge indicator.
84	o_csr_dma_rd_blk_en	OUTPUT	1	i_clk	CSR to DMACTRL - Interface to process second block of read data once first block of read data gets completed.
85	o_csr_dma_beat_select	OUTPUT	2	i_clk	CSR to DMACTRL - Interface for AHB Transmission beat select.
86	o_csr_core_clkgenselect	OUTPUT	1	i_clk	CSR to - Interface for Clk Generator Select.
87	o_csr_dma_sdmanewaddr	OUTPUT	1	i_clk	CSR to DMACTRL - Interface for SDMA Mode New Address Programmed Indication.
88	o_csr_mmc_cmdindexkena	OUTPUT	1	i_clk	CSR to MMCCTRL - Interface for Command Index Check Enable.
89	o_csr_mmc_boottimeout	OUTPUT	32	i_clk	CSR to MMCCTRL - Interface for Boot Timeout Count

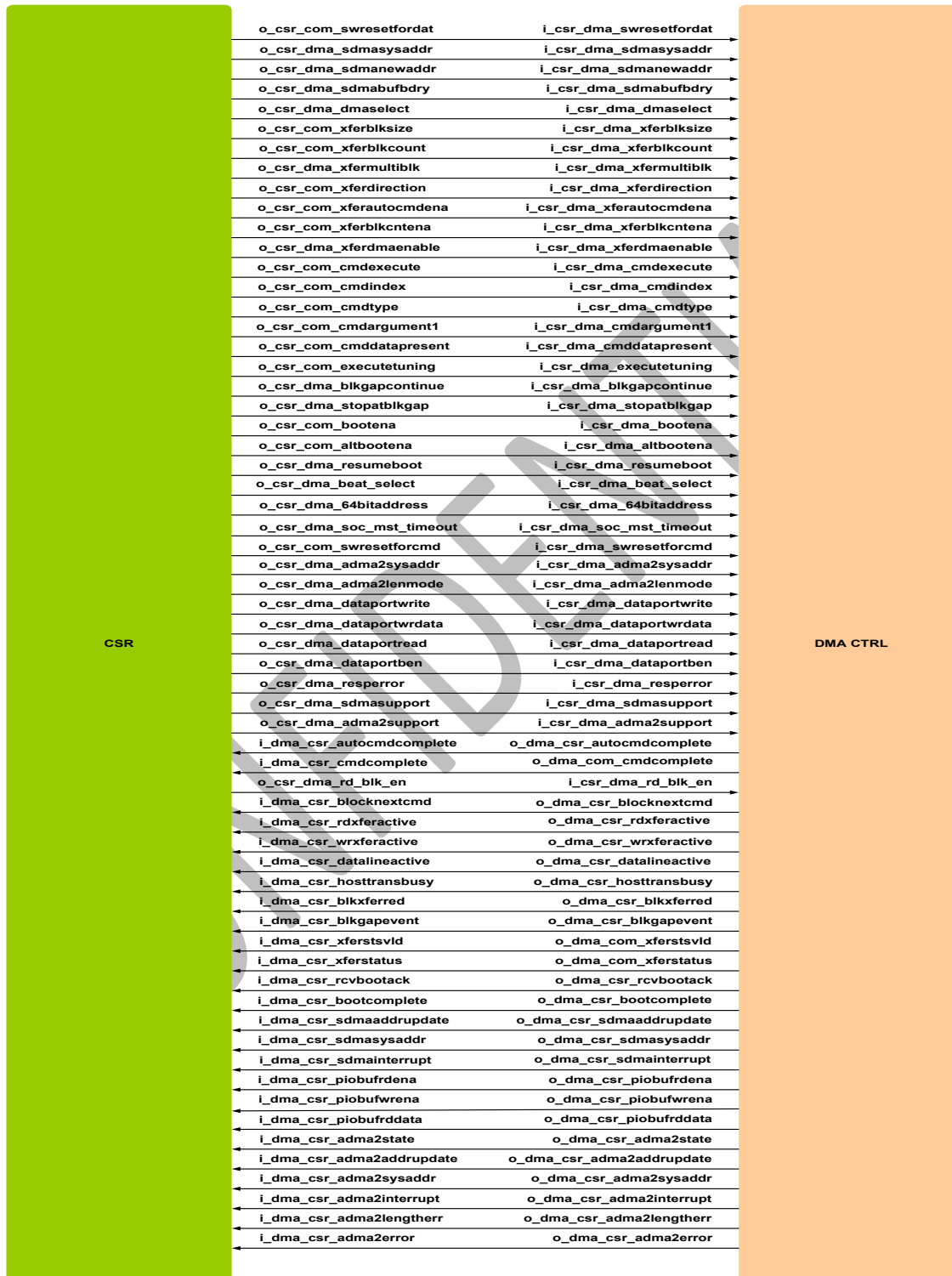
S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
90	o_csr_com_altbootena	OUTPUT	1	i_clk	CSR to COM - Interface for Alternate Boot Mode Enable.
91	o_csr_mmc_bootackena	OUTPUT	1	i_clk	CSR to MMCCTRL - Interface for Boot Acknowledge Enable.
92	o_csr_dma_resumeboot	OUTPUT	1	i_clk	CSR to DMACTRL - Interface for Resume Boot Indication.
93	o_csr_com_bootena	OUTPUT	1	i_clk	CSR to COM - Interface for Normal Boot Mode Enable.
94	o_csr_dma_adma2lengthmode	OUTPUT	1	i_clk	CSR to DMACTRL - Interface for ADMA2 Length Mode.
95	o_csr_com_ddr50mode	OUTPUT	1	i_clk	CSR to COM - Interface for DDR50 Mode Indication.
96	o_csr_cdet_cardintstatusena	OUTPUT	1	i_clk	CSR to CARDDDET - Interface for Interrupt Status Enable on eMMC Interrupt.
97	o_csr_cdet_cardintstatussig	OUTPUT	1	i_clk	CSR to CARDDDET - Interface for eMMC Interrupt Interrupt Status.
98	o_csr_dma_dataportwrite	OUTPUT	1	i_clk	CSR to DMACTRL - Interface for DataPort Write Indication.
99	o_csr_dma_dataportben	OUTPUT	4	i_clk	CSR to DMACTRL - Interface for DataPort Access ByteEnables.
100	o_csr_dma_dataportwdata	OUTPUT	32	i_clk	CSR to DMACTRL - Interface for DataPort Write Data.
101	o_csr_dma_dataportread	OUTPUT	1	i_clk	CSR to DMACTRL - Interface for DataPort Read Indication.
102	o_csr_dma_64bitaddress	OUTPUT	1	i_clk	CSR to DMACTRL - Interface for 64-bit Addressing Enable
103	i_dma_csr_adma2state	INPUT	2	i_clk	DMA to CSR - Interface for DMA_CTRL: ADMA2 State.
104	o_csr_dma_resperror	OUTPUT	1	i_clk	CSR to DMACTRL - Interface for Response Error
105	o_csr_dma_adma2sysaddr	OUTPUT	64	i_clk	CSR to DMACTRL - Interface for ADMA2 System Address (SDMA Sys Addr in 64-bit mode).

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
106	o_csr_mmc_cmdresp type	OUTPUT	2	i_clk	CSR to DMACTRL - Interface for Command Response Type.
107	o_csr_dma_sdmasys addr	OUTPUT	64	i_clk	CSR to DMACTRL - Interface for SDMA System Address.
108	o_csr_dma_xferdma enable	OUTPUT	1	i_clk	CSR to DMACTRL - Interface for Transfer DMA Enable.
109	o_csr_com_executet uning	OUTPUT	1	i_clk	CSR to COM - Interface for Execute Tuning.
110	o_csr_com_xferblkcn tena	OUTPUT	1	i_clk	CSR to COM - Interface for Transfer Block Count Enable.
111	o_csr_com_xferdirect ion	OUTPUT	1	i_clk	CSR to COM - Interface for Transfer Direction (0:Write, 1:Read)
112	o_csr_com_extdataw idth	OUTPUT	1	i_clk	CSR to com - Interface for Extended Data Transfer Width.
113	o_csr_dma_dmasele ct	OUTPUT	2	i_clk	CSR to DMACTRL - Interface for DMA Select from Host Controller#1 Register.
114	o_csr_mmc_samplin gclkse	OUTPUT	1	i_clk	CSR to MMCCTRL - Interface for Sampling Clock Select.
115	o_csr_dma_xfermulti blk	OUTPUT	1	i_clk	CSR to DMACTRL - Interface for Transfer Multi Block Indicaion (0:Single Block).
116	o_csr_dma_blkgapco ntinue	OUTPUT	1	i_clk	CSR to DMACTRL - Interface for Continue after Block Gap Stop.
117	o_csr_com_cmdinde x	OUTPUT	6	i_clk	CSR to COM - Interface for Command Index.
118	o_csr_com_datawidt h	OUTPUT	1	i_clk	CSR to COM - Interface for Data Transfer Width.
119	o_csr_tout_timeoutva lue	OUTPUT	4	i_clk	CSR to TIMEOUT - Interface for Data Time Out Value.
120	o_csr_core_emmchw rst_n	OUTPUT	1	i_clk	CSR to CORE - Interface for eMMC HW Reset.
121	o_csr_com_cmdtype	OUTPUT	2	i_clk	CSR to COM - Interface for Command Type.

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
122	o_csr_com_swresetf orall	OUTPUT	1	i_clk	CSR to COM - Interface for SW Reset for ALL.
123	o_csr_com_swresetf ordat	OUTPUT	1	i_clk	CSR to COM - Interface for SW Reset for DAT.
124	o_csr_dma_stopatblk gap	OUTPUT	1	i_clk	CSR to DMACTRL - Interface for Stop at Block Gap.
125	o_csr_com_xferblkco unt	OUTPUT	32	i_clk	CSR to COM - Interface for Transfer Block Count value to DMA/CMD_CTRL
126	o_csr_com_cmddata present	OUTPUT	1	i_clk	CSR to COM - Interface for Command Data Present Indication.
127	o_csr_cgen_sdclkfre qsel	OUTPUT	10	i_clk	CSR to CLKGEN - Interface for SD Clock Frequency Select.
128	o_csr_cgen_intclken a	OUTPUT	1	i_clk	CSR to CLKGEN - Interface for Internal Clock Enable.
129	o_csr_com_cmdargu ment1	OUTPUT	32	i_clk	CSR to COM - Interface for Command Argument#1 to SD_CTRL Module.
130	o_csr_cgen_newsdc lkfreqsel	OUTPUT	1	i_clk	CSR to CLKGEN - Interface for New SD_CLK Frequency Select Programmed.
131	o_csr_com_xferautoc mdena	OUTPUT	2	i_clk	CSR to COM - Interface for Transfer Auto Command Enable (00:Dis, 01:CMD12, 10:CMD23).
132	o_csr_com_xferblksiz e	OUTPUT	12	i_clk	CSR to COM - Interface for Transfer Block Size value to DMA/CMD_CTRL
133	o_csr_com_swresetf orcnd	OUTPUT	1	i_clk	CSR to COM - Interface for SW Reset for CMD.
134	o_csr_cgen_sdclken a output	OUTPUT	1	i_clk	CSR to CLKGEN - Interface for SD Clock Enable.
135	o_csr_mmc_cmdargu ment2	OUTPUT	32	i_clk	CSR to MMCCTRL - Interface for Command Argument#2 to SD_CTRL Module (CMD23 Block Count).

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
136	o_csr_com_cmdeexecute	OUTPUT	1	i_clk	CSR to COM - Interface for Command Execute Indication.
137	o_csr_mmc_cmdcrcc hkena	OUTPUT	1	i_clk	CSR to MMCCTRL - Interface for Command CRC Check Enable.
138	o_csr_dma_sdmabuf bdry	OUTPUT	3	i_clk	CSR to DMACTRL - Interface for SDMA Buffer Boundary.
139	o_csr_mmc_cfg_tuningcount	OUTPUT	6	i_clk	CSR to CFG - Interface for CFG tuning count
140	o_csr_mmc_cfg_tuning1_5x_disable	OUTPUT	1	i_clk	CSR to CFG - Interface for CFG tuning 1_5x disable
141	o_csr_mmc_spimode	OUTPUT	1	i_clk	CSR to MMCCTRL - Interface for REG_SET: SPI Mode.
142	o_csr_cdet_cd_asyncwakeupena	OUTPUT	1	i_clk	CSR to CARDDDET - Interface for CSR to CD: Asynchronous Wakeup Enable.
143	o_csr_cdet_cd_slottype	OUTPUT	2	i_clk	CSR to CARDDDET - Interface for CSR to CD: Slot Type.
144	o_csr_cdet_cdsigselect	OUTPUT	1	i_clk	CSR to CARDDDET - Interface for Card Detect Signal Select.
145	o_csr_cdet_cdtestlevel	OUTPUT	1	i_clk	CSR to CARDDDET - Interface for Card Detect Test Level.
146	o_csr_cdet_cardremovetssig	OUTPUT	1	i_clk	CSR to CARDDDET - Interface for eMMC Removal Interrupt Status.

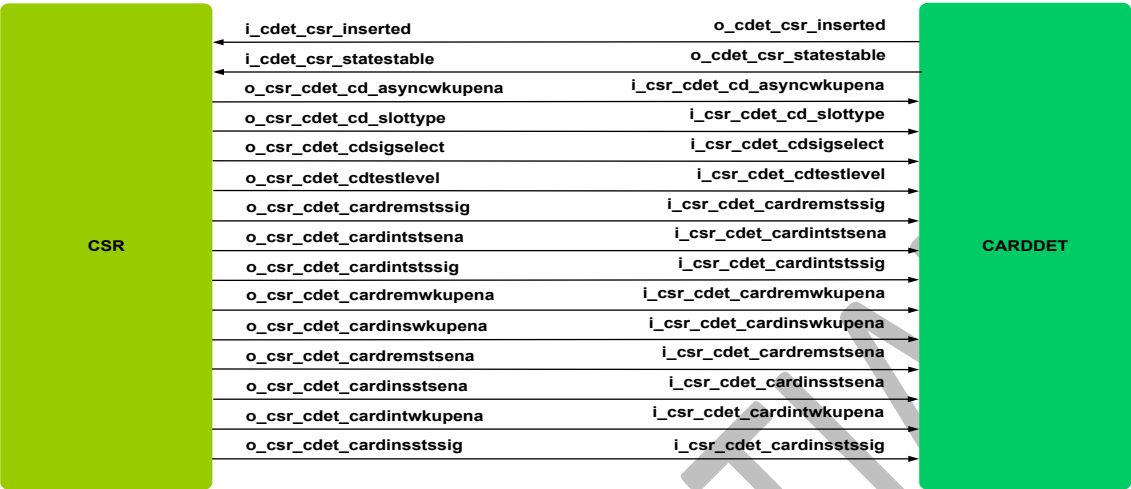
6.2.5.3 CSR and DMA Interface



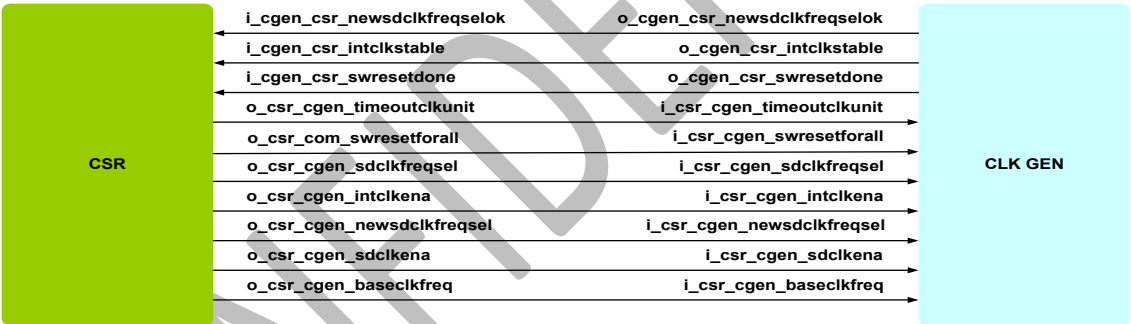
6.2.5.4 CSR and MMC Interface



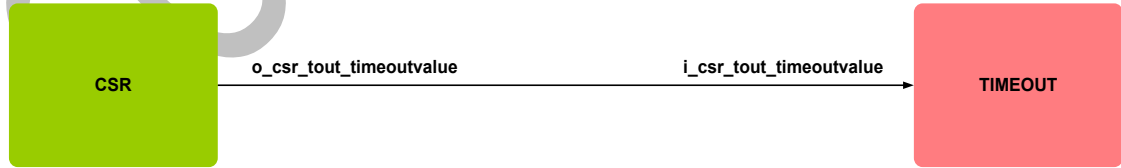
6.2.5.5 CSR and CARDDET Interface



6.2.5.6 CSR and CLKGEN Interface



6.2.5.7 CSR and Timeout Interface



6.2.6 CARDDET Module

6.2.6.1 Functionality

Card Detect State Machine is either the External i_sdif_cd_n or the Internal Signal (Card Detect Test Level) is muxed for Card Detect Logic. The selection is based on the Card Detect Signal Select. The Card Detect State machine monitors the i_sdif_cd_n (SDCD#) signal and performs Debouncing of this signal before setting/resetting "Card Inserted" Signal. Also the The Status Signal for Card Insertion, Card Removal is set based on the state of change in this signal. The Debounce is equal 16 Clock Periods. When the Card Inserted Status Changes from 0 to 1 or from 1 to 0, the Card Insertion or Card Removal Status is Set. This bits are set only when the Corresponding Normal Interrupt Enable is Set. Once set, these bits are cleared when the REG_SET's IntrSts bit is cleared. These Bits are used to generate the Wakeup Event Only. When the Card Insertion/Removal/Interrupt is detected and the corresponding Wakeup Event Enable Bits are set, then the Wakeup Event is generated.

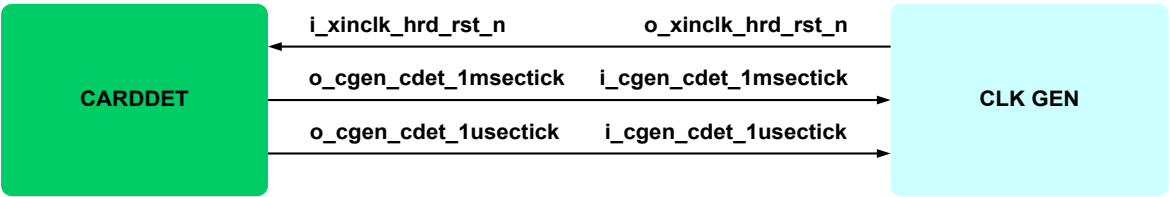
6.2.6.2 Port Descriptions

S.No	PORTS	INPUT/OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
1	i_xin_clk	INPUT	1		XIN CLK - Interface for Crystal Clock for SD.
2	i_xinclk_hrd_rst_n	INPUT	1	i_xin_clk	XIN RESET - Interface for Hard Reset Sync to xin_clk.
3	i_sdif_cd_n	INPUT	1	i_xin_clk	Interface for SD_IF - SD_CD# Input Signal.
4	i_core_cdet_sdif_dat1in	INPUT	1	i_xin_clk	Interface for SD_IF - SD_DAT1 Input Signal.
5	i_csr_cdet_cd_asyncwkupena	INPUT	1	i_clk	CSR to CARDDET - Interface for CSR to CD: Asynchronous Wakeup Enable.
6	i_csr_cdet_cd_slottyp	INPUT	2	i_clk	CSR to CARDDET - Interface for CSR to CD: Slot Type.
7	i_csr_cdet_cdsigselect	INPUT	1	i_clk	CSR to CARDDET - Interface for CSR to CD: Card Detect Signal Select.
8	i_csr_cdet_cdtestlevel	INPUT	1	i_clk	CSR to CARDDET - Interface for CSR to CD: Card Detect Test Level.

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
9	i_csr_cdet_cardremw kupena	INPUT	1	i_clk	CSR to CARDDDET - Interface for CSR to CD: Wakeup Event Enable on SD Card Removal
10	i_csr_cdet_cardinswk upena	INPUT	1	i_clk	CSR to CARDDDET - Interface for CSR to CD: Wakeup Event Enable on SD Card Insertion
11	i_csr_cdet_cardintwk upena	INPUT	1	i_clk	CSR to CARDDDET - Interface for CSR to CD: Wakeup Event Enable on SD Card Interrupt.
12	i_csr_cdet_cardremst sena	INPUT	1	i_clk	CSR to CARDDDET - Interface for CSR to CD: Interrupt Status Enable on SD Card Removal.
13	i_csr_cdet_cardinssts ena	INPUT	1	i_clk	CSR to CARDDDET - Interface for CSR to CD: Interrupt Status Enable on SD Card Insertion.
14	i_csr_cdet_cardintsts ena	INPUT	1	i_clk	CSR to CARDDDET - Interface for CSR to CD: Interrupt Status Enable on SD Card Interrupt
15	i_csr_cdet_cardremst ssig	INPUT	1	i_clk	CSR to CARDDDET - Interface for CSR to CD: SD Card Removal Interrupt Status.
16	i_csr_cdet_cardinssts sig	INPUT	1	i_clk	CSR to CARDDDET - Interface for CSR to CD: SD Card Insertion Interrupt Status.
17	i_csr_cdet_cardintsts sig	INPUT	1	i_clk	CSR to CARDDDET - Interface for CSR to CD: SD Card Interrupt Interrupt Status.
18	o_cdet_csr_statestable	OUTPUT	1	i_xin_clk	CARDDDET to CSR - Interface for Card State Stable Indication.

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
19	o_cdet_csr_inserted	OUTPUT	1	i_xin_clk	CARDDET to CSR - Interface for Card Inserted Indication.
20	o_cdet_core_wakeup event	OUTPUT	1	i_clk	CARDDET to CORE - Interface for Wakeup Event to System Side.
21	i_cgen_cdet_1msecti ck	INPUT	1	i_xin_clk	CLKGEN to CARDDET - Interface for CLK_GEN: 1 MilliSecond Tick.
22	i_cgen_cdet_1usecti ck	INPUT	1	i_xin_clk	CLKGEN to CARDDET - Interface for CLK_GEN: 1 MicroSecond Tick.

6.2.6.3 CARDDET and CLKGEN Interface



6.2.6.4 CARDDET and CSR Interface



6.2.7 Sync True Dpram Module

6.2.7.1 Functionality

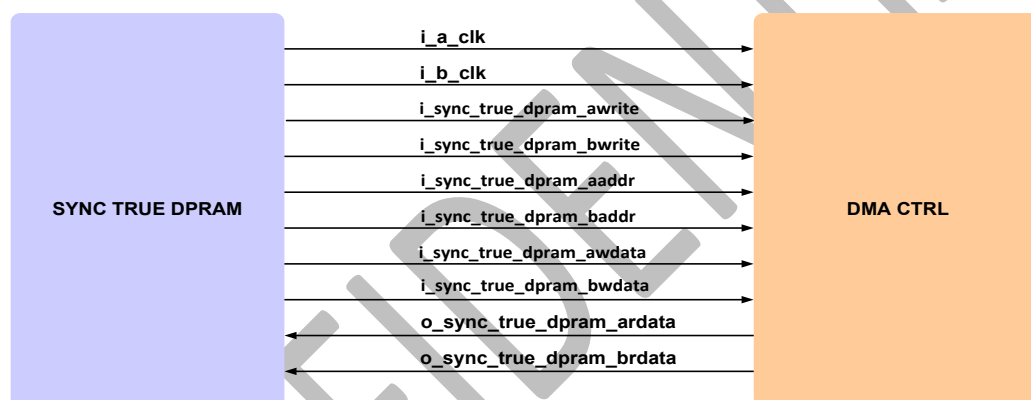
Sync True dpram has two channels, a & b. For channel a it has clock (i_a_clk), input data and output data signals and for channel b it has clock (i_b_clk), input data and output data signals. For channel a, if awrite is one then it will load the awrite data into memory based on address, if it is zero it will read the memory data. For channel b, if bwrite is one then it will load the bwrite data into memory based on baddress if it is zero it will read the memory data.

6.2.7.2 Port Descriptions

S.No	PORTS	INPUT/OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
1	i_a_clk	INPUT	1		SYNC TRUE DPRAM - A port Clock input
2	i_b_clk	INPUT	1		SYNC TRUE DPRAM - B port Clock input
3	i_sync_true_dpram_a_write	INPUT	1	i_clk	SYNC TRUE DPRAM - A port Write control
4	i_sync_true_dpram_b_write	INPUT	1	i_sd_clk	SYNC TRUE DPRAM - B port Write control
5	i_sync_true_dpram_a_addr	INPUT	SYNC_TRUE_DPRAM_DATA_WIDTH	i_clk	SYNC TRUE DPRAM - A port Write pointer input
6	i_sync_true_dpram_b_addr	INPUT	SYNC_TRUE_DPRAM_DATA_WIDTH	i_sd_clk	SYNC TRUE DPRAM - B port Write pointer input
7	i_sync_true_dpram_a_wdata	INPUT	SYNC_TRUE_DPRAM_DATA_WIDTH	i_clk	SYNC TRUE DPRAM - A port Write data input
8	i_sync_true_dpram_b_wdata	INPUT	SYNC_TRUE_DPRAM_DATA_WIDTH	i_sd_clk	SYNC TRUE DPRAM - B port Write data input

S.No	PORTS	INPUT/OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
9	o_sync_true_dpram_ardata	OUTPUT	SYNC_TRUE_DPRAM_DATA_WIDTH	i_clk	SYNC TRUE DPRAM - A port Read data output
10	o_sync_true_dpram_brdata	OUTPUT	SYNC_TRUE_DPRAM_DATA_WIDTH	i_sd_clk	SYNC TRUE DPRAM - B port Read data output

6.2.7.3 Sync True Dpram and DMA Interface



6.2.8 APB Slave Interface Module

6.2.8.1 Functionality

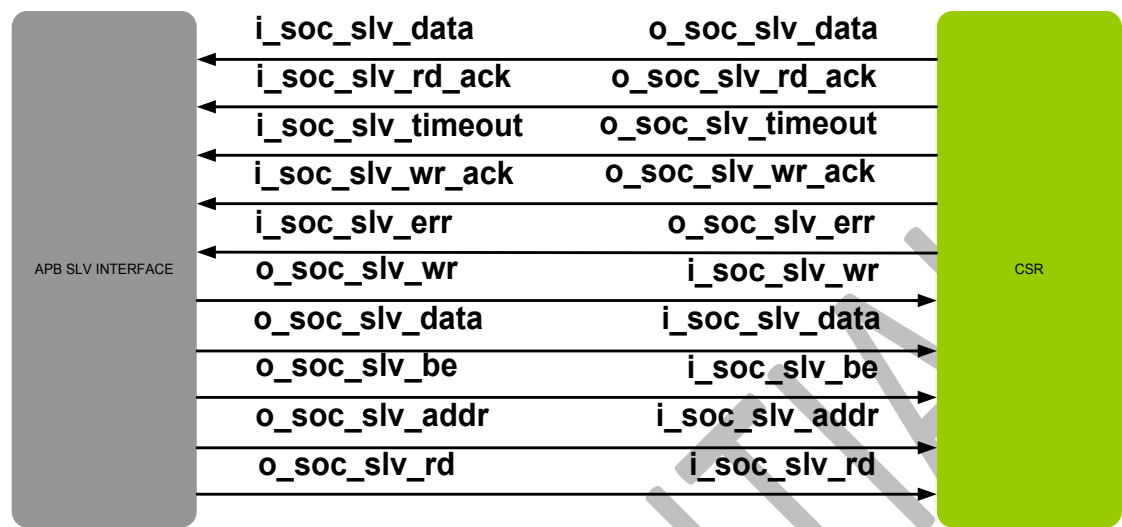
The APB protocol is a low-cost interface, optimized for minimal power consumption and reduced interface complexity. The APB interface is not pipelined and is a simple, synchronous protocol. Every transfer takes at least two cycles to complete. The APB interface is designed for accessing the programmable control registers of peripheral devices. APB peripherals are typically connected to the main memory system using an APB bridge. For example, a bridge from AXI to APB could be used to connect a number of APB peripherals to an AXI memory system. APB transfers are initiated by an APB bridge. APB bridges can also be referred to as a Requester. A peripheral interface responds to requests. APB peripherals can also be referred to as a Completer.

6.2.8.2 Port descriptions

S.No	PORTS	INPUT/OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
1	i_clk	INPUT	1		System - clock
2	i_rst_n	INPUT	1		System - Active low reset input
3	i_apb_slv_addr	INPUT	APB_SLV_ADDR_WIDTH	i_clk	APB Slave - Address input
4	i_apb_slv_sel	INPUT	1	i_clk	APB Slave - Select input
5	i_apb_slv_enable	INPUT	1	i_clk	APB Slave - Enable control input
6	i_apb_slv_write	INPUT	1	i_clk	APB Slave - Read/write control input
7	i_apb_slv_wdata	INPUT	APB_SLV_DATA_WIDTH	i_clk	APB Slave - Write data input
8	o_apb_slv_rdata	OUTPUT	APB_SLV_DATA_WIDTH	i_clk	APB Slave - Read data output
9	o_apb_slv_ready	OUTPUT	1	i_clk	APB Slave - Ready output
10	o_apb_slv_err	OUTPUT	1	i_clk	APB Slave - Error output
11	i_soc_slv_rd_ack	INPUT	1	i_clk	SOC Slave - Read ack input
12	i_soc_slv_wr_ack	INPUT	1	i_clk	SOC Slave - Write ack input
13	i_soc_slv_err	INPUT	1	i_clk	SOC Slave - Error input

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
14	o_soc_slv_wr	OUTPUT	1	i_clk	SOC Slave - Write enable output
15	o_soc_slv_rd	OUTPUT	1	i_clk	SOC Slave - Read enable output
16	o_soc_slv_addr	OUTPUT	APB_SOC_ADDR_WIDTH	i_clk	SOC Slave - Address output
17	o_soc_slv_be	OUTPUT	APB_SOC_BE_WIDTH	i_clk	SOC Slave - Byte enable for write data output
18	o_soc_slv_data	OUTPUT	APB_SOC_DATA_WIDTH	i_clk	SOC Slave - Write data output
19	i_soc_slv_data	INPUT	APB_SOC_DATA_WIDTH	i_clk	SOC Slave - Read data input
20	i_soc_slv_timeout	INPUT	16	i_clk	SOC Slave - Timeout input
21	o_soc_slv_timeout_err	OUTPUT	1	i_clk	SOC Slave - Transfer timeout error
22	o_soc_slv_len	OUTPUT	8	i_clk	SOC Slave - Length of transfer
23	o_soc_slv_id	OUTPUT	8	i_clk	SOC Slave - ID
24	o_soc_slv_ctype	OUTPUT	4	i_clk	SOC Slave - Command type
25	o_soc_slv_width	OUTPUT	2	i_clk	SOC Slave - Width of transfer

6.2.8.3 APB Slave Interface and CSR Interface



6.2.9 AXI Master Module

6.2.9.1 Functionality

AXI master starts a transfer by driving the address and control signals. These signals provide information about the address, direction, width of the transfer, and indicate if the transfer forms part of a burst.

6.2.9.2 Port Descriptions

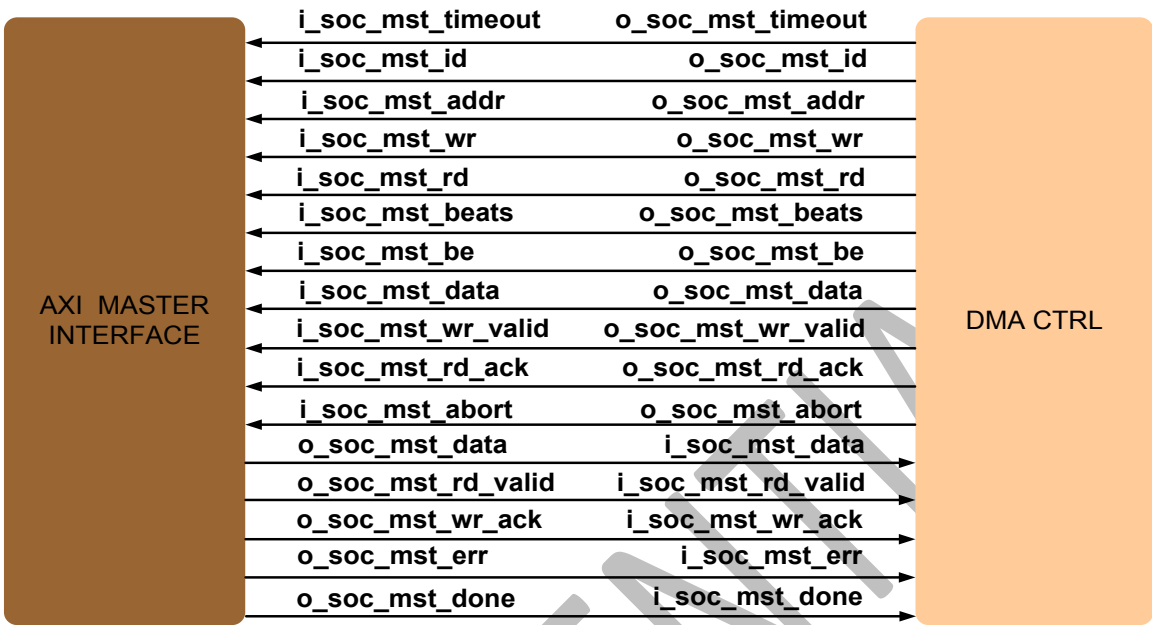
S.No	PORTS	INPUT/OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
1	i_clk	INPUT	1		AXI CLOCK - AXI clock
2	i_rst_n	INPUT	1		AXI RESET - Asynchronous reset, but deassertion synchronous to hclk
3	o_axi_mst_awid	INPUT	AXI_MST_ID_WIDTH	i_dma_clk	AXI Master - Write Address ID Tag to AXI Slave
4	o_axi_mst_awaddr	INPUT	AXI_MST_ADDR_WIDTH	i_dma_clk	AXI Master - Write Address to AXI Slave
5	o_axi_mst_awlen	OUTPUT	AXI_MST_BURST_LENGTH	i_dma_clk	AXI Master - Write Transfer Length to AXI Slave
6	o_axi_mst_awsize	OUTPUT	3	i_dma_clk	AXI Master - Write Transfer Size to AXI Slave
7	o_axi_mst_awburst	OUTPUT	2	i_dma_clk	AXI Master - Write Transfer Burst Type to AXI Slave
8	o_axi_mst_awvalid	OUTPUT	1	i_dma_clk	AXI Master - Write Address/Control Valid Indication to AXI Slave
9	i_axi_mst_awready	INPUT	1	i_dma_clk	AXI Master - Write Address/Control Ready Indication from AXI Slave
10	o_axi_mst_awlock	OUTPUT	2	i_dma_clk	AXI Master - Type of access
11	o_axi_mst_awprot	OUTPUT	3	i_dma_clk	AXI Master - Protection type
12	o_axi_mst_awcache	OUTPUT	4	i_dma_clk	AXI Master - Cache type

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
13	o_axi_mst_wdata	OUTPUT	AXI_MST_DATA_WIDTH	i_dma_clk	AXI Master - Write Data to AXI Slave
14	o_axi_mst_wstrb	OUTPUT	AXI_MST_DATA_BYTE_WIDTH	i_dma_clk	AXI Master - Write Data Strobe to AXI Slave
15	o_axi_mst_wlast	OUTPUT	1	i_dma_clk	AXI Master - Write Data Last Indication to AXI Slave
16	o_axi_mst_wvalid	OUTPUT	1	i_dma_clk	AXI Master - Write Data Valid Indication to AXI Slave
17	i_axi_mst_wready	INPUT	1	i_dma_clk	AXI Master - Write Data Ready Indication from AXI Slave
18	i_axi_mst_bid	INPUT	AXI_MST_ID_WIDTH	i_dma_clk	AXI Master - Write Response ID from AXI Slave
19	i_axi_mst_bresp	INPUT	2	i_dma_clk	AXI Master - Write Response from AXI Slave
20	i_axi_mst_bvalid	INPUT	[AHB_SOC_DATA_WIDTH - 1:0]	i_dma_clk	AXI Master - Write Response Valid Indication from AXI Slave
21	o_axi_mst_bready	OUTPUT	1	i_dma_clk	AXI Master - Write Response Ready Indication to AXI Slave
22	o_axi_mst_arid	OUTPUT	AXI_MST_ID_WIDTH	i_dma_clk	AXI Master - Read Address ID Tag to AXI Slave
23	o_axi_mst_araddr	OUTPUT	AXI_MST_ADDR_WIDTH	i_dma_clk	AXI Master - Read Address to AXI Slave
24	o_axi_mst_arlen	OUTPUT	AXI_MST_BURST_LENGTH	i_dma_clk	AXI Master - Read Transfer Length to AXI Slave
25	o_axi_mst_arsize	OUTPUT	3	i_dma_clk	AXI Master - Read Transfer Size to AXI Slave
26	o_axi_mst_arburst	OUTPUT	2	i_dma_clk	AXI Master - Read Transfer Burst Type to AXI Slave

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
27	o_axi_mst_arlock	OUTPUT	2	i_dma_clk	AXI Master - Type of access
28	o_axi_mst_arprot	OUTPUT	3	i_dma_clk	AXI Master - Protection type
29	o_axi_mst_arcache	OUTPUT	4	i_dma_clk	AXI Master - Cache type
30	o_axi_mst_arvalid	OUTPUT	1	i_dma_clk	AXI Master - Read Address/Control Valid Indication to AXI Slave
31	i_axi_mst_arready	INPUT	1	i_dma_clk	AXI Master - Read Address/Control Ready Indication from AXI Slave
32	i_axi_mst_rid	INPUT	AXI_MS T_ID_WI DTH	i_dma_clk	AXI Master – Read Data ID Tag length from AXI Slave
33	i_axi_mst_rdata	INPUT	AXI_MS T_DATA _WIDTH	i_dma_clk	AXI Master – Read Data from AXI Slave
34	i_axi_mst_rresp	INPUT	2	i_dma_clk	AXI Master - Read Response from AXI Slave
35	i_axi_mst_rlast	INPUT	1	i_dma_clk	AXI Master - Read Data Last from AXI Slave
36	i_axi_mst_rvalid	INPUT	1	i_dma_clk	AXI Master - Read Data/Response Valid Indication from AXI Slave
37	o_axi_mst_rready	OUTPUT	1	i_dma_clk	AXI Master - Read Data/Response Ready Indication to AXI Slave
38	i_soc_mst_beats	INPUT	9	i_dma_clk	SOC Master - Number of beats input
39	i_soc_mst_be	INPUT	AXI_SO C_BE_ WIDTH	i_dma_clk	SOC Master - Byte enable input
40	o_soc_mst_err	OUTPUT	1	i_dma_clk	SOC Master - Error transfer output
41	o_soc_mst_done	OUTPUT	1	i_dma_clk	SOC Master - Transfer done output
42	i_soc_mst_rd_ack	INPUT	1	i_dma_clk	SOC Master - Read ack input
43	i_soc_mst_wr	INPUT	1	i_dma_clk	SOC Master - Write input

S.No	PORTS	INPUT/ OUTPUT	WIDTH	CLOCK DOMAIN	DESCRIPTION
44	i_soc_mst_rd	INPUT	1	i_dma_clk	SOC Master - Read input
45	o_soc_mst_cmd_ack	OUTPUT	1	i_dma_clk	SOC Master - Write/Read command ack
46	o_soc_mst_cmd_full	OUTPUT	1	i_dma_clk	SOC Master - Write command full
47	o_soc_mst_rd_valid	OUTPUT	1	i_dma_clk	SOC Master - Read data valid output
48	o_soc_mst_wr_ack	OUTPUT	1	i_dma_clk	SOC Master - Write data ack output
49	i_soc_mst_wr_valid	INPUT	1	i_dma_clk	SOC Master - Write data valid input
50	i_soc_mst_abort	INPUT	1	i_dma_clk	SOC Master - Abort transfer input
51	i_soc_mst_data	INPUT	AXI_SOC_DATA_WIDTH	i_dma_clk	SOC Master - Write data input
52	o_soc_mst_data	OUTPUT	AXI_SOC_DATA_WIDTH	i_dma_clk	SOC Master - Read data output
53	i_soc_mst_id	INPUT	AXI_SOC_ID_WIDTH	i_dma_clk	SOC Master - ID input
54	i_soc_mst_timeout	INPUT	16	i_dma_clk	SOC Master - Transfer timeout limit
55	o_soc_mst_timeout_err	OUTPUT	1	i_dma_clk	SOC Master - Transfer timeout error
56	i_soc_mst_addr	input	AXI_SOC_ADDR_WIDTH	i_dma_clk	SOC Master - Address input

6.2.9.3 AXI Master and DMA Interface



6.2.10 RST MUX Interface Module

6.2.10.1 Functionality

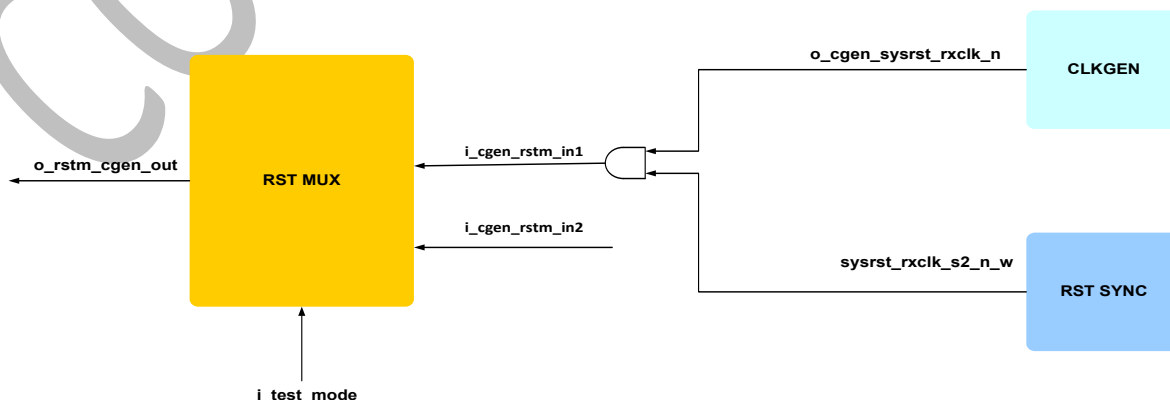
This is 2:1 multiplexer logic. It has two inputs and one output. It will select one of the two inputs using select lines.

6.2.10.2 Port Descriptions

PORTS	TYPE	WIDTH	CLOCK DOMAIN	DESCRIPTION
i_cgen_rstm_in1	INPUT	1	i_xin_clk	IN1 - Interface for First Input
i_cgen_rstm_in2	INPUT	1	i_xin_clk	IN2 - Interface for Second Output
i_cgen_rstm_sel	INPUT	1	i_clk	SELECT - Interface for Select
o_rstm_cgen_out	OUTPUT	1	i_xin_clk	MULTIPLEXED OUT - Interface for Multiplexed output

6.2.10.3 RST MUX Interface

RST MUX has two inputs(*i_cgen_rstm_in1*, *i_cgen_rstm_in2*), one select line(*i_cgen_rstm_sel*) and one output(*o_out*). In RST MUX *i_cgen_rstm_sel* is *i_test_mode* and *i_cgen_rstm_in2* is *i_rst_n*. *o_cgen_sysrst_rxclk_n* from clockgen and *sysrst_rxclk_s2_n_w* from RST SYNC both are given to AND gate, AND gate output is given as *i_cgen_rstm_in1* to RST MUX.

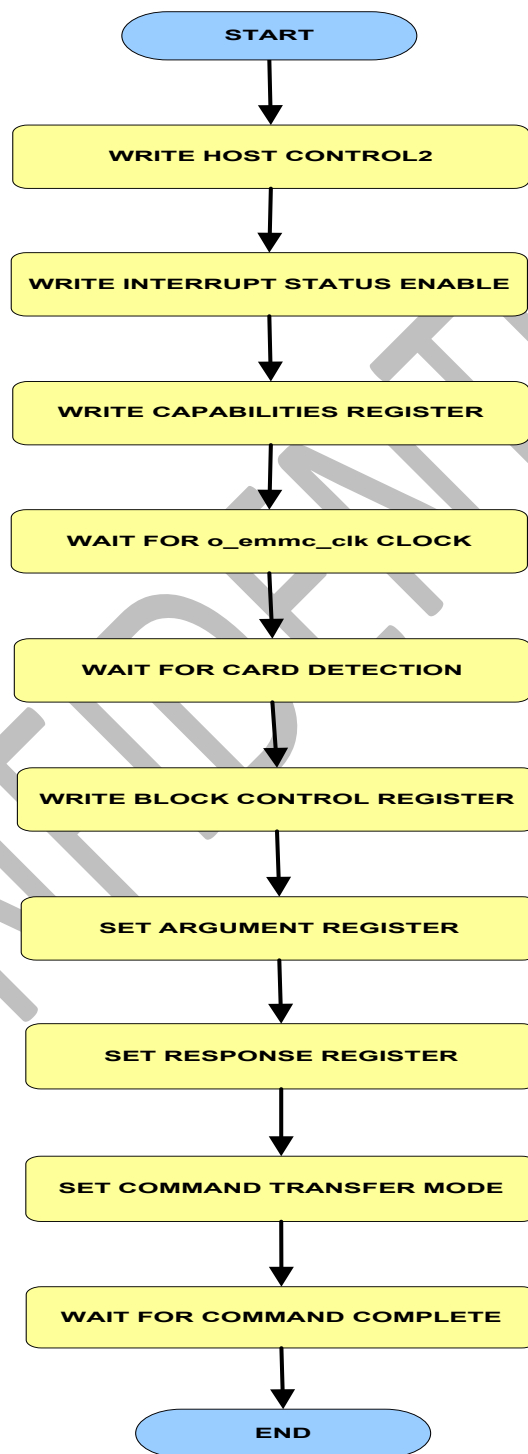


6.3 Flow charts

6.3.1 Command Generation

Below flow chart shows a basic flow of command generation in eMMC Host IIP.

Command Generation

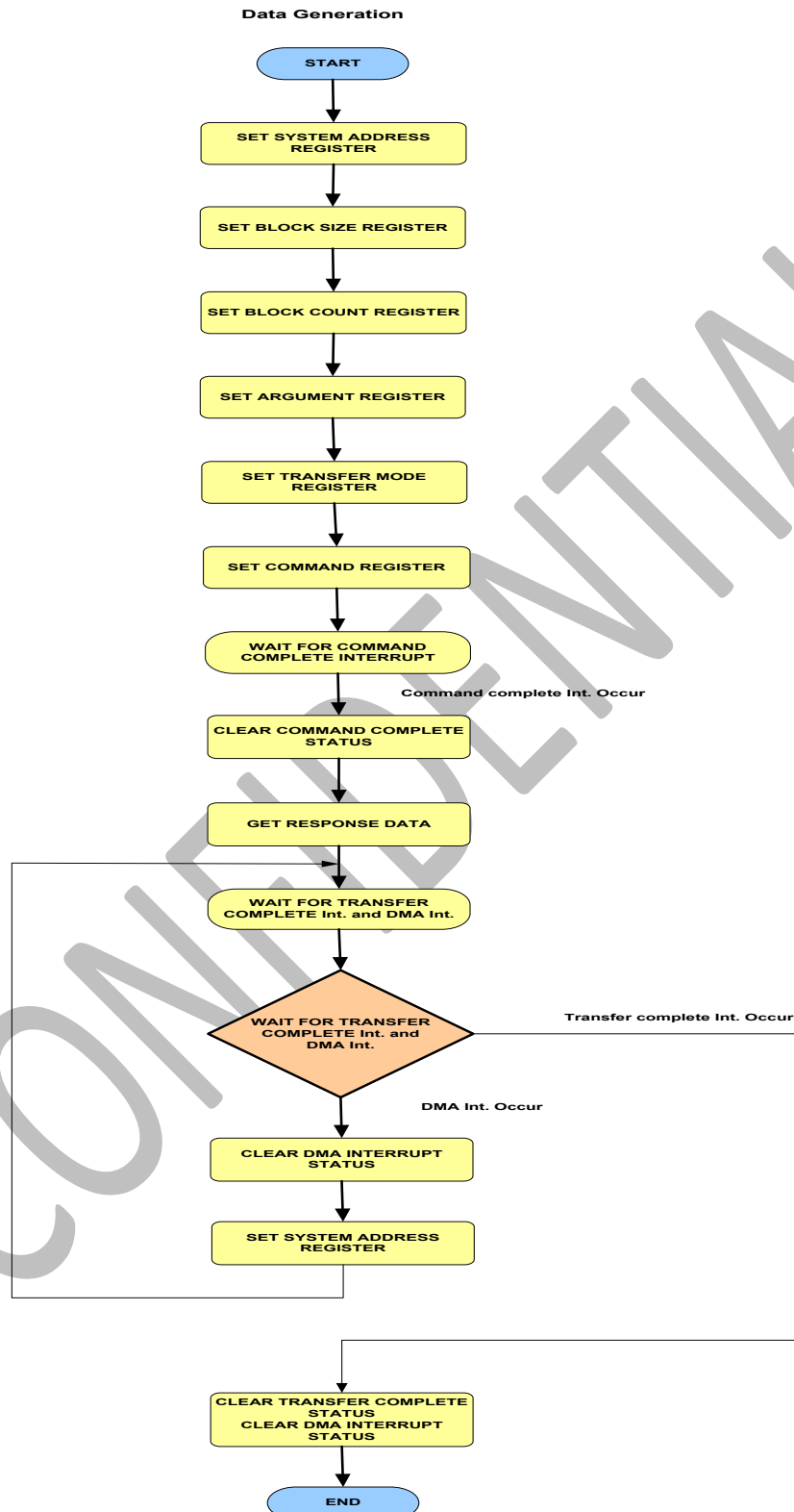


1. Set the Host controller2 for Generating SD clock.
2. Set the Interrupt Status Enable for Enabling the Interrupt status.
3. Set the Capabilities Registers for setting the frequency.
4. Wait for o_emmc_clk
5. Wait for Card detection.
6. Set Block control registers for setting the Block size.
7. Set the Argument value to the Argument register.
8. Set Response registers.
9. Set Command transfer mode register for providing command required settings.
10. Wait for command complete.

6.3.2 Data Generation

1. Data location of system memory is set to the SDMA System Address register.
2. Set the value corresponding to the executed data byte length of one block in the Block Size register.
3. Set the value corresponding to the executed data block count in the Block Count register.
4. Set the argument value to the Argument register.
5. Set the value to the Transfer Mode register. The Host Driver determines Multi / Single Block Select, Block Count Enable, Data Transfer Direction, Multi / Single Block Select and Block Count Enable are determined
6. Set the value to the Command register.
7. Wait for the Command Complete Interrupt.
8. Write 1 to the Command Complete in the Interrupt Status register to command complete status bit.
9. Read Response registers and gets necessary information of the issued command.
10. Wait for the Transfer Complete Interrupt and DMA interrupt.
11. If DMA interrupt occurs, clear dma interrupt and set system address register.
12. If transfer complete interrupt occurs, clear transfer complete status and DMA interrupt status.

The below flow chart shows a basic flow of command generation in eMMC Host IIP.

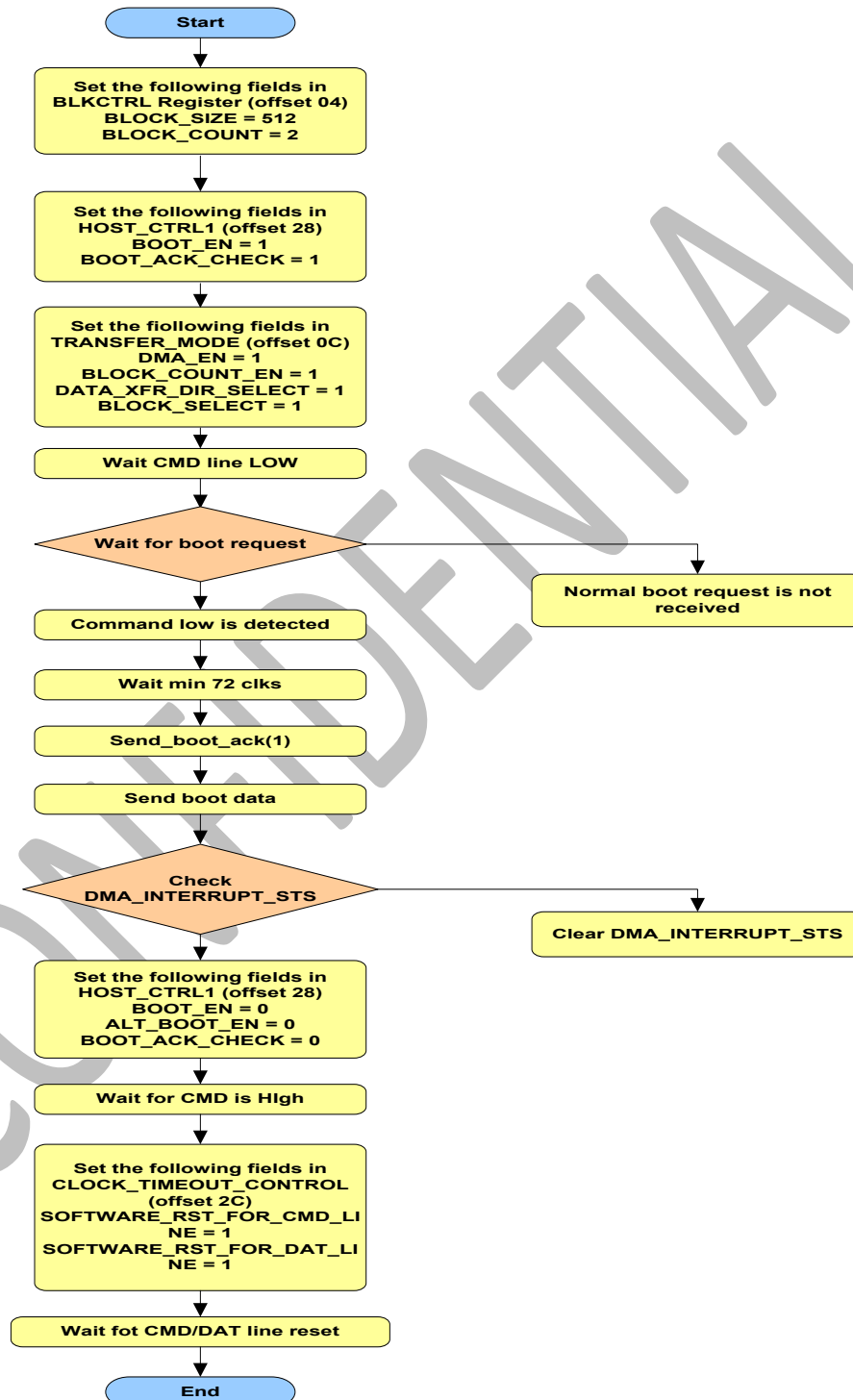


6.3.3 Normal Boot Operation

1. Set block size as 512byte and initial boot block count as 2(For example scenario).
2. Read Host CTRL1 register to change BOOT_EN & BOOT_ACK_CHECK bits as 1.
3. Set DMA_EN, BLOCK_COUNT_EN, BLOCK_SELECT and DATA_XFR_DIR_SELECT for Boot operation.
4. Wait CMD line LOW for normal boot request from Host.
5. If boot request is not received within time limit report error.
6. If CMD load is detected.
7. Wait for min 72 clks CMD line LOW to select normal boot request.
8. Send Boot ACK Block within 1CLK.
9. Send boot data.
10. Check DMA Interrupt.
11. If DMA Interrupt is 1 clear DMA Interrupt.
12. Read Host CTRL1 register to change BOOT_EN & BOOT_ACK_CHECK bits as 0.
13. Wait for CMD is High.
14. Once Boot operation is completed do soft reset on CMD&DAT line.
15. Wait for CMD/DAT line reset.

The below flow chart shows a basic flow of normal boot operation.

Normal boot operation

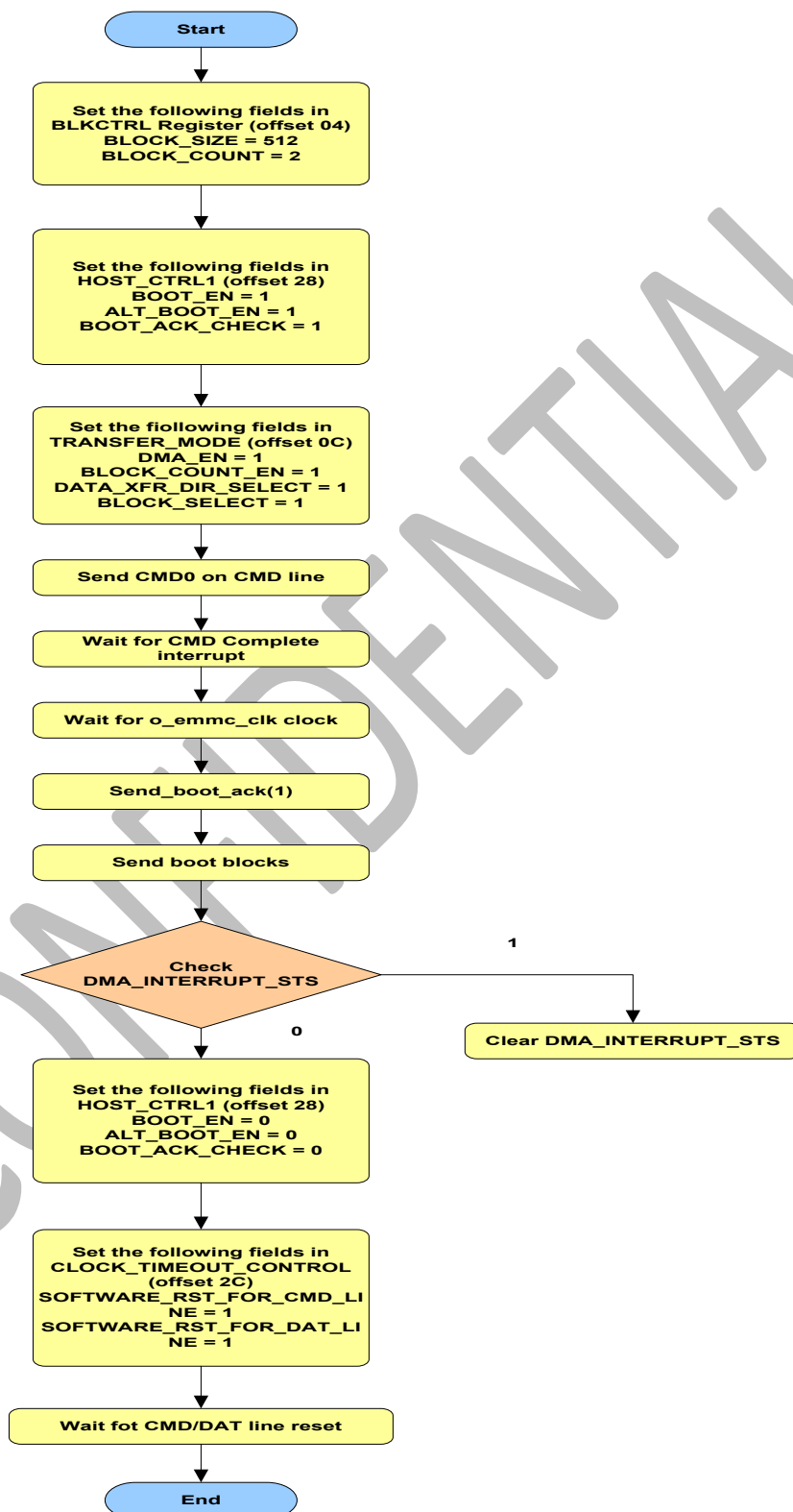


6.3.4 Alternate Boot operation

1. Set block size as 512byte and initial boot block count as 2(For example scenario).
2. Read Host CTRL1 register to change BOOT_EN, ALT_BOOT_EN & BOOT_ACK_CHECK bits as 1.
3. Set DMA_EN, BLOCK_COUNT_EN, BLOCK_SELECT and DATA_XFR_DIR_SELECT for alt Boot operation.
4. Send CMD0 with boot request.
5. Wait for CMD Complete interrupt.
6. Wait for eMMC CLK.
7. Send Boot ACK Block within 1CLK.
8. Send boot data.
9. Check DMA Interrupt.
10. If DMA Interrupt is 1 clear DMA Interrupt.
11. Read Host CTRL1 registers to change BOOT_EN & BOOT_ACK_CHECK bits as 0.
12. Wait for CMD is High.
13. Once Boot operation is completed do soft reset on CMD&DAT line.
14. Wait for CMD/DAT line reset.

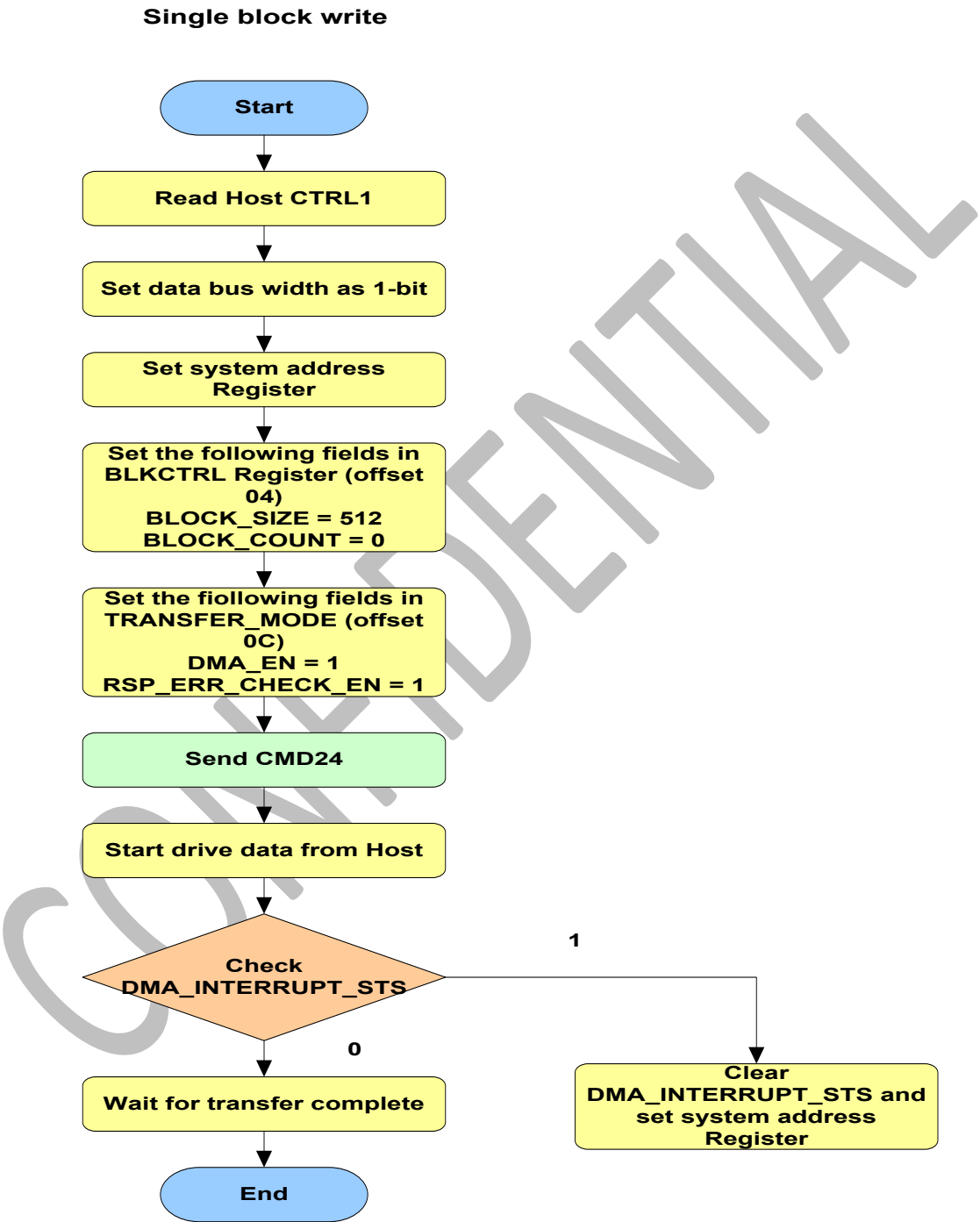
The below flow chart shows a basic flow of alternate boot operation.

Alternate boot operation



6.3.5 Single block Write operation

The below flow chart shows a basic flow of single block write operation.



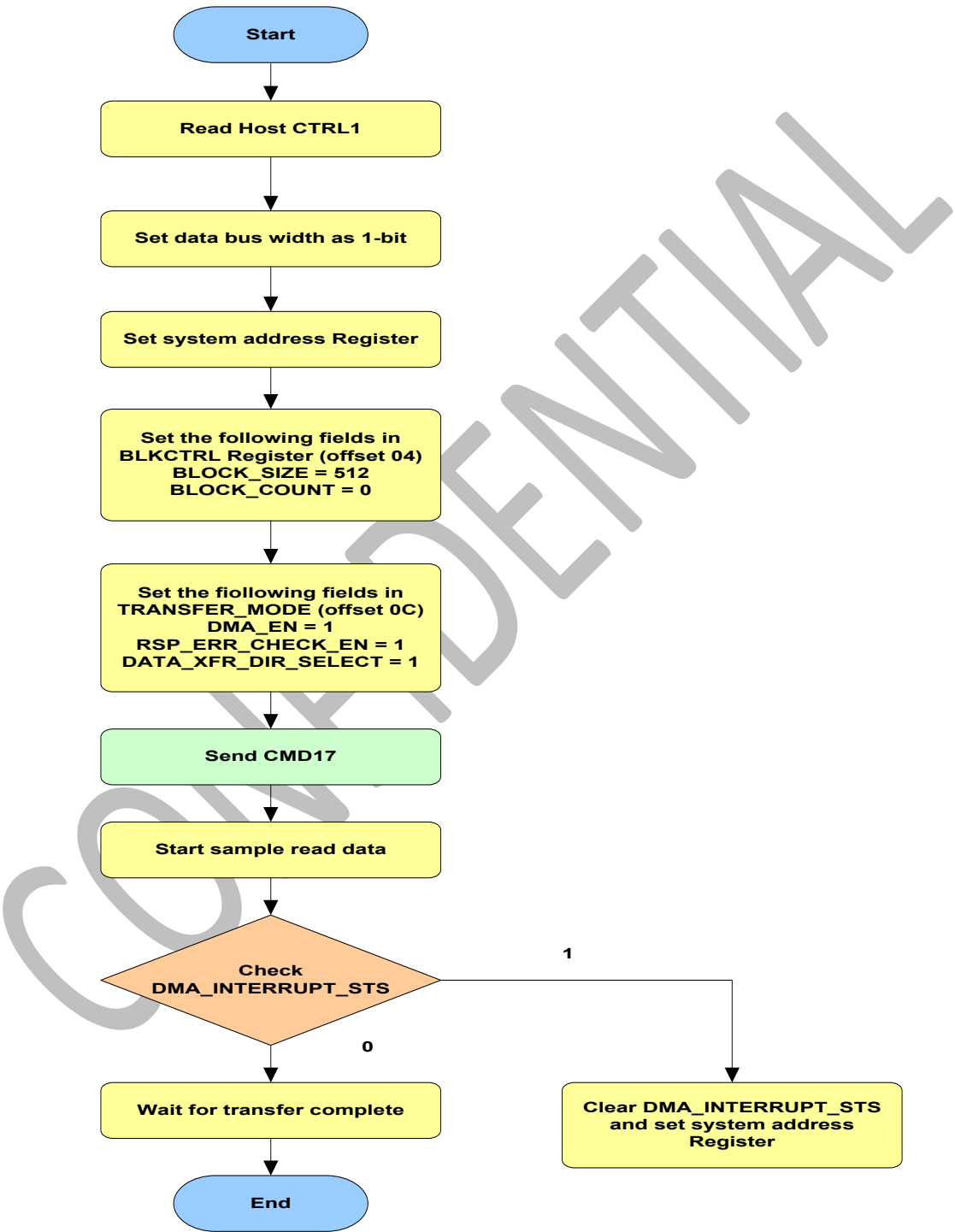
1. Read Host CTRL1 register to change data bus width as 1bit.
2. Set System Address Register.
3. Set block size as 512 byte in BLKCTRL Register.
4. Set DMA_EN and RSP_ERR_CHECK_EN for single block operation.
5. Send CMD24 for Single block write.
6. Drive data from Host.
7. Check DMA Interrupt.
8. If DMA Interrupt is 1 clear DMA Interrupt and set System Address Register.
9. If DMA Interrupt is 0, wait for Transfer complete.

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6.3.6 Single block Read operation

The below flow chart shows a basic flow of single block read operation.

Single block read

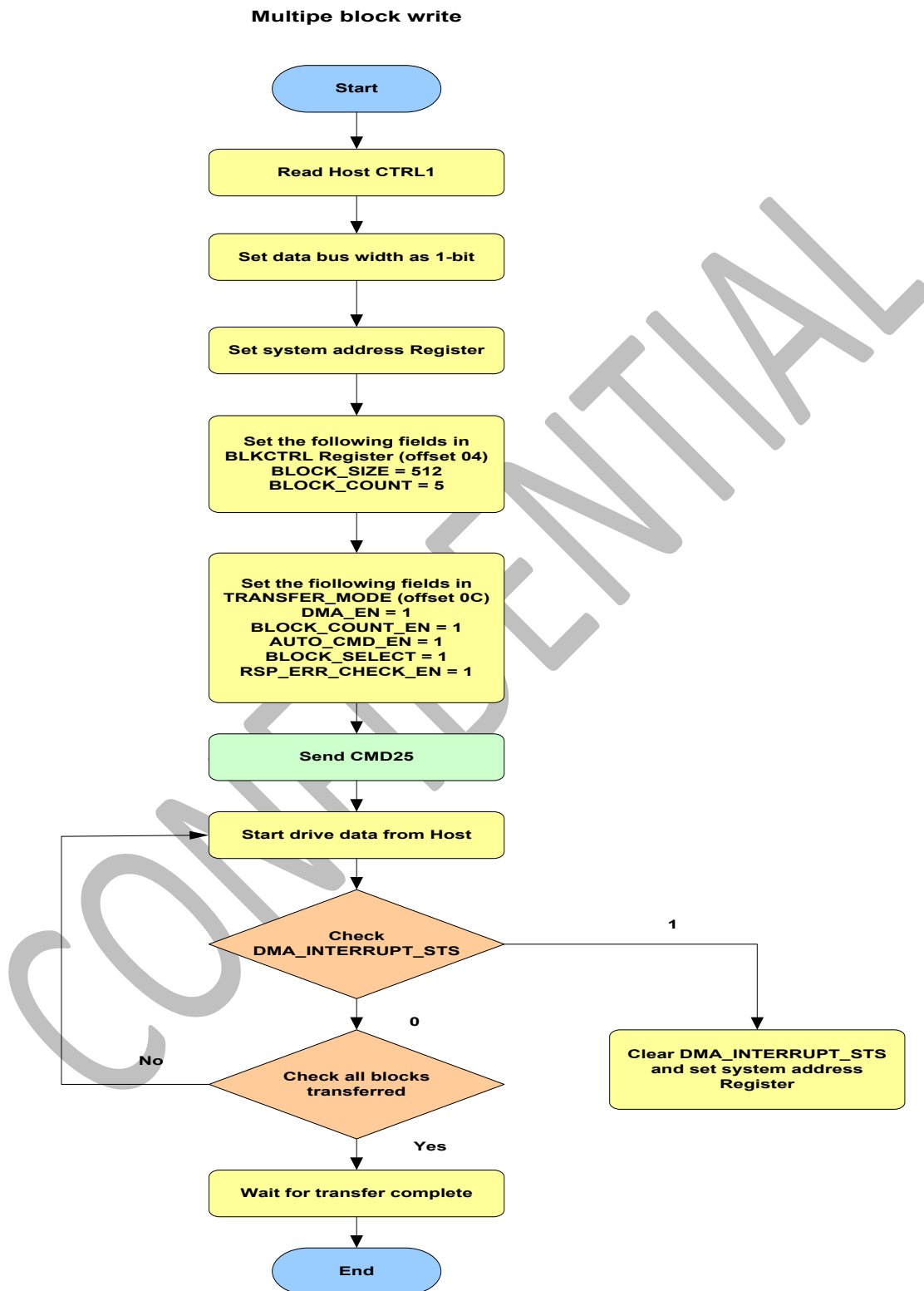


1. Read Host CTRL1 register to change data bus width as 1bit.
2. Set System Address Register.
3. Set block size as 512 byte in BLKCTRL Register.
4. Set DMA_EN, RSP_ERR_CHECK_EN and DATA_XFR_DIR_SELECT for single block read operation.
5. Send CMD17 for Single block Read.
6. Sample read data.
7. Check DMA Interrupt.
8. If DMA Interrupt is 1 clear DMA Interrupt and set System Address Register.
9. If DMA Interrupt is 0, wait for Transfer complete.

6.3.7 Multi block Write operation

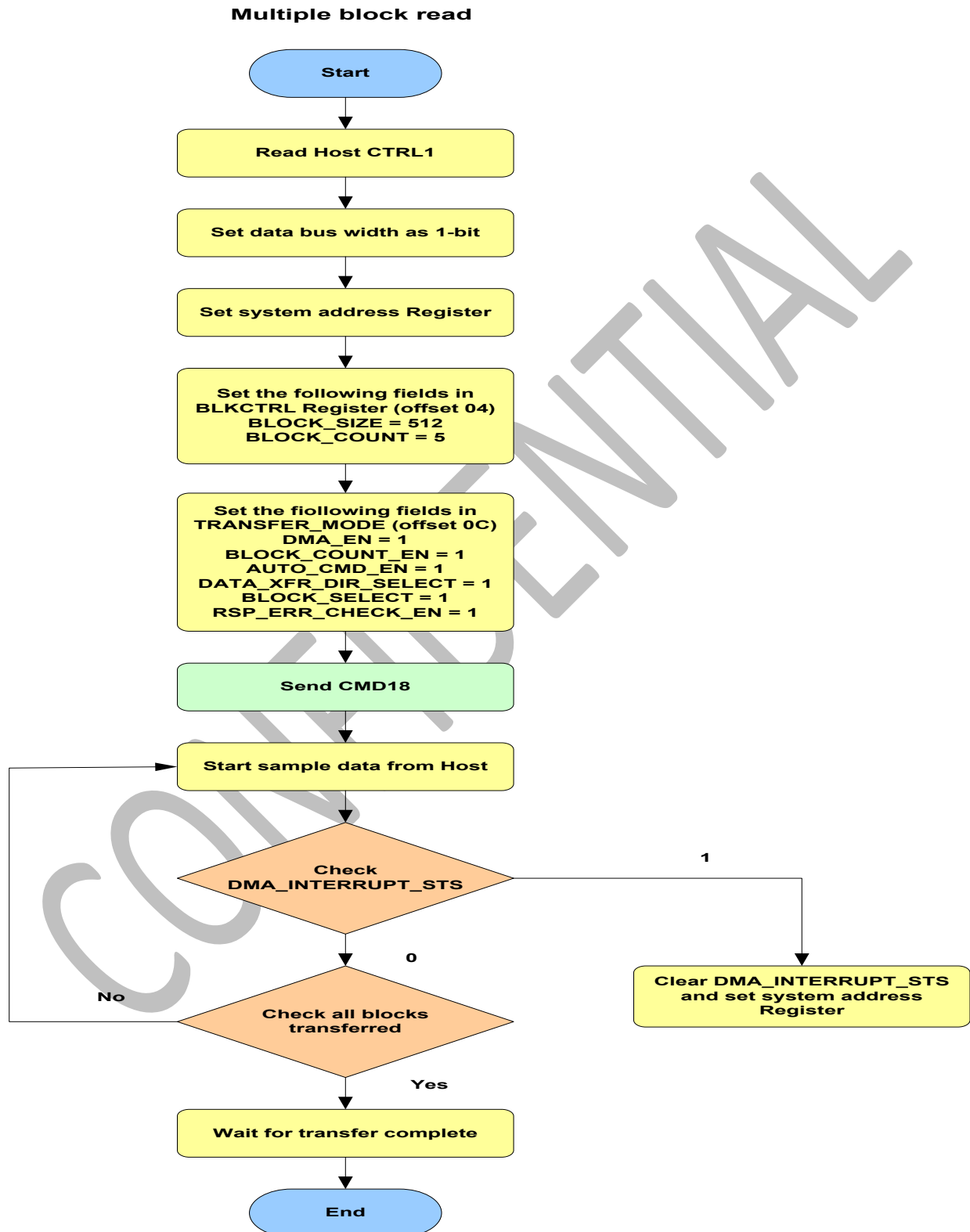
1. Read Host CTRL1 register to change data bus width as 1bit.
2. Set System Address Register.
3. Set block size as 512 byte and block count as 5 byte in BLKCTRL Register.
4. Set DMA_EN, BLOCK_COUNT_EN, AUTO_CMD_EN, BLOCK_SELECT and RSP_ERR_CHECK_EN for multi block operation.
5. Send CMD25 for multi block write.
6. Drive data from Host.
7. Check DMA Interrupt.
8. If DMA Interrupt is 1 clear DMA Interrupt and set System Address Register.
9. If DMA Interrupt is 0, check all blocks are transferred.
10. Wait for transfer complete.

The below flow chart shows a basic flow of multi block write operation.



6.3.8 Multi block Read operation

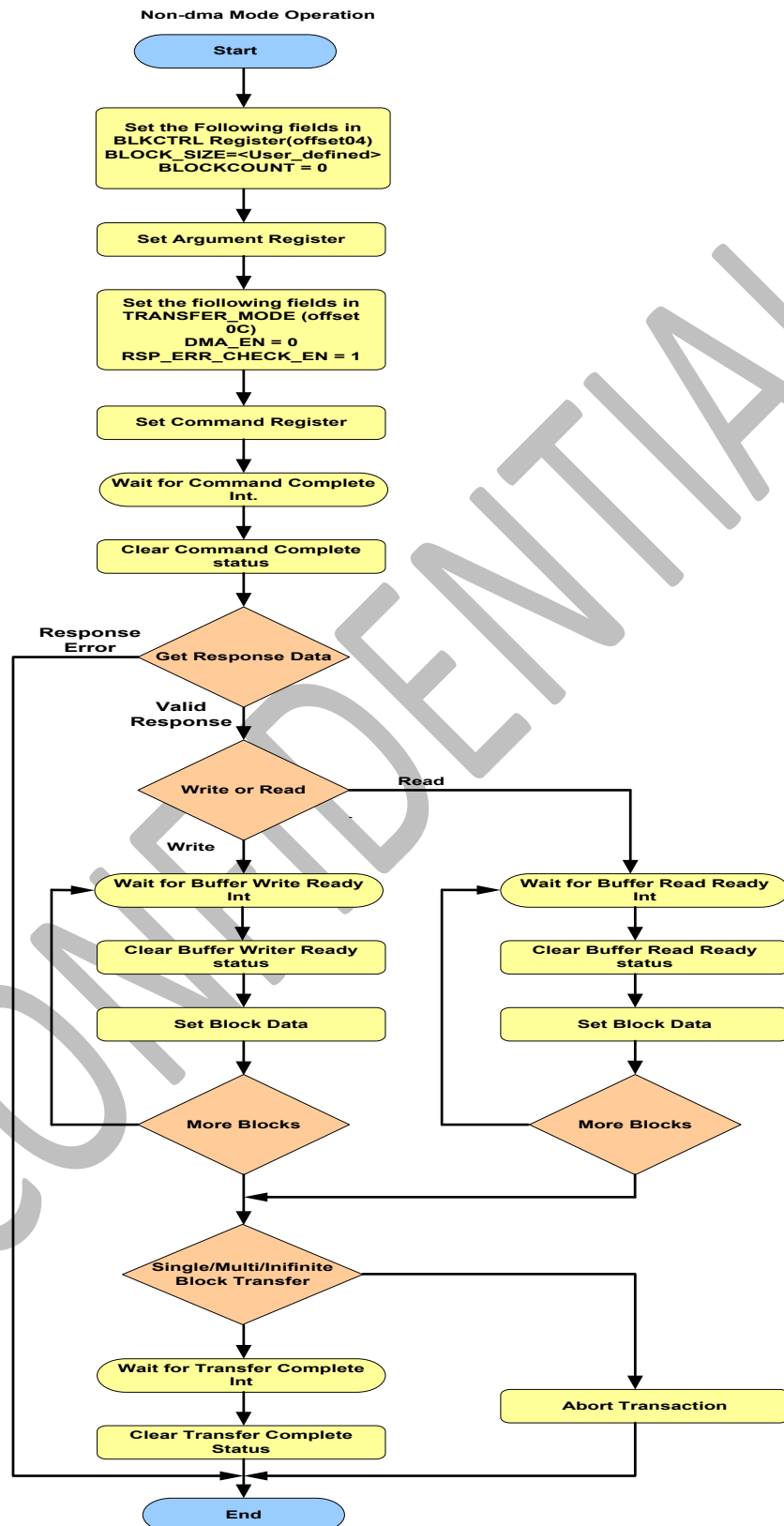
The below flow chart shows a basic flow of multi block read operation.



1. Read Host CTRL1 register to change data bus width as 1bit.
2. Set System Address Register.
3. Set block size as 512 byte and block count as 5 in BLKCTRL Register.
4. Set DMA_EN, BLOCK_COUNT_EN, AUTO_CMD_EN, BLOCK_SELECT, RSP_ERR_CHECK_EN and DATA_XFR_DIR_SELECT for multi block operation.
5. Send CMD18 for multi block read.
6. Sample read data.
7. Check DMA Interrupt.
8. If DMA Interrupt is 1 clear DMA Interrupt and set System Address Register.
9. If DMA Interrupt is 0, check all blocks are transferred.
10. Wait for transfer complete.

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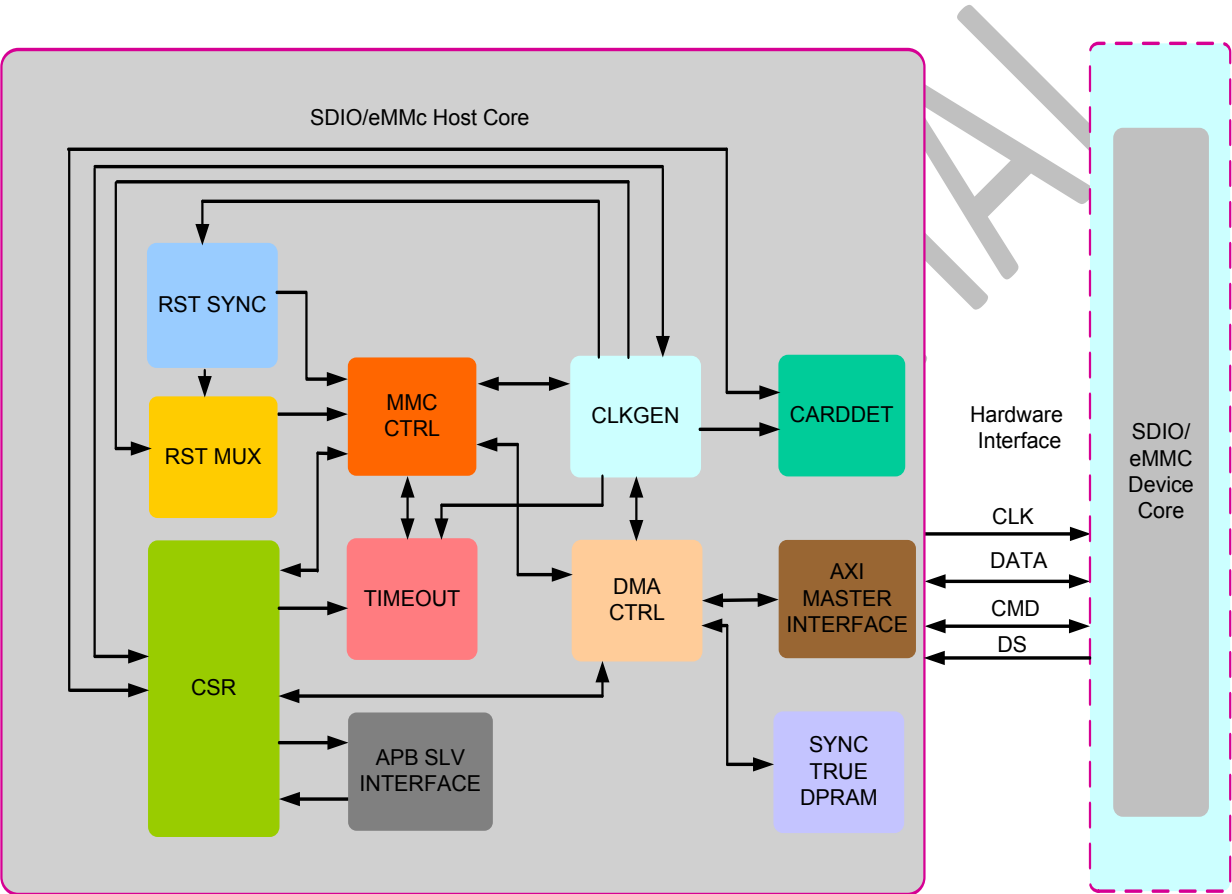
6.3.9 No DMA Mode Operation



7 System Integration

7.1 Block Diagram

Below figure shows a basic system integration block diagram



7.2 Software Interface

This section describes the guidelines for the user to configure the SDIO/eMMC Host core to perform various transactions. SDIO/eMMC Host Core can be initialized by writing appropriate values into the registers in CSR module. The software interface supports interfacing with CSR logic via SOC Bus or interfacing with APB bus interface. The APB Slave signals that resides inside the SDIO/eMMC Host Core are as follows :

- i_apb_slv_addr
- i_apb_slv_wdata
- o_apb_slv_rdata
- i_apb_slv_sel
- i_apb_slv_enable
- i_apb_slv_write
- i_apb_slv_pstrb
- i_apb_slv_pprot
- o_apb_slv_ready
- o_apb_slv_err

7.2.1 CSR register write and read signaling

- **i_apb_slv_addr** -is APB Slave Bus - External interface address input signal. The address value is driven in this port to indicate which register is to be written for write request and which register is to be read for read request.
- **i_apb_slv_wdata** -is APB Slave Bus - External interface write data input signal. The data to be written in CSR registers.
- **o_apb_slv_rdata** -is APB Slave Bus - External interface Read data output signal. The data will be sent out for read command through this port when “i_ahb_slv_write” is low.
- **i_apb_slv_sel** -is APB Slave Bus - External interface transfer type input signal. It selects input signal which ahb slave need to be process.
- **i_apb_slv_enable** -is APB Slave Bus - External interface enable control input.
- **i_apb_slv_write** -is APB Slave Bus - External interface read/write control input signal. The value ‘1’ in this port indicates that requests for write operation.
- **i_apb_slv_pstrb** - APB Slave Bus - External interface strobe input.
- **i_apb_slv_pprot** - APB Slave Bus - External interface protection type input.
- **o_apb_slv_ready** -is APB Slave Bus - External interface Ready output signal. This signal need to high, signal shows transfer done response
- **o_apb_slv_err** -is APB Slave Bus - External interface Slave error output signal.

For a detailed explanation of how to configure each register and to program a command refer Software Registers in [8 Host Software Interface](#)

7.2.2 Interrupt signaling

The “o_irq” is IRQ SOC - interrupt output signal. This is an active high signal.i.e.one set to “o_irq” indicates an interrupt has occurred. The SDIO/eMMC Host IP signals interrupt for the following:

- CMD_COMPLETE_STS
- TRANSFER_COMPLETE_STS
- BLK_GAP_EVENT_STS
- DMA_INTERRUPT_STS
- BUFFER_WRITE_READY_STS
- BUFFER_READ_READY_STS
- CARD_INSERTION_STS
- CARD_REMOVAL_STS
- CARD_INTERRUPT_STS
- INT_A_STS
- INT_B_STS
- INT_C_STS
- RE_TUNNING_EVENT_STS
- BOOT_ACK_RCV
- BOOT_TERMINATE_INTERRUPT
- ERR_INTERRUPT
- CMD_TIMEOUT_ERR_STS
- CMD_CRC_ERR_STS
- CMD_END_BIT_ERR_STS
- CMD_INDEX_ERR_STS
- DATA_TIMEOUT_ERR_STS
- DATA_CRC_ERR_STS
- DATA_END_BIT_ERR_STS
- CURRENT_LIMIT_ERR_STS
- AUTO_CMD_ERR_STS
- ADMA_ERR_STS
- TUNNING_ERR_STS
- RESPONCE_ERR_STS
- TARGET_RESPONCE_ERR_STS
- VENDOR_SPECIFIC_ERR0_STS
- VENDOR_SPECIFIC_ERR1_STS
- VENDOR_SPECIFIC_ERR2_STS

7.3 Hardware Interface

The signals connecting a SDIO/eMMC Host core and SDIO/eMMC Device are:

- **o_emmc_clk**
 - **i_emmc_cmd**
 - **o_emmc_cmd**
 - **i_emmc_data**
 - **o_emmc_data**
-
- **CLK :**
 - **o_emmc_clk** – is Interface for CLK Output signal.
 - **CMD :**
 - **i_emmc_cmd** – is Interface for command input line. The SDIO/eMMC Host samples the command frame in this signal
 - **o_emmc_cmd** – is Interface for command Output line. The SDIO/eMMC Host drives the command frame in this signal.
 - **DATA:**
 - **i_emmc_data** – is Interface for data input line. The SDIO/eMMC Host samples the input data frame in this signal.
 - **o_emmc_data** – is Interface for data Output line. The SDIO/eMMC Host drives the output data frame in this signal.

8 Host Software Interface

8.1 Software Registers

SDIO/eMMC Host needs dedicated registers to support command and data passing. Following table shows details of it.

TABLE 1. Overview of SDIO HOST Register Map

Offset	Register
0x00	BLKCOUNT_SDMA SYSADDR
0x04	BLKCTRL
0x08	ARGUMENT
0x0C	CMD_TRANSFER_MODE
0x10	RESPONSE_0
0x14	RESPONSE_1
0x18	RESPONSE_2
0x1C	RESPONSE_3
0x20	DATA_PORT
0x24	PRESENT_STATE
0x28	HOST_CTRL1
0x2C	CLOCK_TIMEOUT_CONTROL
0x34	INTR_STS_EN
0x30	IRQ_STATUS
0x38	INTR_SIG_EN
0x3C	HOST_CTRL2
0x40	CAPABILITIES_0
0x44	CAPABILITIES_1
0x48	MAX_CURRENT_CAP_0
0x4C	MAX_CURRENT_CAP_1
0x50	FORCE_EVENT_REG
0x54	ADMA_ERR_STATUS
0x58	ADMA_SYS_ADDR0
0x5C	ADMA_SYS_ADDR1
0x60	PRESET_0
0x64	PRESET_1
0x68	PRESET_2
0x6C	PRESET_3
0x0F4	BOOT_TIMEOUT_CNT
0x0F8	VENDOR_SPECIFIC
0x0FC	SLOT_INTR_STS_HOST_CONTROL_VER
0x300	CFG_TUNINGCOUNT
0x310	DRIVING_SAMPLING_EDGE
0x314	RD_BLK_EN

Offset	Register
0x318	BEAT_SELECT
0x31C	SOC_MST_TIMEOUT
0x1000	SOC_TIMEOUT

8.1.1 BLKCOUNT SDMASYSADDR register (Offset Address - 0x00)

The SDMA System address register is described below:

TABLE 2. BLKCOUNT SDMASYSADDR register

Fields	Width	Field bits	Software R/W	Reset Value	Description
BLKCOUNT_S DMASYSADDR	32	31:0	RW	0x00	When Host Version 4 Enable is set to 0 in the Host Control 2 register, DMA uses this register as system address in only 32-bit addressing mode. Auto CMD23 cannot be used with SDMA. When Host Version 4 Enable is set to 1, SDMA uses ADMA System Address register (05Fh-058h) instead of using this register to support both 32-bit and 64-bit addressing. This register is re-assigned to 32-bit Block Count and then SDMA may use Auto CMD23. (1) SDMA System Address (Host Version 4 Enable = 0) This register contains the system memory address for a SDMA Transfer in 32-bit addressing mode. When the Host Controller stops a SDMA transfer, this register shall point to the system address of the next contiguous data position. It can be accessed only if no transaction is executing (i.e., after a transaction has stopped). Reading this register during SDMA transfers may return an invalid value. The Host Driver shall initialize this register before starting a SDMA transaction. After SDMA has

Fields	Width	Field bits	Software R/W	Reset Value	Description
					stopped, the next system address of the next contiguous data position can be read from this register. The SDMA transfer waits at the every boundary specified by the SDMA Buffer Boundary in the Block Size register. The Host Controller generates DMA Interrupt to request the Host Driver to update this register. The Host Driver sets the next system address of the next data position to this register. When the most upper byte of this register (003h) is written, the Host Controller restarts the SDMA transfer. When restarting SDMA by setting Continue Request in the Block Gap Control register, the Host Controller shall start at the next contiguous address stored here in the SDMA System Address register. ADMA does not use this register. (2) 32-bit Block Count (Host Version 4 Enable = 1) Host Controller Version 4.10 re-defines this register as 32-bit BlockCount. In version 4.00, this register may be used as 32-bit blockcount only for Auto CMD23 to set the argument of the CMD23 while executing Auto CMD23. The Host Controller would decrement the block count of this register every block transfer and data transfer stops when the count reaches zero. This register should be accessed only when no transaction is executing. Reading this register

Fields	Width	Field bits	Software R/W	Reset Value	Description
					during data transfers may return invalid value.

8.1.2 BLKCTRL register (Offset Address - 0x04)

The Block control registers is described below:

TABLE 3. BLKCTRL register

Fields	Width	Field bits	Software R/W	Reset Value	Description
BLOCK_SIZE	12	11:0	RW	0x0	This register specifies the block size for block data transfers for CMD17, CMD18, CMD24, CMD25, and CMD53. It can be accessed only if no transaction is executing (i.e after a transaction has stopped). Read operations during transfer return an invalid value and write operations shall be ignored. 0000h - No Data Transfer 0001h - 1 Byte 0002h - 2 Bytes 0003h - 3 Bytes 0004h - 4 Bytes --- --- 01FFh - 511 Bytes 0200h - 512 Bytes --- --- 0800h - 2048 Bytes
BUF_BOUNDARY	3	14:12	RW	0x0	To perform long DMA transfer, System Address register shall be

Fields	Width	Field bits	Software R/W	Reset Value	Description
					updated at every system boundary during DMA transfer. These bits specify the size of contiguous buffer in the system memory. The DMA transfer shall wait at the every boundary specified by these fields and the HC generates the DMA Interrupt to request the HD to update the System Address register. These bits shall support when the DMA Support in the Capabilities register is set to 1 and this function is active when the DMA Enable in the Transfer Mode register is set to 1.
RESERVED	1	15	Reserved	0x0	Used for future purpose.
BLOCK_COUNT	16	31:16	RW	0x0	Use of 16-bit/32-bit Block Count register is enabled when Block Count Enable in the Transfer Mode register is set to 1 and is valid only for multiple block transfers. The Host Driver shall set this register to a value between 1 and the maximum block count. The Host Controller decrements the block count after each block transfer and stops when the count reaches zero. Setting the block count to 0 results in no data blocks is transferred. This register should be accessed only when no transaction is executing (i.e., after transactions are stopped). During data transfer, read operations on this register may return an invalid value and write operations are ignored.

8.1.3 ARGUMENT register (Offset Address - 0x08)

The Argument registers is described below:

TABLE 4. ARGUMENT register

Fields	Width	Field bits	Software R/W	Reset Value	Description
ARGUMENT	32	31:0	RW	0x00	The SD Command Argument is specified as bit39-8 of Command-Format.

8.1.4 CMD TRANSFER MODE register (Offset Address - 0x0C)

The Command and transfer mode registers is described below:

TABLE 5. CMD TRANSFER MODE register

Fields	Width	Field bits	Software R/W	Reset Value	Description
DMA_EN	1	0	RW	0x0	DMA can be enabled only if DMA Support bit in the Capabilities register is set. If this bit is set to 1, a DMA operation shall begin when the HD writes to the upper byte of Command register (00Fh). 0 - Disable 1 - Enable
BLOCK_COUNT_EN	1	1	RW	0x0	This bit is used to enable the Block count register, which is only relevant for multiple block transfers. When this bit is 0, the Block Count register is disabled, which is useful in executing an infinite transfer. Host Driver should set this bit to 0 when ADMA is used. 0 - Disable 1 - Enable
AUTO_CMD_E	2	3:2	RW	0x0	This field determines use

Fields	Width	Field bits	Software R/W	Reset Value	Description
N					of auto command functions 00b - Auto Command Disabled 01b - Auto CMD12 Enable 10b - Auto CMD23 Enable 11b - Reserved There are three methods to stop Multiple-block read/write operation by CMD23 or CMD12. In the other operations (ex. single read/write operation), this field is set to 00b. (1) Auto CMD12 Enable Multiple-block read and write commands for memory require CMD12 to stop the operation. When this field is set to 01b, the Host Controller issues CMD12 automatically when last block transfer is completed. Auto CMD12 error is indicated to the Auto CMD Error Status register. The Host Driver shall not set this bit if the command does not require CMD12. When Host Version 4 Enable =0, CMD12 is issued when 16-bit Block Count is expired. When Host Version 4 Enable =1, CMD12 is issued when 16-bit Block Count or 32-bit Block Count is expired.

Fields	Width	Field bits	Software R/W	Reset Value	Description
					(2) Auto CMD23 Enable When this bit field is set to 10b, the Host Controller issues a CMD23 automatically before issuing a command specified in the Command register. The Host Controller Version 3.00 and later shall support this function. The following conditions are required to use the Auto CMD23. The following conditions are required to use the Auto CMD23. Auto CMD23 Supported (Host Controller Version is 3.00 or later) A memory card that supports CMD23 (SCR[33]=1) If DMA is used, it shall be ADMA. Only when CMD18 or CMD25 is issued Auto CMD23 can be used with or without ADMA. By writing the Command register, the Host Controller issues a CMD23 first and then issues a command specified by the Command Index in Command register. If response errors of CMD23 are detected, the second command is not issued. A CMD23 error is indicated in the Auto CMD Error Status register. 32-bit block

Fields	Width	Field bits	Software R/W	Reset Value	Description
					count value for CMD23 is set to 32-bit Block Count (SDMA System Address) register. (3) Auto CMD Auto Select (Version 4.10) As CMD23 is optional for SD memory card except UHS 104 card, if card supports CMD23, Auto CMD 23 should be used instead of Auto CMD12. Host Controller Version 4.10 defines this "Auto CMD Auto Select" mode. Selection of Auto CMD depends on setting of CMD23 Enable in the Host Control 2 register which indicates whether card supports CMD23. If CMD23 Enable =1, Auto CMD23 is used and if CMD23 Enable =0, Auto CMD12 is used. In case of Version 4.10 or later, use of Auto CMD Auto Select is recommended rather than use of Auto CMD12 Enable or Auto CMD23 Enable.
DATA_XFR_DIR_SELECT	1	4	RW	0x0	This bit defines the direction of data transfers. 0 - Write (Host to Card) 1 - Read (Card to Host)
BLOCK_SELECT	1	5	RW	0x0	This bit enables multiple block data transfers. 0 - Single Block

Fields	Width	Field bits	Software R/W	Reset Value	Description
					1 - Multiple Block
RSP_TYPE_R1_R5	1	6	RW	0x0	When response error check is enabled, this bit selects either R1 or R5 response types. Two types of response checks are supported: R1 for memory and R5 for SDIO. Error Statuses Checked in R1 Bit31 OUT_OF_RANGE Bit30 ADDRESS_ERROR Bit29 BLOCK_LEN_ERROR Bit26 WP_VIOLATION Bit25 CARD_IS_LOCKED Bit23 COM_CRC_ERROR Bit21 CARD_ECC_FAILED Bit20 CC_ERROR Bit19 ERROR Response Flags Checked in R5 Bit07 COM_CRC_ERROR Bit03 ERROR Bit01 FUNCTION_NUMBER Bit00 OUT_OF_RANGE 0 - R1 (Memory) 1 - R5 (SDIO)
RSP_ERR_CH	1	7	RW	0x0	Host Controller Version

Fields	Width	Field bits	Software R/W	Reset Value	Description
ECK_EN					4.00 supports response error check function to avoid overhead of response error check by Host Driver. Only R1 or R5 can be checked. If Host Driver checks response error, this bit is set to 0 and Response Interrupt Disable is set to 0. If Host Controller checks response error, sets this bit to 1 and sets Response Interrupt Disable to 1. Response Type R1 / R5 selects either R1 or R5 response type. If an error is detected, Response Error Interrupt is generated in the Response Error Interrupt Status register. 0 - Response Error Check is disabled 1 - Response Error Check is enabled
RSP_INTERRUPT_PT_DIS	1	8	RW	0x0	Host Controller Version 4.00 supports response error check function to avoid overhead of response error check by Host Driver. Only R1 or R5 can be checked. If Host Driver checks response error, sets this bit to 0 and waits Command Complete Interrupt and then checks the response register. If Host Controller checks response error, sets this bit to 1 and sets Response Error Check Enable to 1. Command Complete Interrupt is disabled by this bit regardless of Command Complete Signal

Fields	Width	Field bits	Software R/W	Reset Value	Description
					Enable. 0 - Response Interrupt is enabled 1 - Response Interrupt is disabled
RESERVED_F OR_XFR_MOD E	7	15:9	RW	Reser ved	Reserved for Future
RSP_TYPE_SE LECT	2	17:16	RW	0x0	Response Type Select 00 - No Response 01 - Response length 136 10 - Response length 48 11 - Response length 48 check Busy after response
SUB_CMD_FLA G	1	18	RW	0x0	This bit is added from Version 4.10 to distinguish a main command or sub command. When issuing a main command, this bit is set to 0 and when issuing a sub command, this bit is set to 1. Setting of this bit is checked by Sub Command Status in the Present State register. Host Driver manages whether main or sub command. Host Controller does not refer to this bit to issue a command. 1 - Sub Command 0 - Main Command
CMD_CRC_CH	1	19	RW	0x0	If this bit is set to 1, the HC

Fields	Width	Field bits	Software R/W	Reset Value	Description
ECK_EN					shall check the CRC field in the response. If an error is detected, it is reported as a Command CRC Error. If this bit is set to 0, the CRC field is not checked. 0 - Disable 1 - Enable
CMD_INDEX_CHECK_EN	1	20	RW	0x0	If this bit is set to 1, the HC shall check the index field in the response to see if it has the same value as the command index. If it is not, it is reported as a Command Index Error. If this bit is set to 0, the Index field is not checked. 0 - Disable 1 - Enable
DATA_PRESENT_SELECT	1	21	RW	0x0	This bit is set to 1 to indicate that data is present and shall be transferred using the DAT line. If is set to 0 for the following: 1. Commands using only CMD line (ex. CMD52) 2. Commands with no data transfer but using busy signal on DAT[0] line (R1b or R5b ex. CMD38) 3. Resume Command 0 - No Data Present 1 - Data Present

Fields	Width	Field bits	Software R/W	Reset Value	Description
CMD_TYPE	2	23:22	RW	0x0	There are three types of special commands. Suspend, Resume and Abort. These bits shall be set to 00b for all other commands. Suspend Command: If the Suspend command succeeds, the HC shall assume the SD Bus has been released and that it is possible to issue the next command which uses the DAT line. The HC shall de-assert Read Wait for read transactions and stop checking busy for write transactions. The Interrupt cycle shall start, in 4-bit mode. If the Suspend command fails, the HC shall maintain its current state. and the HD shall restart the transfer by setting Continue Request in the Block Gap Control Register. Resume Command: The HD restarts the data transfer by restoring the registers in the range of 000-00Dh. The HC shall check for busy before starting write transfers. Abort Command: If this command is set when executing a read transfer, the HC shall stop reads to the buffer. If this command is set when executing a write transfer, the HC shall stop driving the DAT line. After issuing the Abort command, the HD

Fields	Width	Field bits	Software R/W	Reset Value	Description
					should issue a software reset 00b - Normal 01b - Suspend 10b - Resume 11b - Abort
CMD_INDEX	6	29:24	RW	0x0	This bit shall be set to the command number (CMD0-63, ACMD0- 63).
RESERVED_CMD	2	31:30	RW	0x0	Reserved for Future

8.1.5 RESPONSE 0 register (Offset Address - 0x10)

The Response registers [15:0] is described below:

TABLE 6. RESPONSE 0 register

Fields	Width	Field bits	Software R/W	Reset Value	Description
RESPONSE_0	32	31:0	R	0x00	It refers to a bit range within the response data as transmitted on the SD Bus.

8.1.6 RESPONSE 1 register (Offset Address - 0x14)

The Response registers [31:16] is described below:

TABLE 7. RESPONSE 1 register

Fields	Width	Field bits	Software R/W	Reset Value	Description
RESPONSE_1	32	31:0	R	0x00	It refers to a bit range within the response data as transmitted on the SD Bus.

8.1.7 RESPONSE 2 register (Offset Address - 0x18)

The Response registers [47:32] is described below:

TABLE 8. RESPONSE 2 register

Fields	Width	Field bits	Software R/W	Reset Value	Description
RESPONSE_2	32	31:0	R	0x00	It refers to a bit range within the response data as transmitted on the SD Bus.

8.1.8 RESPONSE 3 register (Offset Address - 0x1C)

The Response registers [63:48] is described below:

TABLE 9. RESPONSE 3 register

Fields	Width	Field bits	Software R/W	Reset Value	Description
RESPONSE_3	32	31:0	R	0x00	It refers to a bit range within the response data as transmitted on the SD Bus.

8.1.9 DATA PORT register (Offset Address - 0x20)

The Buffer Data Port Register is described below:

TABLE 10. DATA PORT register

Fields	Width	Field bits	Software R/W	Reset Value	Description
DATA_PORT	32	31:0	RW	0x00	Used to set Buffer Data Port Register The Host Controller Buffer can be accessed through this 32-bit Data Port Register used in Non DMA mode(PIO).

8.1.10 PRESENT STATE register (Offset Address - 0x24)

The Present State Register is described below:

TABLE 11. PRESENT STATE register

Fields	Width	Field bits	Software R/W	Reset Value	Description
COMMAND_IN HIBIT_CMD	1	0	R	0x0	If this bit is 0, it indicates the CMD line is not in use and the HC can issue a SD command

Fields	Width	Field bits	Software R/W	Reset Value	Description
					using the CMD line. This bit is set immediately after the Command register (00Fh) is written. This bit is cleared when the command response is received. Even if the Command Inhibit (DAT) is set to 1, Commands using only the CMD line can be issued if this bit is 0. Changing from 1 to 0 generates a Command complete interrupt in the Normal Interrupt Status register. If the HC cannot issue the command because of a command conflict error or because of Command Not Issued By Auto CMD12 Error, this bit shall remain 1 and the Command Complete is not set. Status issuing Auto CMD12 is not read from this bit. Auto CMD12 and Auto CMD23 consist of two responses. In this case, this bit is not cleared by the response of CMD12 or CMD23 but cleared by the response of a read/write command. Status issuing Auto CMD12 is not read from this bit. So if a command is issued during Auto CMD12 operation, Host Controller shall manage to issue two commands: CMD12 and a command set by Command register.
COMMAND_INHIBIT_DAT	1	1	R	0x0	This status bit is generated if either the DAT Line Active or the Read transfer Active is set to 1. If this bit is 0, it indicates the HC can issue the next SD command. Commands with busy signal belong to Command Inhibit (DAT) (ex. R1b, R5b type). Changing from 1 to 0 generates a Transfer Complete interrupt in the Normal interrupt status register.

Fields	Width	Field bits	Software R/W	Reset Value	Description
					Note: The SD Host Driver can save registers in the range of 000-00Dh for a suspend transaction after this bit has changed from 1 to 0. 1 - cannot issue command which uses the DAT line 0 - Can issue command which uses the DAT line
DAT_LINE_ACTIVE	1	2	R	0x0	This bit indicates whether one of the DAT line on SD bus is in use. 1 - DAT line active 0 - DAT line inactive
RE_TUNNING_REQUEST	1	3	R	0x0	The Host Controller may request Host Driver to execute retuning sequence by setting this bit when the data window is shifted by temperature drift and a tuned sampling point does not have a good margin to receive correct data. This bit is cleared when a command is issued with setting Execute Tuning in the Host Control 2 register. Changing of this bit from 0 to 1 generates ReTuning Event. Refer to Normal Interrupt registers for more detail. This bit isn't set to 1 if Sampling Clock Select in the Host Control 2 register is set to 0 (using fixed sampling clock). 1 Sampling clock needs retuning 0 Fixed or well tuned sampling clock
DAT_07_04_LINE_SIGNAL_LEVEL	4	7:4	R	0x0	This status is used to check the DAT line level to recover from errors, and for debugging. D07 DAT[7] D06 DAT[6] D05 DAT[5] D04 DAT[4]
WRITE_TRANSFER_ACTIVE	1	8	R	0x0	This status indicates a write transfer is active. If this bit is 0, it

Fields	Width	Field bits	Software R/W	Reset Value	Description
					means no valid write data exists in the HC. This bit is set in either of the following cases: After the end bit of the write command. When writing a 1 to Continue Request in the Block Gap Control register to restart a write transfer. This bit is cleared in either of the following cases: After getting the CRC status of the last data block as specified by the transfer count (Single or Multiple) After getting a CRC status of any block where data transmission is about to be stopped by a Stop At Block Gap Request. During a write transaction, a Block Gap Event interrupt is generated when this bit is changed to 0, as a result of the Stop At Block Gap Request being set. This status is useful for the HD in determining when to issue commands during write busy. 1 - transferring data 0 - No valid data
READ_TRANSFER_ACTIVE	1	9	R	0x0	This status is used for detecting completion of a read transfer. This bit is set to 1 for either of the following conditions: After the end bit of the read command When writing a 1 to continue Request in the Block Gap Control register to restart a read transfer This bit is cleared to 0 for either of the following conditions: When the last data block as specified by block length is

Fields	Width	Field bits	Software R/W	Reset Value	Description
					transferred to the system. When all valid data blocks have been transferred to the system and no current block transfers are being sent as a result of the Stop At Block Gap Request set to 1. A transfer complete interrupt is generated when this bit changes to 0. 1 - Transferring data 0 - No valid data
BUFFER_WRITE_EN	1	10	R	0x0	This status is used for non-DMA write transfers. This read only flag indicates if space is available for write data. If this bit is 1, data can be written to the buffer. A change of this bit from 1 to 0 occurs when all the block data is written to the buffer. A change of this bit from 0 to 1 occurs when top of block data can be written to the buffer and generates the Buffer Write Ready Interrupt. 0 - Write Disable 1 - Write Enable.
BUFFER_READ_EN	1	11	R	0x0	This status is used for non-DMA read transfers. This read only flag indicates that valid data exists in the host side buffer status. If this bit is 1, readable data exists in the buffer. A change of this bit from 1 to 0 occurs when all the block data is read from the buffer. A change of this bit from 0 to 1 occurs when all the block data is ready in the buffer and generates the Buffer Read Ready Interrupt. 0 - Read Disable 1 - Read Enable.
RESERVED_P RESET_STAT E_15_12	4	15:12	Reserved	0x0	Reserved for Future use
CARD_INSERT	1	16	R	0x0	This bit indicates whether a

Fields	Width	Field bits	Software R/W	Reset Value	Description
ED					card has been inserted. Changing from 0 to 1 generates a Card Insertion interrupt in the Normal Interrupt Status register and changing from 1 to 0 generates a Card Removal Interrupt in the Normal Interrupt Status register. The Software Reset For All in the Software Reset register shall not affect this bit. If a Card is removed while its power is on and its clock is oscillating, the HC shall clear SD Bus Power in the Power Control register and SD Clock Enable in the Clock control register. In addition the HD should clear the HC by the Software Reset For All in Software register. The card detect is active regardless of the SD Bus Power. 0 - Reset or Debouncing or No Card 1 - Card Inserted
CARD_STATE_STABLE	1	17	R	0x0	This bit is used for testing. If it is 0, the Card Detect Pin Level is not stable. If this bit is set to 1, it means the Card Detect Pin Level is stable. The Software Reset For All in the Software Reset Register shall not affect this bit. 0 - Reset of Debouncing 1 - No Card or Inserted
CARD_DETECT_PIN_LEVEL	1	18	R	0x0	This bit reflects the inverse value of the i_sdif_cd_n pin. 0 - No Card present (i_sdif_cd_n = 1) 1 - Card present (i_sdif_cd_n = 0)
WRITE_PROTECT_SWITCH_PIN_LEVEL	1	19	R	0x0	The Write Protect Switch is supported for memory and combo cards. This bit reflects the i_sdif_wp pin. 0 - Write protected (i_sdif_wp =

Fields	Width	Field bits	Software R/W	Reset Value	Description
					1) 1 - Write enabled (i_sdif_wp = 0)
DAT_03_00_LINE_SIGNAL_LEVEL	4	23:20	R	0x0	DAT[3:0] Line Signal Level (SD Mode only) This status is used to check DAT line level to recover from errors, and for debugging. This is especially useful in detecting the busy signal level from DAT[0]. D23 - DAT[3] D22 - DAT[2] D21 - DAT[1] D20 - DAT[0]
CMD_LINE_SIGNAL_LEVEL	1	24	R	0x0	This status is used to check CMD line level to recover from errors, and for debugging.
HOST_REGULATOR_VOLTAGE_STABLE	1	25	R	0x0	This status is added from Version 4.10 and is used to check whether host regulator voltage is stable for switching signal voltage of UHS-I mode. Support of this function is checked by reading this status after that Software Reset For All in the Software Reset register is cleared by the Host Controller in initialization. Setting this status to 1 means that this function is supported by the Host Controller. This status may be related to 1.8V Signaling Enable in the Host Control 2 register. Changing 1.8V Signaling Enable causes unstable of host regulator voltage for I/O cell. Then once this status is set to 0 and retrieved to 1 when host regulator voltage is stable again. When executing power cycle, Host Driver also executes Software Reset For All and it clears 1.8V Signaling Enable to go back signal voltage to 3.3V. If this status is not supported, Host Driver should

Fields	Width	Field bits	Software R/W	Reset Value	Description
					take more than 5ms for stable time of host voltage regulator from changing 1.8V Signaling Enable. Specific Host Driver may use a specific time, which is provided by Host System, instead of using 5ms. 1 - Host Regulator Voltage is stable 0 - Host Regulator Voltage is not stable
RESERVED_P RESENT_STAT E	1	26	Reserved	0x0	Reserved for future use
CMD_NOT_ISS UED_BY_ERR	1	27	R	0x0	Setting of this status indicates that a command cannot be issued due to an error except Auto CMD12 error. (Equivalent error status by Auto CMD12 error is defined as Command Not Issued By Auto CMD12 Error in the Auto CMD Error Status register.) This status is set to 1 when Host Controller cannot issue a command after setting Command register. Sub Command Status (D28) indicates which command is not issued (main or sub). 1 - Command cannot be issued 0 - No error for issuing a command
SUB_COMMAND D_STS	1	28	R	0x0	The Command register and Response register are commonly used for main command and sub command. This status is used to distinguish which response error statuses, main command or sub command, indicated in the Error Interrupt Status register. Just before reading of this register, the Sub Command Flag of the Command register is copied to this status. This status is effective

Fields	Width	Field bits	Software R/W	Reset Value	Description
					not only when Response Error interrupt is generated but also when data error interrupt is generated with Command Not Issued by Error (D27 of this register) or Auto CMD Error interrupt is generated with Command Not Issued by Error by Auto CMD12 in the Auto CMD Error Status register. 1 - Sub Command Status 0 - Main Command Status
PRESENT_STATUS_RFU	3	31:29	Reserved	0x0	Reserved for future use

8.1.11 HOST CTRL1 register (Offset Address - 0x28)

The Control registers is described below:

TABLE 12. HOST CTRL1 register

Fields	Width	Field bits	Software R/W	Reset Value	Description
LED_CONTROL	1	0	RW	0x0	This bit is used to caution the user not to remove the card while the SD card is being accessed. If the software is going to issue multiple SD commands, this bit can be set during all transactions. It is not necessary to change for each transaction. 1 - LED on 0 - LED off
DATA_TRANSFER_WIDTH	1	1	RW	0x0	This bit selects the data width of the HC. The HD shall select it to match the data width of the SD card. 1 - 4 bit mode 0 - 1 bit mode
HIGH_SPEED_EN	1	2	RW	0x0	This bit is optional. Before setting this bit, the HD shall check the High Speed Support in the capabilities register. If this bit

Fields	Width	Field bits	Software R/W	Reset Value	Description
					is set to 0 (default), the HC outputs CMD line and DAT lines at the falling edge of the SD clock (up to 25 MHz/20MHz for MMC). If this bit is set to 1, the HC outputs CMD line and DAT lines at the rising edge of the SD clock (up to 50 MHz for SD/52MHz for MMC)/ 208Mhz (for SD3.0) If Preset Value Enable in the Host Control 2 register is set to 1, Host Driver needs to reset SD Clock Enable before changing this field to avoid generating clock glitches. After setting this field, the Host Driver sets SD Clock Enable again. 1 - High Speed Mode 0 - Normal Speed Mode
DMA_SELECT	2	4:3	RW	0x0	This field is used to select DMA type. The Host Driver shall check support of DMA modes by referring the Capabilities register. Selected DMA is enabled by DMA Enable of the Transfer Mode register in SD mode. (1) Up to Version 3.00 When Host Version 4 Enable is set to 0, setting of this field is compatible to Host Controller Version 3.00. SDMA is initiated by writing to the Command register when this field is set to 00b and the SDMA System Address register (32-bit) is used. SDMA does not support 64-bit addressing. ADMA2 is initiated by writing to the Command register when this field is set to 10b or 11b. Lower 32-bit of the ADMA System Address register is used when this field is set to 10b and 64-bit of the ADMA System Address register is used when this field is set to

Fields	Width	Field bits	Software R/W	Reset Value	Description
					11b. Support of 64-bit System Addressing is indicated by 64-bit System Address Support for V3 in the Capabilities register. 64-bit AMDA2 uses 96-bit Descriptor. 00 - SDMA is selected 01 - 32-bit Address ADMA1 is selected 10 - 32-bit Address ADMA2 is selected 11 - 64-bit Address ADMA2 is selected (Optional) (2) Version 4.00 or later When Host Version 4 Enable is set to 1, setting of this field is changed as follows. SDMA is initiated by Host Driver writes to the Command register when this field is set to 00b. ADMA2 is initiated by Host Driver writes to the Command register when this field is set to 10b or 11b and by ADMA3 sets to the ADMA System Address register when this field is set to 11b. ADMA3 is initiated by Host Driver writes to the ADMA3 ID Address register when this field is set to 11b. 00 - SDMA is selected 01 - Not Used (New assignment is not allowed) 10 - ADMA2 is selected (AMDA3 is not supported or disabled) 11 - ADMA2 or ADMA3 is selected Support of 64-bit DMA and 128-bit Descriptor is indicated by 64-bit System Address Support for V4 in the Capabilities register. If the support bit is set to 1, all supported DMAs (depends on Support, ADMA2 Support and ADMA3 Support) shall support 64-bit addressing. 64-bit

Fields	Width	Field bits	Software R/W	Reset Value	Description
					Addressing in the Host Controller 2 register selects either 32-bit or 64-bit system addressing of DMAs.
EXTENDED_DATA_TRANSFER_WIDTH	1	5	RW	0x0	This bit controls 8-bit bus width mode for embedded device. Support of this function is indicated in 8-bit Support for Embedded Device in the Capabilities register. If a device supports 8-bit bus mode, this bit may be set to 1. If this bit is 0, bus width is controlled by Data Transfer Width in the Host Control 1 register. This bit is not effective when multiple devices are installed on a bus slot (Slot Type is set to 10b in the Capabilities register). In this case, each device bus width is controlled by Bus Width Preset field in the embedded control register. 1 - 8-bit Bus Width 0 - Bus Width is Selected by Data Transfer Width
CARD_DETECT_TEST_LEVEL	1	6	RW	0x0	This bit is enabled while the Card Detect Signal Selection is set to 1 and it indicates card inserted or not. Generates (card ins or card removal) interrupt when the normal int sts enable bit is set. 1 - Card Inserted 0 - No Card
CARD_DETECT_SIGNAL_DETECTION	1	7	RW	0x0	This bit selects source for card detection. 1 - The card detect test level is selected 0 - i_sdif_cd_n is selected (for normal use)
SD_BUS_POWER_FOR_VDD1	1	8	RW	0x0	Before setting this bit, the SD host driver shall set SD Bus Voltage Select. If the HC detects

Fields	Width	Field bits	Software R/W	Reset Value	Description
					the No Card State, this bit shall be cleared. If this bit is cleared, the Host Controller should immediately stop driving CMD and DAT[3:0] (tri-state), and drive SDCLK to low level. If card is connected to Host Controller, Host Controller shall set these lines to low before stopping to supply VDD1. 1 - Power on 0 - Power off
SD_BUS_VOLTAGE_SELECT_FOR_VDD1	3	11:9	RW	0x0	By setting these bits, the HD selects the voltage level for the SD card. Before setting this register, the HD shall check the voltage support bits in the capabilities register. If an unsupported voltage is selected, the Host System shall not supply SD bus voltage 111b - 3.3 Flattop.) 110b - 3.0 V (Typ.) 101b - 1.8 V (Typ.) for Embedded 100b - 000b - Reserved
SD_BUS_POWER_FOR_VDD2	1	12	RW	0x0	Setting this bit enables providing VDD2. It is used only for UHS-II Reserved for Future 1 - Power on 0 - Power off
SD_BUS_VOLTAGE_SELECT_FOR_VDD2	3	15:13	RW	0x0	This field determines supply voltage range to VDD2. This field can be set to 101b if 1.8V VDD2 Support in the Capabilities register is set to 1. It is used only for UHS-II Reserved for Future 111b - Not used 110b - Not used 101b - 1.8V 100b - Reserved for 1.2V 011b – 001b - Reserved 000b - VDD2 Not Supported

Fields	Width	Field bits	Software R/W	Reset Value	Description
STOP_AT_BLK_GAP_REQUEST	1	16	RW	0x0	This bit is used to stop executing a transaction at the next block gap for non-DMA, SDMA and ADMA transfers. Until the transfer complete is set to 1, indicating a transfer completion the HD shall leave this bit set to 1. Clearing both the Stop At Block Gap Request and Continue Request shall not cause the transaction to restart. Read Wait is used to stop the read transaction at the block gap. The HC shall honour Stop At Block Gap Request for write transfers, but for read transfers it requires that the SD card support Read Wait. Therefore the HD shall not set this bit during read transfers unless the SD card supports Read Wait and has set Read Wait Control to 1. In case of write transfers in which the HD writes data to the Buffer Data Port register, the HD shall set this bit after all block data is written. If this bit is set to 1, the HD shall not write data to Buffer data port register. This bit affects Read Transfer Active, Write Transfer Active, DAT line active and Command Inhibit (DAT) in the Present State register. 1 - Stop 0 - Transfer
CONTINUE_REQUEST	1	17	RW	0x0	This bit is used to restart a transaction which was stopped using the Stop At Block Gap Request. To cancel stop at the block gap, set Stop At block Gap Request to 0 and set this bit to restart the transfer. The Host Controller automatically clears this bit when the transaction

Fields	Width	Field bits	Software R/W	Reset Value	Description
					restarts. If Stop At Block Gap Request is set to 1, any write to this bit is ignored. In SD mode, this bit is cleared in either of the following cases: 1) In the case of a read transaction, the DAT Line Active changes from 0 to 1 as a read transaction restarts. 2) In the case of a write transaction, the Write transfer active changes from 0 to 1 as the write transaction restarts. Therefore it is not necessary for Host driver to set this bit to 0. If Stop At Block Gap Request is set to 1, any write to this bit is ignored. 1 - Restart 0 - Ignored
READ_WAIT_CONTROL	1	18	RW	0x0	The read wait function is optional for SDIO cards. If the card supports read wait, set this bit to enable use of the read wait protocol to stop read data using DAT[2] line. Otherwise the HC has to stop the SD clock to hold read data, which restricts commands generation. When the HD detects an SD card insertion, it shall set this bit according to the CCCR of the SDIO card. If the card does not support read wait, this bit shall never be set to 1 otherwise DAT line conflict may occur. If this bit is set to 0, Suspend / Resume cannot be supported 1 - Enable Read Wait Control 0 - Disable Read Wait Control
INTERRUPT_A T_BLK_GAP	1	19	RW	0x0	This bit is valid only in 4-bit mode of the SDIO card and selects a sample point in the interrupt cycle. Setting to 1 enables

Fields	Width	Field bits	Software R/W	Reset Value	Description
					interrupt detection at the block gap for a multiple block transfer. If the SD card cannot signal an interrupt during a multiple block transfer, this bit should be set to 0. When the HD detects an SD card insertion, it shall set this bit according to the CCCR of the SDIO card.
SPIMODE_BLK_GAP_CTRL	1	20	RW	0x0	To Initiate SPI Mode operation, this register needs to set as HIGH
BOOT_EN	1	21	RW	0x0	To start boot code access 1 - To start boot code access 0 - To stop boot code access
ALT_BOOT_EN	1	22	RW	0x0	To start boot code access in alternative mode. 1 - To start alternate boot mode access 0 - To stop alternate boot mode access
BOOT_ACK_CHECK	1	23	RW	0x0	To check for the boot acknowledge in boot operation. 1 - wait for boot ack from eMMC card 0 - Will not wait for boot ack from eMMC card
WAKEUP_EV_EN_ON_CARD_INTERRUPT	1	24	RW	0x0	Wakeup event enable on card interrupt
WAKEUP_EV_EN_ON_CARD_INSERTION	1	25	RW	0x0	Wakeup event enable on card insertion
WAKEUP_EV_EN_ON_CARD_REMOVAL	1	26	RW	0x0	Wakeup event enable on card removal
RESERVED_WAKEUP_CTRL	5	31:27	Reserved	0x0	Reserved for Future use

8.1.12 CLOCK TIMEOUT CONTROL register (Offset Address - 0x2C)

The Control registers is described below:

TABLE 13. CLOCK TIMEOUT CONTROL register

Fields	Width	Field bits	Software R/W	Reset Value	Description
INTERNAL_CLK_EN	1	0	RW	0x0	This bit is set to 0 when the HD is not using the HC or the HC awaits a wakeup event. The HC should stop its internal clock to go very low power state. Still, registers shall be able to be read and written. Clock starts to oscillate when this bit is set to 1. When clock oscillation is stable, the HC shall set Internal Clock Stable in this register to 1. This bit shall not affect card detection. 1 - Oscillate 0 - Stop
INTERNAL_CLK_STABLE	1	1	RW	0x0	This bit is set to 1 when internal clock is stable after writing to Internal Clock Enable in this register to 1. 1 - Ready 0 - Not Ready
SD_CLK_EN	1	2	RW	0x0	The Host Controller shall stop providing SDCLK or RCLK when writing this bit to 0. SDCLK/RCLK Frequency Select can be changed when this bit is 0. Then, the Host Controller shall maintain the same clock frequency until SDCLK is stopped (Stop at SDCLK=0). If the Card Inserted in the Present State register is

Fields	Width	Field bits	Software R/W	Reset Value	Description
					cleared, this bit shall be cleared. 1 - Enable providing SDCLK or RCLK 0 - Disable providing SDCLK or RCLK
RFU_PLL_EN	1	3	RW	0x0	PLL enable
RESERVED_CLK_CONTROL_4	1	4	RW	0x0	Reserved for Future use
CLK_GENERATOR_SELECT	1	5	RW	0x0	This bit is used to select the clock generator mode in SDCLK/RCLK Frequency Select. If the Programmable Clock Mode is supported (non-zero value is set to Clock Multiplier in the Capabilities register), this bit attribute is RW, and if not supported, this bit attribute is RO and zero is read. This bit depends on the setting of Preset Value Enable in the Host Control 2 register. If the Preset Value Enable = 0, this bit is set by Host Driver. If the Preset Value Enable = 1, this bit is automatically set to a value specified in one of Preset Value registers. 1 Programmable Clock Mode 0 Divided Clock Mode Reserved for Future use. Currently, Clock is generated using Divided clock mode.

Fields	Width	Field bits	Software R/W	Reset Value	Description
U_SDCLK_FREQ_SELECT	2	7:6	RW	0x0	Host Controller Version 1.00 and 2.00 do not support these bits and they are treated as 00b fixed value (ROC). Host Controller Version 3.00 shall support these bits to expand SDCLK/RCLK Frequency Select to 10-bit. Bit 07-06 is assigned to bit 09-08 of clock divider in SDCLK/RCLK Frequency Select.
L_SDCLK_FREQ_SELECT	8	15:8	RW	0x0	This register is used to select the frequency of the SDCLK pin. The frequency is not programmed directly; rather this register holds the divisor of the Base Clock Frequency For SD clock in the capabilities register. Only the following settings are allowed. 80h - base clock divided by 256 40h - base clock divided by 128 20h - base clock divided by 64 10h - base clock divided by 32 08h - base clock divided by 16 04h - base clock divided by 8 02h - base clock divided by 4 01h - base clock divided by 2 00h - base clock Setting 00h specifies the highest frequency of the SD

Fields	Width	Field bits	Software R/W	Reset Value	Description
					Clock.
DATA_TIME_OUT_CNTR_VALUE	4	19:16	RW	0x0	This value determines the interval by which DAT line time-outs are detected. Refer to the Data Time-out Error in the Error Interrupt Status register for information on factors that dictate time-out generation. Time-out clock frequency will be generated by dividing the sdclockTMCLK by this value.
RESERVED_TIMEOUT_CTRL_7_4	4	23:20	Reserved	0x0	Reserved for Future
SOFTWARE_RESET_FOR_ALL	1	24	RW	0x0	This reset affects the entire HC except for the card detection circuit. Register bits of type ROC, RW, RW1C, RWAC are cleared to 0. During its initialization, the HD shall set this bit to 1 to reset the HC. The HC shall reset this bit to 0 when capabilities registers are valid and the HD can read them. Additional use of Software Reset For All may not affect the value of the Capabilities registers. If this bit is set to 1, the SD card shall reset itself and must be re initialized by the HD. 1 - Reset 0 - Work
SOFTWARE_RESET_FOR_CMD_LINE	1	25	RW	0x0	Software Reset For CMD Line Only part of command circuit is reset to be able to issue a command. This reset is effective

Fields	Width	Field bits	Software R/W	Reset Value	Description
					only command issuing circuit (including response error statuses related to Command Inhibit (CMD) control) and does not affect data transfer circuit. Host Controller can continue data transfer even this reset is executed during handling of sub command response errors. The following registers and bits are cleared by this bit: Present State register Command Inhibit (CMD) Normal Interrupt Status register Command Complete Error Interrupt Status (from Version 4.10) Response error statuses related to Command Inhibit (CMD) 1 - Reset 0 - Work
SOFTWARE_RST_FOR_DATA_LINE	1	26	RW	0x0	Only part of data circuit is reset. The following registers and bits are cleared by this bit: Buffer Data Port Register Buffer is cleared and Initialized. Present State register Buffer read Enable

Fields	Width	Field bits	Software R/W	Reset Value	Description
					Buffer write Enable Read Transfer Active Write Transfer Active DAT Line Active Command Inhibit (DAT) Block Gap Control register Continue Request Stop At Block Gap Request Normal Interrupt Status register Buffer Read Ready Buffer Write Ready Block Gap Event Transfer Complete 1 - Reset 0 - Work
RESERVED_S SOFTWARE_RS T_7_3	5	31:27	Reserved	0x0	Reserved for Future

8.1.13 INTR STS EN register (Offset Address - 0x34)

The Interrupt status enable registers is described below:

TABLE 14. INTR STS EN register

Fields	Width	Field bits	Software R/W	Reset Value	Description
CMD_COMPLE TE_STS	1	0	RW	0x00	This bit is set when we get the end bit of the command response (Except Auto CMD12 and Auto CMD23) Note: Command Time-

Fields	Width	Field bits	Software R/W	Reset Value	Description
					out Error has higher priority than Command Complete. If both are set to 1, it can be considered that the response was not received correctly. Version 4.00 defines response check function for R1 and R5. If Response Interrupt Disable in the Transfer Mode register is set to 1, generation of this interrupt is prohibited regardless of Command Complete Signal Enable.
TRANSFER_C COMPLETE_STS	1	1	RW	0x00	This bit is set when a read / write transaction is completed. Read Transaction: This bit is set at the falling edge of Read Transfer Active Status. There are two cases in which the Interrupt is generated. The first is when a data transfer is completed as specified by data length (After the last data has been read to the Host System). The second is when data has stopped at the block gap and completed the data transfer by setting the Stop At Block Gap Request in the Block Gap Control Register (After valid data has been read to the Host System). Write Transaction: This bit is set at the falling edge of the DAT Line Active Status. There are two cases in which the Interrupt is generated. The first is

Fields	Width	Field bits	Software R/W	Reset Value	Description
					when the last data is written to the card as specified by data length and Busy signal is released. The second is when data transfers are stopped at the block gap by setting Stop At Block Gap Request in the Block Gap Control Register and data transfers completed. (After valid data is written to the SD card and the busy signal is released). Note: Transfer Complete has higher priority than Data Time-out Error. If both bits are set to 1, the data transfer can be considered complete Note: While performing tuning procedure (Execute Tuning is set to 1), Transfer Complete is not set to 1 Command with Busy: This bit is set when busy is de-asserted. Refer to DAT Line Active and Command Inhibit (DAT) in the Present State register. UHS-I mode While performing tuning procedure (Execute Tuning is set to 1), Transfer Complete is not set to 1. 0 - No Data Transfer Complete 1 - Data Transfer Complete
BLK_GAP_EVE NT_STS	1	2	RW	0x00	If the Stop At Block Gap Request in the Block Gap Control

Fields	Width	Field bits	Software R/W	Reset Value	Description
					Register is set, this bit is set. Read Transaction: This bit is set at the falling edge of the DAT Line Active Status (When the transaction is stopped at SD Bus timing. The Read Wait must be supported inorder to use this function). Write Transaction: This bit is set at the falling edge of Write Transfer Active Status (After getting CRC status at SD Bus timing). 0 - No Block Gap Event 1 - Transaction stopped at Block Gap
DMA_INTERRUPT_STS	1	3	RW	0x00	This status is set if the HC detects the Host DMA Buffer Boundary in the Block Size register. 0 - No DMA Interrupt 1 - DMA Interrupt is Generated
BUFFER_WRITE_READY_STS	1	4	RW	0x00	This status is set if the Buffer Write Enable changes from 0 to 1. 0 - Not Ready to Write Buffer. 1 - Ready to Write Buffer.
BUFFER_READ_READY_STS	1	5	RW	0x00	This status is set if the Buffer Read Enable changes from 0 to 1. Buffer Read Ready is set to 1

Fields	Width	Field bits	Software R/W	Reset Value	Description
					for every CMD19 execution in tuning procedure. 0 - Not Ready to read Buffer. 1 - Ready to read Buffer.
CARD_INSERTION_STS	1	6	RW	0x00	This status is set if the Card Inserted in the Present State register changes from 0 to 1. When the HD writes this bit to 1 to clear this status the status of the Card Inserted in the Present State register should be confirmed. Because the card detect may possibly be changed when the HD clear this bit an Interrupt event may not be generated. 0 - Card State Stable or Debouncing 1 - Card Inserted
CARD_REMOVAL_STS	1	7	RW	0x00	This status is set if the Card Inserted in the Present State register changes from 1 to 0. When the HD writes this bit to 1 to clear this status the status of the Card Inserted in the Present State register should be confirmed. Because the card detect may possibly be changed when the HD clear this bit an Interrupt event may not be generated. 0 - Card State Stable or Debouncing

Fields	Width	Field bits	Software R/W	Reset Value	Description
					1 - Card Removed
CARD_INTERRUPT_STS	1	8	RW	0x00	When this status has been set and the Host Driver needs to start this interrupt service, Card Interrupt Status Enable in the Normal Interrupt Status Enable register may be set to 0 in order to clear the card interrupt status latched in the Host Controller and to stop driving the interrupt signal to the Host System. After completion of the card interrupt service (It should reset interrupt factors in the SD card and the interrupt signal may not be asserted), set Card Interrupt Status Enable to 1 and start sampling the interrupt signal again. Writing this bit to 1 does not clear this bit. It is cleared by resetting the SD card interrupt factor. (1) DAT[1] Interrupt Input in SD Mode In 1-bit mode, the Host Controller shall detect the Card Interrupt without SD Clock to support wakeup. In 4-bit mode, the card interrupt signal is sampled during the interrupt cycle, so there are some sample delays between the interrupt signal from the SD card and the interrupt to the Host System. Interrupt detected by DAT[1] is supported when there is a card per slot. In case of UHS-I mode, switching time of Interrupt Period

Fields	Width	Field bits	Software R/W	Reset Value	Description
					is relaxed for 2 clock cycles. Then Host Controller needs to delay start of interrupt sampling at least 2 clocks and should sample interrupt while Interrupt Period is stable. 1 - Generate Card Interrupt 0 - No Card Interrupt
INT_A_STS	1	9	RW	0x00	Init A status enable
INT_B_STS	1	10	RW	0x00	Init B status enable
INT_C_STS	1	11	RW	0x00	Init C status enable
RE_TUNNING_EVENT_STS	1	12	RW	0x00	This status is set if Re-Tuning Request in the Present State register changes from 0 to 1. Host Controller requests Host Driver to perform re-tuning for next data transfer. Current data transfer (not large block count) can be completed without re-tuning. 1 - Re-Tuning should be performed 0 - Re-Tuning is not required
BOOT_ACK_RCV	1	13	RW	0x00	This status is set if the boot acknowledge is received from device. 0 - Boot ack is not received. 1 - Boot ack is received.
BOOT_TERMIN	1	14	RW	0x0	This status is set if the boot

Fields	Width	Field bits	Software R/W	Reset Value	Description
ATE_INTERRUPT					operation get terminated 0 - Boot operation is not terminated. 1 - Boot operation is terminated
ERR_INTERRUPT	1	15	RW	0x00	If any of the bits in the Error Interrupt Status Register are set, then this bit is set. Therefore the HD can test for an error by checking this bit first. 0 - No Error. 1 - Error.
CMD_TIMEOUT_ERR_STS	1	16	RW	0x00	Occurs only if the no response is returned within 64 SDCLK cycles from the end bit of the command. If the HC detects a CMD line conflict, in which case Command CRC Error shall also be set. This bit shall be set without waiting for 64 SDCLK cycles because the command will be aborted by the HC. 0 - No Error 1 - Timeout
CMD_CRC_ERR_STS	1	17	RW	0x00	Command CRC Error is generated in two cases. 1. If a response is returned and the Command Time-out Error is set to 0, this bit is set to 1 when detecting a CRT error in the command response 2. The HC detects a CMD line conflict by monitoring the CMD line when a command is issued.

Fields	Width	Field bits	Software R/W	Reset Value	Description
					If the HC drives the CMD line to 1 level, but detects 0 level on the CMD line at the next SDCLK edge, then the HC shall abort the command (Stop driving CMD line) and set this bit to 1. The Command Timeout Error shall also be set to 1 to distinguish CMD line conflict. 0 - No Error 1 - CRC Error Generated
CMD_END_BIT_ERR_STS	1	18	RW	0x00	Occurs when detecting that the end bit of a command response is 0. 0 - No Error 1 - End Bit Error Generated
CMD_INDEX_ERR_STS	1	19	RW	0x00	Occurs if a Command Index error occurs in the Command Response. 0 - No Error 1 - Error
DATA_TIMEOUT_ERR_STS	1	20	RW	0x00	Occurs when detecting one of following timeout conditions. 1. Busy Timeout for R1b, R5b type. 2. Busy Timeout after Write CRC status 3. Write CRC status Timeout 4. Read Data Timeout 0 - No Error

Fields	Width	Field bits	Software R/W	Reset Value	Description
					1 - Timeout
DATA_CRC_ERR_STS	1	21	RW	0x00	Occurs when detecting CRC error when transferring read data which uses the DAT line or when detecting the Write CRC Status having a value of other than "010". 0 - No Error 1 - Error
DATA_END_BIT_ERR_STS	1	22	RW	0x00	Occurs when detecting 0 at the end bit position of read data which uses the DAT line or the end bit position of the CRC status. 0 - No Error 1 - Error
CURRENT_LIMIT_ERR_STS	1	23	RW	0x00	By setting the SD Bus Power bit in the Power Control Register, the HC is requested to supply power for the SD Bus. If the HC supports the Current Limit Function, it can be protected from an Illegal card by stopping power supply to the card in which case this bit indicates a failure status. Reading 1 means the HC is not supplying power to SD card due to some failure. Reading 0 means that the HC is supplying power and no error has occurred. This bit shall always set to be 0, if the HC does not support this function. 0 - No Error

Fields	Width	Field bits	Software R/W	Reset Value	Description
					1 - Power Fail
AUTO_CMD_ERR_STS	1	24	RW	0x00	Auto CMD12 and Auto CMD23 use this error status. This bit is set when detecting that any of the bits D00 to D05 in Auto CMD Error Status register has changed from 0 to 1. D07 is effective in case of Auto CMD12. Auto CMD Error Status register is valid while this bit is set to 1 and may be cleared with clearing of this bit (another implementation is also allowed). 0 - No Error 1 - Error
ADMA_ERR_STS	1	25	RW	0x00	This bit is set when the Host Controller detects errors during ADMA based data transfer. The state of the ADMA at an error occurrence is saved in the ADMA Error Status Register. 1 - Error 0 -No error
TUNNING_ERR_STS	1	26	RW	0x00	This bit is set when an unrecoverable error is detected in a tuning circuit except during tuning procedure (Occurrence of an error during tuning procedure is indicated by Sampling Select). By detecting Tuning Error, Host Driver needs to abort a command executing and perform tuning. To reset tuning circuit, Sampling Clock shall be set to 0 before executing tuning procedure. The

Fields	Width	Field bits	Software R/W	Reset Value	Description
					Tuning Error is higher priority than the other error interrupts generated during data transfer. By detecting Turning Error, the Host Driver should discard data transferred by a current read/write command and retry data transfer after the Host Controller retrieved from tuning circuit error. 1 - Error 0 -No error
RESPONCE_ERROR_STS	1	27	RW	0x00	Host Controller Version 4.00 supports response error check function to avoid overhead of response error check by Host Driver during DMA execution. If Response Error Check Enable is set to 1 in the Transfer Mode register, Host Controller Checks R1 or R5 response. If an error is detected in a response, this bit is set to 1. 1 - Error 0 - No Error
TARGET_RESPONSE_ERROR_STS	1	28	RW	0x00	Occurs when detecting ERROR in m_hresp(dma transaction) 0 - no error 1 - error
VENDOR_SPECIFIC_ERROR0_STS	1	29	RW	0x0	Vendor specific error 0 status enable
VENDOR_SPECIFIC_ERROR1_STS	1	30	RW	0x0	Vendor specific error 1 status

Fields	Width	Field bits	Software R/W	Reset Value	Description
CIFIC_ERR1_STS					enable
VENDOR_SPECIFIC_ERR2_STS	1	31	RW	0x0	Vendor specific error 2 status enable

8.1.14 IRQ STATUS register (Offset Address - 0x30)

The Interrupt status registers is described below:

TABLE 15. IRQ STATUS register

Fields	Width	Field bits	Software R/W	Reset Value	Description
CMD_COMPLETE_STS	1	0	RW	0x0	Command complete status enable
TRANSFER_COMPLETE_STS	1	1	RW	0x0	Transfer complete status enable
BLK_GAP_EVENT_STS	1	2	RW	0x0	Block gap event status enable
DMA_INTERRUPT_STS	1	3	RW	0x0	DMA interrupt status enable
BUFFER_WRITE_READY_STS	1	4	RW	0x0	Buffer write ready status enable
BUFFER_READ_READY_STS	1	5	RW	0x0	Buffer read ready status enable
CARD_INSERTION_STS	1	6	RW	0x0	Card Insertion status enable
CARD_REMOVAL_STS	1	7	RW	0x0	Card Removal status enable
CARD_INTERRUPT_STS	1	8	RW	0x0	Card interrupt status enable
INT_A_STS	1	9	RW	0x0	Init A status enable
INT_B_STS	1	10	RW	0x0	Init B status enable
INT_C_STS	1	11	RW	0x0	Init C status enable
RE_TUNNING_EVENT_STS	1	12	RW	0x0	Re tuning event status enable
BOOT_ACK_RCVD	1	13	RW	0x0	Set if the boot acknowledge is received from device
BOOT_TERMINATE_INTERRUPT	1	14	RW	0x0	Set if the boot operation get terminated

Fields	Width	Field bits	Software R/W	Reset Value	Description
PT					
ERR_INTERRUPT	1	15	RW	0x0	Error Interrupt Status
CMD_TIMEOUT_ERR_STS	1	16	RW	0x0	Command timeout error status enable
CMD_CRC_ERR_STS	1	17	RW	0x0	Command CRC error status enable
CMD_END_BIT_ERR_STS	1	18	RW	0x0	Command Bit error status enable
CMD_INDEX_ERR_STS	1	19	RW	0x0	Command index error status enable
DATA_TIMEOUT_ERR_STS	1	20	RW	0x0	Data time out error status enable
DATA_CRC_ERR_STS	1	21	RW	0x0	Data CRC error status enable
DATA_END_BIT_ERR_STS	1	22	RW	0x0	Data End bit error status enable
CURRENT_LIMIT_ERR_STS	1	23	RW	0x0	Current limit error status enable
AUTO_CMD_ERR_STS	1	24	RW	0x0	Auto command error status enable
ADMA_ERR_STS	1	25	RW	0x0	ADMA error status enable
TUNNING_ERR_STS	1	26	RW	0x0	Tunning error status enable
RESPONSE_ERR_STS	1	27	RW	0x0	Response error status enable
TARGET_RESPONSE_ERR_STS	1	28	RW	0x0	Target Response error status enable
VENDOR_SPECIFIC_ERR0_STS	1	29	RW	0x0	Vendor specific error 0 status enable
VENDOR_SPECIFIC_ERR1_STS	1	30	RW	0x0	Vendor specific error 1 status enable
VENDOR_SPECIFIC_ERR2_STS	1	31	RW	0x0	Vendor specific error 2 status enable

8.1.15 INTR SIG EN register (Offset Address - 0x38)

The Interrupt signal enable registers is described below:

TABLE 16. INTR SIG EN register

Fields	Width	Field bits	Software R/W	Reset Value	Description
CMD_COMPLETE_SIGNAL_EN	1	0	RW	0x0	Command complete Signal Enable
TRANSFER_COMPLETE_SIGNAL_EN	1	1	RW	0x0	Transfer complete Signal Enable
BLK_GAP_EVENT_SIGNAL_EN	1	2	RW	0x0	Block gap event Signal Enable
DMA_INTERRUPT_SIGNAL_EN	1	3	RW	0x0	DMA Interrupt Signal Enable
BUFFER_WRITE_READY_SIGNAL_EN	1	4	RW	0x0	Buffer write ready Signal Enable
BUFFER_READ_READY_SIGNAL_EN	1	5	RW	0x0	Buffer read ready Signal Enable
CARD_INSERTION_SIGNAL_EN	1	6	RW	0x0	Card insertion Signal Enable
CARD_REMOVAL_SIGNAL_EN	1	7	RW	0x0	Card removal Signal Enable
CARD_INTERRUPT_SIGNAL_EN	1	8	RW	0x0	Card interrupt Signal Enable
INT_A_SIGNAL_EN	1	9	RW	0x0	Init A Signal Enable
INT_B_SIGNAL_EN	1	10	RW	0x0	INit B Signal Enable
INT_C_SIGNAL_EN	1	11	RW	0x0	Init C Signal Enable
RE_TUNNING_EVENT_SIGNAL_EN	1	12	RW	0x0	Re tuning event Signal Enable
BOOT_ACK_RCVD_SIGNAL_EN	1	13	RW	0x0	Set if the boot acknowledge is received from device
BOOT_TERMIN	1	14	RW	0x0	Set if the boot operation get

Fields	Width	Field bits	Software R/W	Reset Value	Description
ATE_INTERRUPT_SIGNAL_EN					terminated
FIXED_TO_0_NORMAL_INTR_SIGNAL_EN	1	15	R	0x0	Error Interrupt Status
CMD_TIMEOUT_ERR_SIGNAL_EN	1	16	RW	0x0	Command time out Signal Enable
CMD_CRC_ERR_SIGNAL_EN	1	17	RW	0x0	Command CRC Signal Enable
CMD_END_BIT_ERR_SIGNAL_EN	1	18	RW	0x0	Command end bit Signal Enable
CMD_INDEX_ERR_SIGNAL_EN	1	19	RW	0x0	Command Index Signal Enable
DATA_TIMEOUT_ERR_SIGNAL_EN	1	20	RW	0x0	Data timeout Signal Enable
DATA_CRC_ERR_SIGNAL_EN	1	21	RW	0x0	Data CRC Signal Enable
DATA_END_BIT_ERR_SIGNAL_EN	1	22	RW	0x0	Data end bit Signal Enable
CURRENT_LIMIT_ERR_SIGNAL_EN	1	23	RW	0x0	Current limit error Signal Enable
AUTO_CMD_ERR_SIGNAL_EN	1	24	RW	0x0	Auto command error Signal Enable
ADMA_ERR_SIGNAL_EN	1	25	RW	0x0	ADMA error Signal Enable
TUNNING_ERR_SIGNAL_EN	1	26	RW	0x0	Tunning error Signal Enable
RESPONSE_ERR_SIGNAL_EN	1	27	RW	0x0	Response error Signal Enable
VENDOR_SPECIFIC_ERR_SIGNAL_EN	4	31:28	RW	0x0	Vendor specific error Signal

8.1.16 HOST CTRL2 register (Offset Address - 0x3C)

The Control registers is described below:

TABLE 17. HOST CTRL2 register

Fields	Width	Field bits	Software R/W	Reset Value	Description
ACMD_12_NOT_EXECUTED	1	0	RW	0x0	If memory multiple block data transfer is not started due to command error, this bit is not set because it is not necessary to issue Auto CMD12. Setting this bit to 1 means the HC cannot issue Auto CMD12 to stop memory multiple block transfer due to some error. If this bit is set to 1, other error status bits (D04 - D01) are meaningless. This bit is set to 0 when Auto CMD Error is generated by Auto CMD23 0 - Executed 1 - Not Executed
ACMD_TIMEOUT_ERR	1	1	RW	0x0	Occurs if the no response is returned within 64 SDCLK cycles from the end bit of the command. If this bit is set to 1, the other error status bits (D04 - D02) are meaningless. 0 - No Error 1 - Timeout
ACMD_CRC_ERROR	1	2	RW	0x0	Occurs when detecting a CRC error in the command response. 0 - No Error 1 - CRC Error Generated
ACMD_END_BIT_ERR	1	3	RW	0x0	Occurs when detecting that the end bit of command response is

Fields	Width	Field bits	Software R/W	Reset Value	Description
					0. 0 - No Error 1 - End Bit Error Generated
ACMD_INDEX_ERR	1	4	RW	0x0	Occurs if the Command Index error occurs in response to a command. 0 - No Error 1 - Error
ACMD_RESPONCE_ERR	1	5	RW	0x0	This bit is set when Response Error Check Enable in the Transfer Mode register is set to 1 and an error is detected in either R1 response of either Auto CMD12 or Auto CMD23. 0 - No error 1 - Error
HOST_CTRL2_RESERVED	1	6	Reserved	0x0	Reserved for future
CMD_NOT_ISSUED_BY_AUTO_CMD12_ERR	1	7	RW	0x0	Setting this bit to 1 means CMD_wo_DAT is not executed due to an Auto CMD12 error (D04 - D01) in this register. This bit is set to 0 when Auto CMD Error is generated by Auto CMD23 0 - No Error 1 - Not Issued
RESERVED_AUTO_CMD_ERROR_STS_15_8	8	15:8	RW	0x0	Reserved for future use

Fields	Width	Field bits	Software R/W	Reset Value	Description
MODE_SELECT	3	18:16	RW	0x0	This field is used to select one of UHS-I modes. In case of UHS-I mode, this field is effective when 1.8V Signaling Enable is set to 1. Setting of this field is used to select one of preset values in UHS-Imode. If Preset Value Enable in the Host Control 2 register is set to 1, Host Controller sets SDCLK/RCLK Frequency Select, Clock Generator Select in the Clock Control register and Driver Strength Select according to Preset Value registers. In this case, one of preset value registers is selected by this field. Host Driver needs to reset SD Clock Enable before changing this field to avoid generating clock glitch. After setting this field, Host Driver sets SD Clock Enable again. 000b - SDR12 001b - SDR25 010b - SDR50 011b - SDR104 100b - DDR50 101b - Reserved 110b - Reserved 111b - Reserved When SDR50, SDR104 or DDR50 is selected for SDIO

Fields	Width	Field bits	Software R/W	Reset Value	Description
					card, interrupt detection at the block gap shall not be used. Read Wait timing is changed for these modes
V1_8_SIGNALLING_EN	1	19	RW	0x0	This bit controls voltage regulator for I/O cell. 3.3V is supplied to the card regardless of signaling voltage. Setting this bit from 0 to 1 starts changing signal voltage from 3.3V to 1.8V. 1.8V regulator output shall be stable within 5ms. Host Controller clears this bit if switching to 1.8V signaling fails. Clearing this bit from 1 to 0 starts changing signal voltage from 1.8V to 3.3V. Refer to Host Regulator Voltage Stable in the Present State register how to determine stable of host regulator voltage. 3.3V regulator output shall be stable within 5ms. Host Driver can set this bit to 1 when Host Controller supports 1.8V signaling (One of support bits is set to 1: SDR50, SDR104 or DDR50 in the Capabilities register) and the card or device supports UHS-I 1 - 1.8V Signaling 0 - 3.3V Signaling
DRIVER_STRENGTH_SELECT	2	21:20	RW	0x0	Host Controller output driver in 1.8V signaling is selected by this bit. In 3.3V signaling, this field is not effective. This field can be set depends on Driver Type A, C and D support bits in the Capabilities

Fields	Width	Field bits	Software R/W	Reset Value	Description
					register. This bit depends on setting of Preset Value Enable. If Preset Value Enable = 0, this field is set by Host Driver. If Preset Value Enable = 1, this field is automatically set by a value specified in the one of Preset Value registers. 00b Driver Type B is Selected (Default) 01b Driver Type A is Selected 10b Driver Type C is Selected 11b Driver Type D is Selected
EXECUTE_TUNING	1	22	RW	0x0	This bit is set to 1 to start tuning procedure and automatically cleared when tuning procedure is completed. The result of tuning is indicated to Sampling Clock Select. Tuning procedure is aborted by writing 0 for more detail about tuning procedure. 1 - Execute Tuning, 0 Not Tuned or Tuning Completed
SAMPLING_CLOCK_SELECT	1	23	RW	0x0	This bit is set by tuning procedure when Execute Tuning is cleared. Writing 1 to this bit is meaningless and ignored. Setting 1 means that tuning is completed successfully and setting 0 means that tuning is failed. Host

Fields	Width	Field bits	Software R/W	Reset Value	Description
					Controller uses this bit to select sampling clock to receive CMD and DAT. This bit is cleared by writing 0. Change of this bit is not allowed while the Host Controller is receiving response or a read data block. 1 Tuned clock is used to sample data 0 Fixed clock is used to sample data
HOST_CTRL2_RSV2	2	25:24	Reserved	0x0	Reserved for future use
ADMA2_LENGT_H_MODE	1	26	RW	0x0	This bit selects one of ADMA2 Length Modes either 16-bit or 26-bit. 1 - 26-bit Data Length Mode 0 - 16-bit Data Length Mode
CMD23_EN	1	27	RW	0x0	In memory card initialization, Host Driver Version 4.10 checks whether card supports CMD23 by checking a bit SCR[33]. If the card supports CMD23 (SCR[33]=1), this bit is set to 1. This bit is used to select Auto CMD23 or Auto CMD12 for ADMA3 data transfer. Refer to Auto CMD Enable in the Transfer Mode register.
HOST_VERSION_4_EN	1	28	RW	0x0	This bit selects either Version 3.00 compatible mode or Ver4. mode. In Version 4.00, support of 64-bit System Addressing is modified. All DMAs support 64-bit System Addressing. In Version

Fields	Width	Field bits	Software R/W	Reset Value	Description
					4.10, supported 32-bit Block Count for all operations. Functions of following fields are modified. SDMA Address SDMA uses ADMA System Address register (05Fh-058h) instead of SDMA System Address register (Offset 003-000h) ADMA2 / ADMA3 Selection ADMA3 is selected by DMA Select in the Host Control 1 register. 64bit ADMA Descriptor Size 128bit descriptor is used instead of 96-bit descriptor when 64-bit Addressing is set to 1. Selection of 32-bit / 64-bit System Addressing Either 32-bit or 64-bit system addressing is selected by 64-bit Addressing bit in this register instead of DMA Select in the Host Control 1 register. 32-bit Block Count SDMA System Address register (003h-000h) is modified to 32-bit Block Count register. 0 - Version 3.00 Compatible Mode 1 - Version 4.00 Mode
64BIT_ADDRESSING	1	29	RW	0x0	This field is effective when Host Version 4.00 Enable is set to 1. Host Controller selects either of 32-bit or 64-bit addressing modes to access system memory. Whether 32-bit or 64-bit is determined by OS installed in a host system. Host Driver sets this bit depends on

Fields	Width	Field bits	Software R/W	Reset Value	Description
					addressing mode of installed OS. Refer to 64-bit System Address Support in the Capabilities register. 0 - 32 bits addressing 1 - 64 bits addressing
ASYNCHRONOUS_INTERRUPT_EN	1	30	RW	0x0	This bit can be set to 1 if a card support asynchronous interrupt and Asynchronous Interrupt Support is set to 1 in the Capabilities register. Asynchronous interrupt is effective when DAT[1] interrupt is used in 4-bit SD mode (and zero is set to Interrupt Pin Select in the Embedded Control register). If this bit is set to 1, the Host Driver can stop the SDCLK during asynchronous interrupt period to save power. During this period, the Host Controller continues to deliver Card Interrupt to the host when it is asserted by the Card. 0 - Disabled 1 - Enabled,
PRESET_VALUE_EN	1	31	RW	0x0	Host Controller Version 3.00 supports this bit. As the operating SDCLK/RCLK frequency and I/O driver strength depend on the Host System implementation, it is difficult to determine these parameters in the Standard Host Driver. When Preset Value Enable is set to automatic. This bit enables the functions defined in the Preset Value registers. 1

Fields	Width	Field bits	Software R/W	Reset Value	Description
					Automatic Selection by Preset Value are Enabled 0 SDCLK and Driver Strength are controlled by Host Driver If this bit is set to 0, SDCLK/RCLK Frequency Select, Clock Generator Select in the Clock Control register and Driver Strength Select in Host Control 2 register are set by Host Driver. If this bit is set to 1, SDCLK /RCLK Frequency Select, Clock Generator Select in the Clock Control register and Driver Strength Select in Host Control 2 register are set by Host Controller as specified in the Preset Value registers.

8.1.17 CAPABILITIES 0 register (Offset Address - 0x40)

The Capability register [31:0] is described below:

TABLE 18. CAPABILITIES 0 register

Fields	Width	Field bits	Software R/W	Reset Value	Description
TIMEOUT_CLK_FREQ	6	5:0	RW	0x0	This bit shows the base clock frequency used to detect Data Timeout Error.
RESERVED_CAPABILITIES_07_00	1	6	Reserved	0x0	Reserved for future
TIMEOUT_CLK_UNIT	1	7	RW	0x0	This bit shows the unit of base clock frequency used to detect Data Timeout Error
BASE_CLK_FREQ_FOR_SD_CLK	8	15:8	RW	0x0	This mode is supported by the Host Controller Version 3.00. Unit values are 1MHz. The supported clock range is 10MHz to 255MHz.
MAX_BLK_LENGTH	2	17:16	RW	0x0	This value indicates the maximum block size that the HD can read and write to the buffer in the HC. The buffer shall transfer this block size without wait

Fields	Width	Field bits	Software R/W	Reset Value	Description
					cycles. Three sizes can be defined as indicated below. 00 - 512 byte 01 - 1024 byte 10 - 2048 byte 11 - 4096 byte
EMBEDDED_DEVICE_8BIT	1	18	RW	0x1	This bit indicates whether the Host Controller is capable of using 8-bit bus width mode. This bit is not effective when Slot Type is set to 10b. In this case, refer to Bus Width Preset in the Embedded Control register. Version 4.10 Host Controller shall support ADMA2 if ADMA3 is supported. 1 - 8-bit Bus Width Supported 0 - 8-bit Bus Width not Supported
ADMA2_SUPPORT	1	19	RW	0x0	ADMA2 support 1 - ADMA2 support. 0 - ADMA2 not support
RESERVED_CAPABILITIES_23_16	1	20	Reserved	0x0	Reserved for future
HIGH_SPEED_SUPPORT	1	21	RW	0x0	This bit indicates whether the HC and the Host System support High Speed mode and they can supply SD Clock frequency from 25Mhz to 50 Mhz (for SD)/ 20MHz to 52MHz (for MMC). 0 - High Speed Not Supported 1 - High Speed Supported
SDMA_SUPPORT	1	22	RW	0x1	This bit indicates whether the HC is capable of using DMA to transfer data between system memory and the HC directly. Version 4.10 Host Controller shall support SDMA if ADMA2 is supported. 0 - SDMA Not Supported 1 - SDMA Supported.
SUSPEND_RESUME_SUPPORT	1	23	RW	0x0	This bit indicates whether the HC supports Suspend / Resume functionality. If this bit is 0, the

Fields	Width	Field bits	Software R/W	Reset Value	Description
					Suspend and Resume mechanism are not supported and the HD shall not issue either Suspend / Resume commands. 0 - Not Supported 1 - Supported
VOLTAGE_SUPPORT_3_3V	1	24	RW	0x0	Voltage support for 3.3V 0 - 3.3 V Not Supported 1 - 3.3 V Supported
VOLTAGE_SUPPORT_3_0V	1	25	RW	0x0	Voltage support for 3.0V 0 - 3.0 V Not Supported 1 - 3.0 V Supported
VOLTAGE_SUPPORT_1_8V	1	26	RW	0x0	Voltage support for 1.8V 0 - 1.8 V Not Supported 1 - 1.8 V Supported
SYS_ADDR_SUPPORT_FOR_V4_64BIT	1	27	RW	0x0	This bit is added from Version 4.10. Setting 1 to this bit indicates that the Host Controller supports 64-bit System Addressing of Version 4 mode. When this bit is set to 1, full or a part of 64-bit address should be used to decode Host Controller Registers so that Host Controller Registers can be placed above system memory area. 64-bit address decode of Host Controller Registers is effective regardless of setting to 64bit Addressing in Host Control 2. If this bit is set to 1, 64-bit DMA Addressing for Version 4 is enabled by setting Host Version 4 Enable =1, 64-bit Addressing =1 in the Host Control 2 register. SDMA can be used and ADMA2 uses 128-bit Descriptor. 1 - 64-bit System Address for V4 is Supported 0 - 64-bit System Address for V4 is not Supported
SYS_ADDR_SU	1	28	RW	0x0	Meaning of this bit is different

Fields	Width	Field bits	Software R/W	Reset Value	Description
PPOINT_FOR_V3_64BIT					depends on Versions. Host Controller Version 3.00 and Ver4.10 use this bit as 64-bit System Address support for V3 mode. Host Controller Version 4.00 uses this bit as 64-bit System Address support for both V3 and V4 modes. SDMA cannot be used in 64-bit Addressing in Version 3 mode. If this bit is set to 1, 64-bit ADMA2 with using 96-bit Descriptor may be enabled as follows: In case of Host Controller Version 3, 64-bit ADMA2 is enabled by DMA Select =11b in the Host Control 1 register. In case of Host Controller Version 4, 64-bit ADMA2 for Version 3 is enabled by setting Host Version 4 Enable =0 and DMA Select = 11b. 1 - 64-bit System Address for V3is Supported 0 - 64-bit System Address for V3is not Supported
ASYNCHRONOUS_INTERRUPT_SUPPORT	1	29	RW	0x0	Asynchronous Interrupt support 1 Asynchronous Interrupt Supported 0 Asynchronous Interrupt Not Supported
SLOT_TYPE	2	31:30	RW	0x1	A specific Host Driver is required developed by a Host System. 00b Removable Card Slot 01b Embedded Slot for One Device 10b Shared Bus Slot (SD Mode) 11b UHS-II Multiple Embedded Devices

8.1.18 CAPABILITIES 1 register (Offset Address - 0x44)

The Capability register [31:0] is described below:

TABLE 19. CAPABILITIES 1 register

Fields	Width	Field bits	Software R/W	Reset Value	Description
SDR50_SUPPORT	1	0	RW	0x0	SDR50 support
SDR104_SUPPORT	1	1	RW	0x0	SDR104 support
DDR_SUPPORT	1	2	RW	0x0	DDR support
CAPABILITIES_1_RSV1	1	3	Reserved	0x0	Reserved for future use
DRIVER_TYPE_A_SUPPORT	1	4	RW	0x0	Driver type A support
DRIVER_TYPE_C_SUPPORT	1	5	RW	0x0	Driver type C support
DRIVER_TYPE_D_SUPPORT	1	6	RW	0x0	Driver type D support
RFU_DRIVER_TYPE_4_SUPPORT	1	7	RW	0x0	Reserved for future use
TIMER_COUNT_FOR_RE_TUNNING	4	11:8	RW	0x0	Timer count for retuning
RESERVED_CAPABILITIES_4_4	1	12	Reserved	0x0	Reserved for future
USE_TUNNING_FOR_SDR	1	13	RW	0x0	Use tuning for SDR
RE_TUNNING_MODES	2	15:14	RW	0x0	Re tuning modes
CLK_MULTIPLIER	8	23:16	RW	0x0	Clock Multiplier
SPI_MODE	1	24	RW	0x0	SPI Mode enable
SPI_BLOCK_MODE	1	25	RW		SPI Block Mode
RESERVED_CAPABILITIES_5_8	1	26	Reserved	0x0	Reserved for future
RESERVED_ADMA3_SUPPORT	1	27	RW	0x0	RESERVED ADMA3 support
V1_8V_VDD2_	1	28	RW	0x0	1.8V VDD2 support

Fields	Width	Field bits	Software R/W	Reset Value	Description
SUPPORT					
RESERVED_F OR_FUTURE_V DD2	1	29	Reserved	0x0	Reserved for future
RESERVED2_C APABILITIES_6 2	1	30	Reserved	0x0	Reserved for future
RESERVED_ HS_400_SUP PORT	1	31	Reserved	0x0	RESERVED HS_400_support

8.1.19 MAX CURRENT CAP 0 register (Offset Address - 0x48)

The Max Current Capabilities Register [31:0] is described below:

TABLE 20. MAX CURRENT CAP 0 register

Fields	Width	Field bits	Software R/W	Reset Value	Description
MAX_CURREN T_CAP_0	32	31:0	RW	0x00	Max Current Capabilities Register [31:0]

8.1.20 MAX CURRENT CAP 1 register (Offset Address - 0x4C)

The Max Current Capabilities Register [31:0] is described below:

TABLE 21. MAX CURRENT CAP 1 register

Fields	Width	Field bits	Software R/W	Reset Value	Description
MAX_CURREN T_CAP_1	32	31:0	RW	0x00	Max Current Capabilities Register [31:0]

8.1.21 FORCE EVENT REG register (Offset Address - 0x50)

The Force Event Registers is described below:

TABLE 22. FORCE EVENT REG register

Fields	Width	Field bits	Software R/W	Reset Value	Description
FE_FOR_ACM D_12_NOT_EX ECUTED	1	0	RW	0x0	Force Event for Auto Command 12 not executed

Fields	Width	Field bits	Software R/W	Reset Value	Description
					1 – Set 0 – Not Affected
FE_FOR_ACM D_TIMEOUT_E RR	1	1	RW	0x0	Force Event for Auto Command time out error 1 – Set 0 – Not Affected
FE_FOR_ACM D_CRC_ERR	1	2	RW	0x0	Force Event for Auto Command CRC error 1 – Set 0 – Not Affected
FE_FOR_ACM D_END_BIT_E RR	1	3	RW	0x0	Force Event for Auto Command Endbit error 1 – Set 0 – Not Affected
FE_FOR_ACM D_INDEX_ERR	1	4	RW	0x0	Force Event for Auto Command Index error 1 – Set 0 – Not Affected
FE_FOR_ACM D_RESPONCE _ERR	1	5	RW	0x0	Force Event for Auto Command Responce error 1 – Set 0 – Not Affected
RESERVED_FE _7	1	6	Reserved	0x0	Reserved for future
FE_FOR_CMD_ NOT_ISSUED_ BY_ACMD12_E RR	1	7	RW	0x0	Force Event for Command not issued by ACMD 12 error 1 – Set 0 – Not Affected
RESERVED_FE _AUTOCMD_E RR_STS	8	15:8	Reserved	0x0	Reserved for future

Fields	Width	Field bits	Software R/W	Reset Value	Description
FE_FOR_CMD_TIMEOUT_ERR	1	16	RW	0x0	Force Event for Auto Command time out error 1 – Set 0 – Not Affected
FE_FOR_CMD_CRC_ERR	1	17	RW	0x0	Force Event for Auto Command CRC error 1 – Set 0 – Not Affected
FE_FOR_CMD_END_BIT_ERR	1	18	RW	0x0	Force Event for Auto Command Endbit error 1 – Set 0 – Not Affected
FE_FOR_CMD_INDEX_ERR	1	19	RW	0x0	Force Event for Auto Command Index error 1 – Set 0 – Not Affected
FE_FOR_DATA_TIMEOUT_ERR	1	20	RW	0x0	Force Event for data timeout error 1 – Set 0 – Not Affected
FE_FOR_DATA_CRC_ERR	1	21	RW	0x0	Force Event for data CRC error 1 – Set 0 – Not Affected
FE_FOR_DATA_END_BIT_ERR	1	22	RW	0x0	Force Event for data Endbit error 1 – Set 0 – Not Affected
FE_FOR_CURRENT_LIMIT_ERR	1	23	RW	0x0	Force Event for current limit error 1 – Set 0 – Not Affected
FE_FOR_ACM	1	24	RW	0x0	Force Event for Auto Command

Fields	Width	Field bits	Software R/W	Reset Value	Description
D_ERR					error 1 – Set 0 – Not Affected
FE_FOR_ADM A_ERR_INTR_ STS	1	25	RW	0x0	Force Event for ADMA error 1 – Set 0 – Not Affected
FE_FOR_TUNN ING_ERR	1	26	RW	0x0	Force Event for Tunning error 1 – Set 0 – Not Affected
FE_FOR_RESP ONCE_ERR	1	27	RW	0x0	Force Event for Responce error 1 – Set 0 – Not Affected
FE_FOR_VEND OR_SPC_ERR_ STS	4	31:28	RW	0x0	Force Event for Vendor specific error status 1 – Set 0 – Not Affected

8.1.22 ADMA ERR STATUS register (Offset Address - 0x54)

The ADMA Error/Status Register is described below:

TABLE 23. ADMA ERR STATUS register

Fields	Width	Field bits	Software R/W	Reset Value	Description
ADMA_ERR_S TATES	2	1:0	R	0x0	This field indicates the state of ADMA when error is occurred during ADMA data transfer. This field never indicates "10" because ADMA never stops in this state. D01 – D00 : ADMA Error State when error occurred Contents of SYS_SDR register

Fields	Width	Field bits	Software R/W	Reset Value	Description
					00 - ST_STOP (Stop DMA) Points to next of the error descriptor 01 - ST_FDS (Fetch Descriptor) Points to the error descriptor 10 - Never set this state (Not used) 11 - ST_TFR (Transfer Data) Points to the next of the error descriptor
ADMA_LENTH_MISMATCH_ERR	1	2	R	0x0	match error This error occurs in the following 2 cases. While Block Count Enable being set, the total data length specified by the Descriptor table is different from that specified by the Block Count and Block Length. Total data length can not be divided by the block length. 1 - Error 0 - No error
RESERVED_ADMA_ERR_STS_7_3	5	7:3	Reserved	0x0	Reserved for future

8.1.23 ADMA SYS ADDR0 register (Offset Address - 0x58)

The ADMA System Address [31:00] is described below:

TABLE 24. ADMA SYS ADDR0 register

Fields	Width	Field bits	Software R/W	Reset Value	Description
ADMA_SYS_A_DDR0	32	31:0	RW	0x00	The 32-bit addressing Host Driver uses lower 32-bit of this

Fields	Width	Field bits	Software R/W	Reset Value	Description
					register (upper 32-bit should be set to 0) and shall program Descriptor Table on 32-bit boundary and set 32-bit boundary address to this register. DMA2/3 ignores lower 2-bit of this register and assumes it to be 00b. DMA in 64-bit addressing. The 64-bit addressing Host Driver uses all bits of this register and shall program Descriptor Table on 64-bit boundary and set 64-bit boundary address to this register. DMA2/3 ignores lower 3-bit of this register and assumes it to be 000b. SDMA If Host Version 4.00 Enable is set to 1, SDMA use this register to indicate System Address of data location instead of using SDMA System Address register (Offset 003-000h). SDMA can be used in 32-bit and 64-bit addressing in Version 4.00. ADMA2 This register holds byte address of executing command of the Descriptor table. At the start of ADMA2, the Host Driver shall set start address of the Descriptor table. The ADMA increments this register address, which points to next line, when every fetching a Descriptor line. When the ADMA Error Interrupt is generated, this register shall hold the Descriptor address depending on the ADMA state. ADMA3 Host Driver is not necessary to set this register. The ADMA3 increments address of this register, which points to next line,

Fields	Width	Field bits	Software R/W	Reset Value	Description
					when every time fetching a Descriptor line. When Error Interrupt is generated, this register shall hold the Descriptor address depending on the ADMA state. Register Value - 00000000_xxxxxxxh Addressing Mode - 32-bit System Address Register Value - xxxxxxxx_xxxxxxxh Addressing Mode - 64-bit System Address

8.1.24 ADMA SYS ADDR1 register (Offset Address - 0x5C)

The ADMA System Address [63:32] is described below:

TABLE 25. ADMA SYS ADDR1 register

Fields	Width	Field bits	Software R/W	Reset Value	Description
ADMA_SYS_A DDR1	32	31:0	RW	0x00	ADMA System Address [63:32]

8.1.25 PRESET 0 register (Offset Address - 0x60)

The Preset value register for initialization and default speed is described below:

TABLE 26. PRESET 0 register

Fields	Width	Field bits	Software R/W	Reset Value	Description
INT_SDCLK_F REQ_SELECT_ VALUE_9_0	10	9:0	RW	0x0	Initialization SDCLK Frequency select value [09:08] 10-bit preset value to set SDCLK/RCLK Frequency Select in the Clock Control Register is described by a host system.
INT_CLK_GEN ERATOR_SELE CT_VALUE	1	10	RW	0x0	Initialization Clock generator select value This bit is effective when Host Controller supports programmable clock generator. 1 - Programmable Clock

Fields	Width	Field bits	Software R/W	Reset Value	Description
					Generator 0 - Host Controller Ver2.00 Compatible Clock Generator
INT_RESERVE D_PRESET_13_11	3	13:11	Reserved	0x0	Initialization Reserved for future
INT_DRIVER_SELECT_STRENGTH_VALUE	2	15:14	RW	0x0	Initialization Driver select strength value Driver Strength is supported by 1.8V signaling bus speed modes. This field is meaningless for 3.3V signaling. 11b - Driver Type D is Selected 10b - Driver Type C is Selected 01b - Driver Type A is Selected 00b - Driver Type B is Selected
DS_SDCLK_FREQUENCY_SELECT_VALUE_9_0	10	25:16	RW	0x0	Default speed SDCLK Frequency select value [09:08] 10-bit preset value to set SDCLK/RCLK Frequency Select in the Clock Control Register is described by a host system.
DS_CLK_GENERATOR_SELECT_VALUE	1	26	RW	0x0	Default speed Clock generator select value This bit is effective when Host Controller supports programmable clock generator. 1 - Programmable Clock Generator 0 - Host Controller Ver2.00 Compatible Clock Generator
DS_RESERVE_D_PRESET_13_11	3	29:27	Reserved	0x0	Default speed Reserved for future
DS_DRIVER_SELECT_STRENGTH_VALUE	2	31:30	RW	0x0	Default speed Driver select strength value Driver Strength is supported by 1.8V signaling bus speed modes. This field is meaningless for 3.3V signaling. 11b - Driver Type D is Selected 10b - Driver Type C is Selected 01b - Driver Type A is Selected 00b - Driver Type B is Selected

8.1.26 PRESET 1 register (Offset Address - 0x64)

The Preset value register for high speed is described below:

TABLE 27. PRESET 1 register

Fields	Width	Field bits	Software R/W	Reset Value	Description
HS_SDCLK_FREQ_SELECT_VALUE_9_0	10	9:0	RW	0x0	High Speed SDCLK Frequency select value [09:08]
HS_CLK_GENERATOR_SELECT_VALUE	1	10	RW	0x0	High Speed Clock generator select value
HS_RESERVED_PRESET_13_11	3	13:11	RW	0x0	High Speed Reserved for future
HS_DRIVER_SELECT_STRENGTH_VALUE	2	15:14	RW	0x0	High Speed Driver select strength value
SDR12_SDCLK_FREQ_SELECT_VALUE_9_0	10	25:16	RW	0x0	SDR12 SDCLK Frequency select value [09:08]
SDR12_CLK_GENERATOR_SELECT_VALUE	1	26	RW	0x0	SDR12 Clock generator select value
SDR12_RESERVED_PRESET_13_11	3	29:27	RW	0x0	SDR12 Reserved for future
SDR12_DRIVER_SELECT_STRENGTH_VALUE	2	31:30	RW	0x0	SDR12 Driver select strength value

8.1.27 PRESET 2 register (Offset Address - 0x6C)

The Preset value register for DDR50 is described below:

TABLE 28. PRESET 2 register

Fields	Width	Field bits	Software R/W	Reset Value	Description
SDR25_SDCLK_FREQ_SELECT_VALUE_9_0	10	9:0	RW	0x0	SDR25 SDCLK Frequency select value [09:08]
SDR25_CLK_GENERATOR_SELECT_VALUE	1	10	RW	0x0	SDR25 Clock generator select value

SDR25_RESERVED_PRESET_13_11	3	13:11	Reserved	0x0	SDR25 Reserved for future
SDR25_DRIVER_SELECT_STRENGTH_VALUE	2	15:14	RW	0x0	SDR25 Driver select strength value
SDR50_SDCLK_FREQ_SELECT_VALUE_9_0	10	25:16	RW	0x0	SDR50 SDCLK Frequency select value [09:08]
SDR50_CLK_GENERATOR_SELECT_VALUE	1	26	RW	0x0	SDR50 Clock generator select value
SDR50_RESERVED_PRESET_13_11	3	29:27	Reserved	0x0	SDR50 Reserved for future
SDR50_DRIVER_SELECT_STRENGTH_VALUE	2	31:30	RW	0x0	SDR50 Driver select strength value

8.1.28 PRESET 3 register (Offset Address - 0x6C)

The Preset value register for DDR50 is described below:

TABLE 29. PRESET 3 register

Fields	Width	Field bits	Software R/W	Reset Value	Description
SDR104_SDCLK_FREQ_SELECT_VALUE_9_0	10	9:0	RW	0x0	SDR104 SDCLK Frequency select value [09:08]
SDR104_CLK_GENERATOR_SELECT_VALUE	1	10	RW	0x0	SDR104 Clock generator select value
SDR104_RESERVED_PRESET_13_11	3	13:11	RW	0x0	SDR104 Reserved for future
SDR104_DRIVER_SELECT_STRENGTH_VALUE	2	15:14	RW	0x0	SDR104 Driver select strength value
DDR50_SDCLK_FREQ_SELECT	10	25:16	RW	0x0	DDR50 SDCLK Frequency select value [09:08]

Fields	Width	Field bits	Software R/W	Reset Value	Description
T_VALUE_9_0					
DDR50_CLK_GENERATOR_SELECT_VALUE	1	26	RW	0x0	DDR50 Clock generator select value
DDR50_RESERVED_PRESET_13_11	3	29:27	RW	0x0	DDR50 Reserved for future
DDR50_DRIVER_SELECT_STRENGTH_VALUE	2	31:30	RW	0x0	DDR50 Driver select strength value

8.1.29 BOOT TIMEOUT CNT register (Offset Address - 0x0F4)

The Boot Timeout Control is described below:

TABLE 30. BOOT TIMEOUT CNT register

Fields	Width	Field bits	Software R/W	Reset Value	Description
BOOT_TIMEOUT_CNT	32	31:0	RW	0x1000	Boot Timeout Control

8.1.30 VENDOR SPECIFIC register (Offset Address - 0x0F8)

The Vendor specific register is described below:

TABLE 31. VENDOR SPECIFIC register

Fields	Width	Field bits	Software R/W	Reset Value	Description
VENDOR_SPECIFIC_RFU0	1	0	RW	0x0	Vendor specific reserved
VENDOR_EMMC_HWRESET	1	1	RW	0x0	Vendor specific hw reset
VENDOR_SPECIFIC_RFU1	14	15:2	RW	0x0	Vendor specific reserved

8.1.31 SLOT INTR STATUS HOST CONTROLLER VER register (Offset Address - 0x0FC)

The Register for Intr/status and Host Controller Version is described below:

TABLE 32. SLOT INTR STATUS HOST CONTROLLER VER register

Fields	Width	Field bits	Software R/W	Reset Value	Description
SLOT_INTR_STATUS_07_00	16	15:0	RW	0x00	Slot Interrupt Status [07:00]
HOST_CNTRL_VERSION_07_00	16	31:16	RW	0x00	Host Controller Version [07:00]

8.1.32 CFG TUNINGCOUNT register (Offset Address - 0x300)

The Registers for Tuning is described below:

TABLE 33. CFG TUNINGCOUNT register

Fields	Width	Field bits	Software R/W	Reset Value	Description
CFG_TUNING_COUNT	6	5:0	RW	0x0	CFG Tuning count Based on this register the number of tuning counts will be decided
CFG_TUNING_1_5X_DIS	1	6	RW	0x0	CFG Tuning 1.5 times disable, 0 - It will perform 1.5 times of tuning count 1 - It will perform the number of times of tuning count

8.1.33 DRIVING SAMPLING EDGE register (Offset Address - 0x310)

The Data and cmd driving and sampling edge Register is described below:

TABLE 34. DRIVING SAMPLING EDGE register

Fields	Width	Field bits	Software R/W	Reset Value	Description
WR_DATA_START_DRIVING_EDGE	1	0	RW	0x0	Write data start bit driving edge 0 - Negedge, 1 - posedge for Normal mode
CMD_START_DRIVING_EDGE	1	1	RW	0x1	CMD start bit driving edge 0 - Negedge, 1 - posedge for Normal mode
RD_DATA_START_SAMPLING_EDGE	1	2	RW	0x1	Read data start sampling edge 0 - Negedge, 1 - posedge

Fields	Width	Field bits	Software R/W	Reset Value	Description
RESP_START_SAMPLING_EDGE	1	3	RW	0x1	Resp start sampling edge 0 - Negedge, 1 - posedge for and Normal mode

8.1.34 RD BLK EN register (Offset Address - 0x314)

The Enable register to process second block of read data, once DMA transaction of first block data gets completed, untill that emmc clock will not be driven. Enabled when Buffer size is 256. is described below:

TABLE 35. RD BLK EN register

Fields	Width	Field bits	Software R/W	Reset Value	Description
RD_BLK_EN	1	0:0	RW	0x1	Enable register to process second block of read data, once DMA transaction of first block data gets completed, untill that emmc clock will not be driven. Enabled when Buffer size is 256.

8.1.35 BEAT SELECT register (Offset Address - 0x318)

The Transaction Beat selector. 0 -> 1 beat transfer, 1 -> 4 beat transfer, 2 -> 8 beat transfer and 3 -> 16 beat transfer, 4 -> 32 beat transfer, 5-> 64 beat transfer, 6-> 128 beat transfer, 7-> 256 beat transfer is described below:

TABLE 36. BEAT SELECT register

Fields	Width	Field bits	Software R/W	Reset Value	Description
BEAT_SELECT	3	2:0	RW	0x0	Transaction Beat selector. 0 -> 1 beat transfer, 1 -> 4 beat transfer, 2 -> 8 beat transfer and 3 -> 16 beat transfer, 4 -> 32 beat transfer, 5-> 64 beat transfer, 6-> 128 beat transfer, 7-> 256 beat transfer,.

8.1.36 SOC MST TIMEOUT register (Offset Address - 0x31C)

The SOC Master Timeout register is described below:

TABLE 37. SOC MST TIMEOUT register

Fields	Width	Field bits	Software R/W	Reset Value	Description
SOC_MST_TIMEOUT	16	15:0	RW	0xFFF	SoC Master Timeout register

8.1.37 SOC TIMEOUT register (Offset Address - 0x1000)

The SOC timeout register is described below:

TABLE 38. SOC TIMEOUT register

Fields	Width	Field bits	Software R/W	Reset Value	Description
SOC_TIMEOUT	16	15:0	RW	0xF	SOC timeout register

9 C Driver Code

9.1 eMMC Host Methods

The eMMC/SDIO Host functionality is described in this section. Names are proposed along with parameters and return values.

sc_init ()

- In this method, all the following initialization will be done depending on the configuration. To make basic setting we call this method

write_byte_reg (p_addr_12b, p_data_8b)

- This method is used to update the required CSR register fields using slave interface
- The p_addr_12b parameter will be given with the address value of the CSR register to be configured
- The p_data_8b parameter will be given with the data value to be configured in the CSR

write_reg (p_addr_12b, p_data_32b)

- This method is used to update the required CSR register fields using slave interface
- The p_addr_12b parameter will be given with the address value of the CSR register to be configured
- The p_data_32b parameter will be given with the data value to be configured in the CSR address

read_reg (p_addr_12b)

- This method is used to read the data from the given CSR address
- The p_addr_12b parameter will be given with the address value of the CSR register to be read
- Return Value – The data from the given address field

init_chip (p_config_st)

- This method is used to initialize the IP.
- The config parameter points to the config object

irq_handler (p_cmd_st)

- This method is used to handle the interrupts generated by core
- The p_cmd_st parameter points to the command object

enable_chip (p_config_st,p_enable_b)

- This method is used to enable/disable chip
- The p_config_st points to the config structure

mask_irq (p_config_st,p_mask_32b)

- This method is used to mask the interrupt status bits
- The p_config_st points to the config structure
- The p_mask_32b is the mask value written to the Interrupt status registers

unmask_irq (p_config_st,p_mask_32b)

- This method is used to unmask the interrupt status bits
- The p_config_st points to the config structure
- The p_mask_32b is the unmask value written to the Interrupt status registers

reset_chip (p_config_st)

- This method is used to reset the whole chip
- The p_config_st points to the config structure

write_memory (p_addr_32b, p_data_32b)

- This method is used to Write data into memory
- The p_addr_32b parameter will be given with the address value
- The p_data_32b parameter will be given with the data value.

read_memory (p_addr_32b)

- This method is used to read the data from the given memory
- The p_addr_32b parameter will be given with the address value.
- Return Value(32bit) – The data from the given address field

print_config (p_config_st)

- This method is used to print all config registers
- p_config_st is a pointer to the config object

print_command (p_cmd_st)

- This method is used to print all the command structure fields
- p_cmd_st - Pointer to command object

test (p_cmd_st,p_config_st)

- The p_cmd_st parameter is used to point command object
- The p_config_st parameter is used to point config object

eMMC_handler

- eMMC_handler method is to show the status of IRQ status.
- This method is used to check the IRQ status bits.

send_command (p_cmd_st,p_cmd_arg_32b, p_rsp_type_2b, p_crc_chken_b, p_cidx_chken_b, p_cmd_type_2b)

- This method is used for sending command
- The p_cmd_st parameter is used to point command object
- The p_cmd_arg_32b parameter is used to write command argument for emmc command
- The p_rsp_type_2b parameter is used to write the response type of command
- The p_crc_chken_b parameter is used to enable the crc check
- The p_cidx_chken_b parameter is used command index check enable
- The p_cmd_type_2b parameter is used for command type

send_data_cmd (p_cmd_st, p_config_st, p_rw_b, p_data_2d_8b)

- This method is used for sending command with data
- The p_config_st is a pointer to the config object
- The p_cmd_st parameter is used to point command object
- The p_rw_b parameter is used to indicate write/read data transfer
- The p_data_2d_8b parameter is used for loading data

block_write (p_cmd_st, p_config_st, p_data_2d_8b)

- This method is used to write the block data
- The p_config_st is a pointer to the config object
- The p_cmd_st parameter is used to point command object
- The p_data_2d_8b parameter is used for loading data

block_read (p_cmd_st, p_config_st, p_data_2d_8b)

- This method is used to read the block data
- The p_cmd_st parameter is used to point command object
- The p_data_2d_8b parameter is used to for loading data

do_emmc_init (p_cmd_st, p_config_st)

- This method is used to do emmc initialization process
- The p_config_st is a pointer to the config object
- The p_cmd_st parameter is used to point command object

update_rsp_data (p_cmd_st)

- This method is used to update response data
- The p_cmd_st parameter is used to point command object

change_bus_speed (p_cmd_st, p_bus_speed_b)

- This method is used for setting default speed or high speed
- The p_cmd_st parameter is used to point command object
- The p_bus_speed_b parameter is used to setting default speed or high speed

change_bus_width (p_cmd_st, p_bus_width_b)

- This method is used for buswidth setting as 4bit or 8bit
- The p_cmd_st parameter is used to point command object
- The p_bus_width_b parameter is used to buswidth setting as 4bit or 8bit

9.2 Write operation

Methods	Description
write_blkcount_sdmasysaddr_csr	This is used to write SDIPS_EMMC_HOST_BLKCOUNT_SDMA SYSADDR
write_blkctrl_csr	This is used to write SDIPS_EMMC_HOST_BLKCTRL
write_argument_csr	This is used to write SDIPS_EMMC_HOST_ARGUMENT
write_data_port_csr	This is used to write SDIPS_EMMC_HOST_DATA_PORT
write_host_ctrl1_csr	This is used to write SDIPS_EMMC_HOST_HOST_CTRL1
write_intr_sts_en_csr	This is used to write SDIPS_EMMC_HOST_INTR_STS_EN
write_intr_sig_en_csr	This is used to write SDIPS_EMMC_HOST_INTR_SIG_EN
write_capabilities_0_csr	This is used to write SDIPS_EMMC_HOST_CAPABILITIES_0
write_capabilities_1_csr	This is used to write SDIPS_EMMC_HOST_CAPABILITIES_1
write_max_current_cap_0_csr	This is used to write SDIPS_EMMC_HOST_MAX_CURRENT_CAP_0
write_max_current_cap_1_csr	This is used to write SDIPS_EMMC_HOST_MAX_CURRENT_CAP_1
write_force_event_reg_csr	This is used to write SDIPS_EMMC_HOST_FORCE_EVENT_REG
write_adma_sys_addr1_csr	This is used to write SDIPS_EMMC_HOST_ADMA_SYS_ADDR1
write_preset_0_csr	This is used to write SDIPS_EMMC_HOST_PRESET_0
write_preset_1_csr	This is used to write SDIPS_EMMC_HOST_PRESET_1
write_preset_2_csr	This is used to write SDIPS_EMMC_HOST_PRESET_2
write_preset_3_csr	This is used to write SDIPS_EMMC_HOST_PRESET_3
write_preset_4_csr	This is used to write SDIPS_EMMC_HOST_PRESET_4
write_boot_timeout_cnt_csr	This is used to write SDIPS_EMMC_HOST_BOOT_TIMEOUT_CNT
write_vendor_specific_csr	This is used to write SDIPS_EMMC_HOST_VENDOR_SPECIFIC
write_slot_intr_sts_host_contro	This is used to write

Methods	Description
l_ver_csr	SDIPS_EMMC_HOST_SLOT_INTR_STS_HOST_CONTROL_VER
write_cfg_tuningcount_csr	This is used to write SDIPS_EMMC_HOST_CFG_TUNINGCOUNT
write_driving_sampling_edge_csr	This is used to write SDIPS_EMMC_HOST_DRIVING_SAMPLING_EDGE
write_rd_blk_en_csr	This is used to write SDIPS_EMMC_HOST_RD_BLK_EN
write_beat_select_csr	This is used to write SDIPS_EMMC_HOST_BEAT_SELECT
write_soc_mst_timeout_csr	This is used to write SDIPS_EMMC_HOST_SOC_MST_TIMEOUT
write_soc_timeout_csr	This is used to write SDIPS_EMMC_HOST_SOC_TIMEOUT

- The method test() is used to do write operation with data
- The method init_chip() is used to initialize the chip with the above configurations

9.3 Read operation

- The method test() is used to do Master read operation
- This method is used for read transfer of an instruction or data

10 Automotive Safety

This core employ fault tolerance design techniques, such as ECC SRAM protection, parity protection, and their deliverables include Safety Manual (SAM), a Failure Modes, Effects and Diagnostics Analysis (FMEDA).

SOC Bus (AHB/APB/AXI) are protected with parity on Data and address signals.

Internal FIFO's protected with Parity bits.

Internal RAM's are protected with ECC/Parity

This core also has failure detection at all FIFO's/RAM's/SOC Bus, these generate interrupt when failure is generated and CPU/Firmware can take the corrective action.

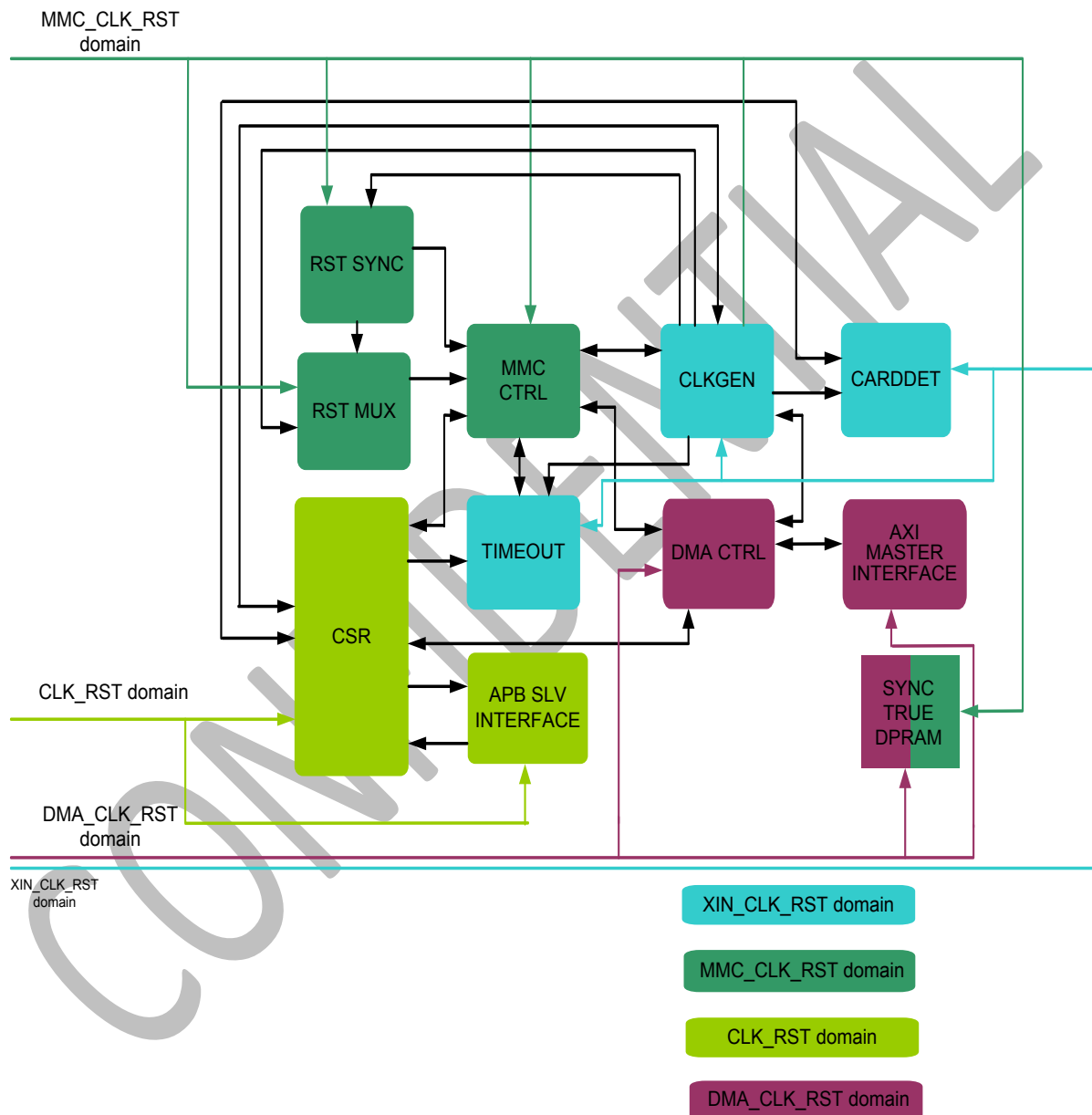
This core comes with testcases to simulate failure analysis.

This core achieves ASIL B and can be made to achieve ASIL D.

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11 Reset Diagram

Below figure shows the reset diagram of SDIO/eMMC host core

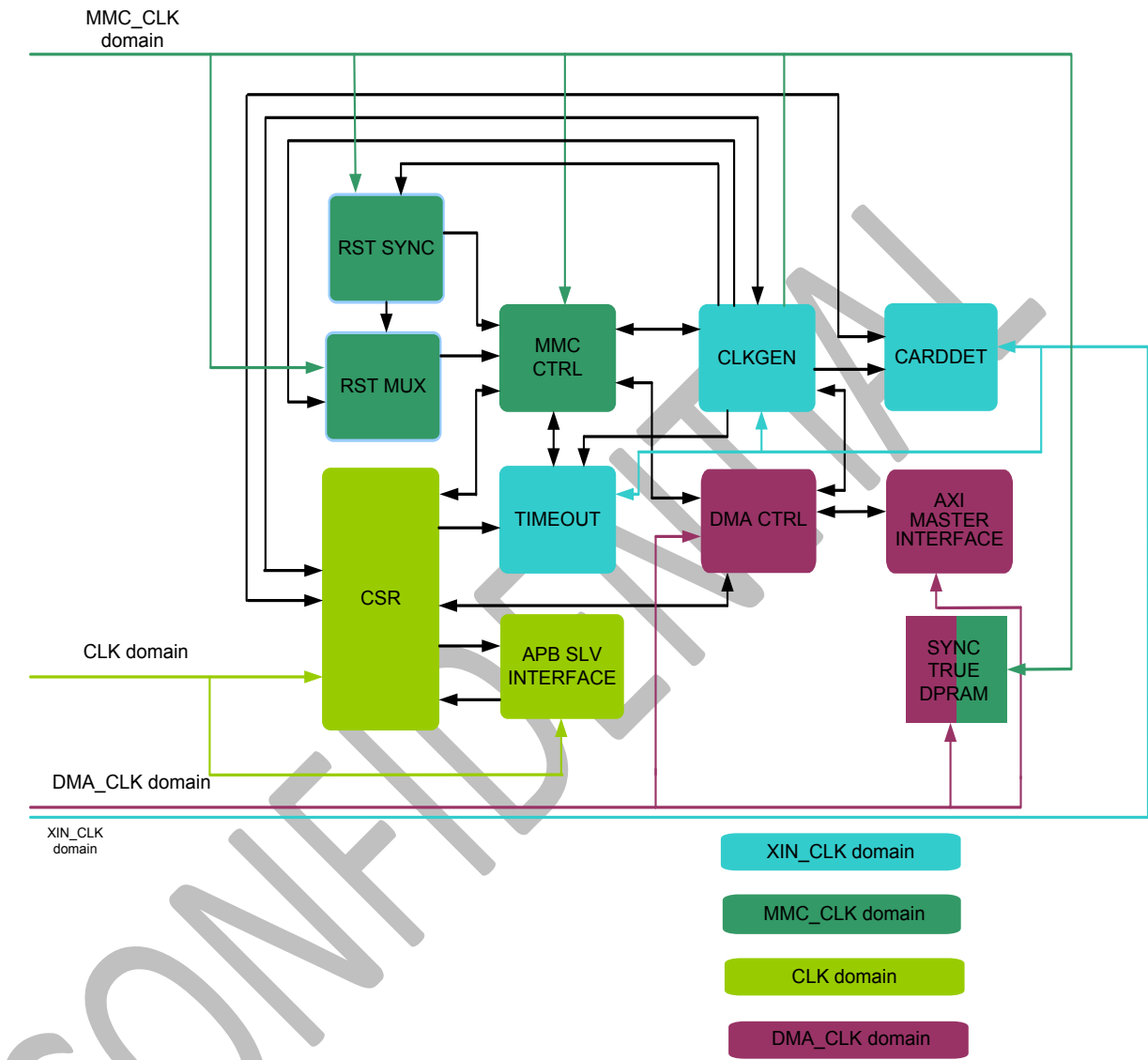


12 Clock Diagram

SDIO/eMMC Host core has 4 clock domains as shown in below figure.

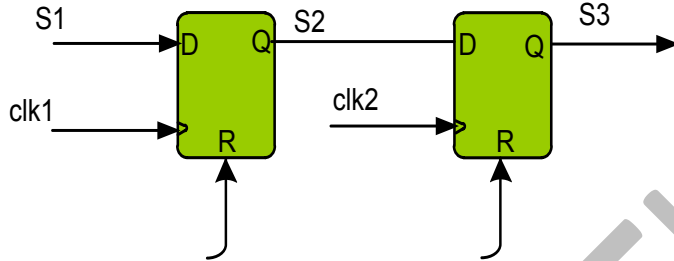
- **MMC CLOCK**
 - o_emmc_clk is Interface for CLK Output in MMC clock domain. emmc clock is generated by dividing the xin clock.
 - The o_emmc_clk will be generated depending on the divider value(L_SDCLK_FREQ_SELECT and U_SDCLK_FREQ_SELECT) of CLOCK_TIMEOUT_CONTROL register(offset - 0x2C)
 - To get "o_emmc_clk" as same as "i_xin_clk", then L_SDCLK_FREQ_SELECT and U_SDCLK_FREQ_SELECT of CLOCK_TIMEOUT_CONTROL register(offset - 0x2C) needs to be zero.
 - The "o_emmc_clk" is synchronous with "i_xin_clk".
 - The "o_emmc_clk" is asynchronous to "i_dma_clk, i_clk".
- **CLK CLOCK**
 - i_clk is Interface for System Clock. It is used for APB and CSR interface. The i_clk has the frequency range upto 200 MHz.
 - The i_clk" is asynchronous to "i_dma_clk, o_emmc_clk and i_xin_clk".
- **AXI CLOCK**
 - i_dma_clk is Interface for AXI master Clock domain. It is used for AXI and DMACTRL interface. The i_dma_clk has the frequency range upto 200 MHz.
 - The "i_dma_clk" is asynchronous to "o_emmc_clk, i_clk and i_xin_clk".
- **XIN CLOCK**
 - i_xin_clk is a base clock with range up to the 200Mhz. Based on this clock, "o_emmc_clk" will be generated.
 - The "o_emmc_clk" is synchronous to "i_xin_clk".
 - The "i_xin_clk" is asynchronous to "i_dma_clk, i_clk".

Below figure shows clock diagram of SDIO/eMMC host core



13 Clock Domain Crossing

Clock domain crossing occurs whenever data is transferred from a flop driven by one clock to a flop driven by another clock. The below diagram depicts a basic clock domain crossing scenario:

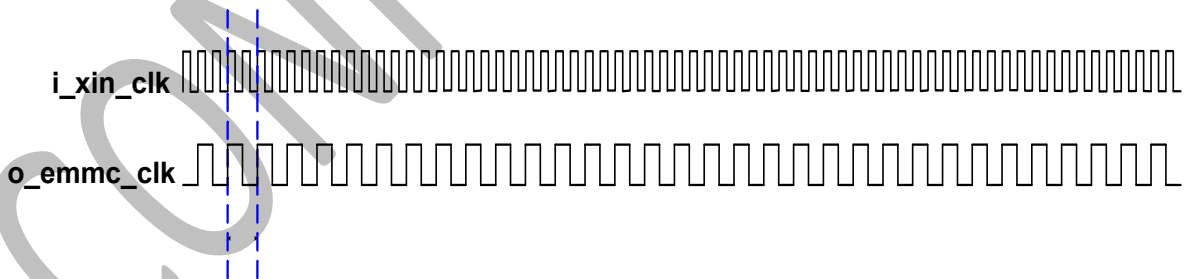


Here, the signal S2 launched by clk1 clock domain should be properly captured by clk2 clock domain. Thus, the data transfer from source clock to destination clock causes different types of problems and there exists multiple solutions to synchronize data between these clock domains.

13.1 Prescaled clock

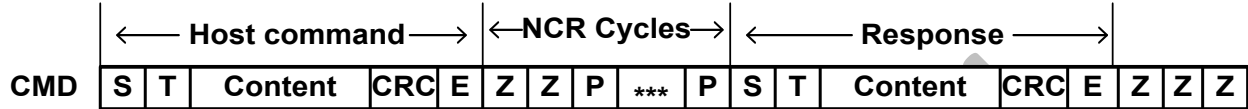
The o_emmc_clk frequency for transfer is generated by dividing the xin clock frequency by a prescaler value programmed. The prescaled clock will be in phase with the xin clock and it doesn't cause CDC violations.

An example waveform for prescaler value equals 1 is given below:



14 Timing Diagram

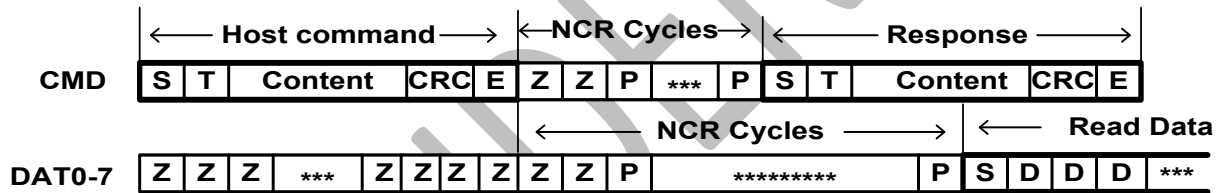
14.1 Command and response



Command response timing (data transfer mode)

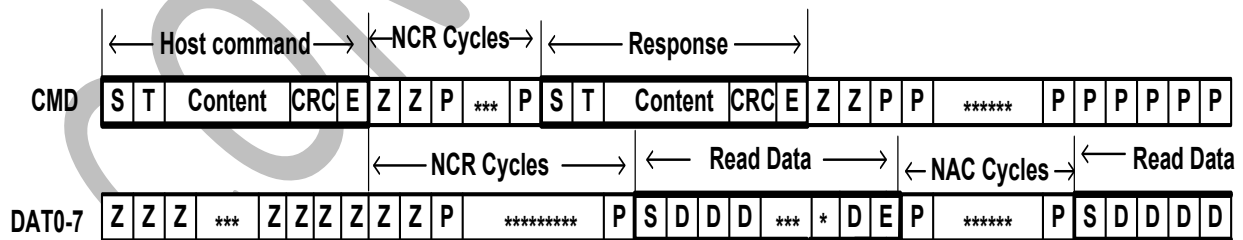
14.2 Data read

14.2.1 Single block read

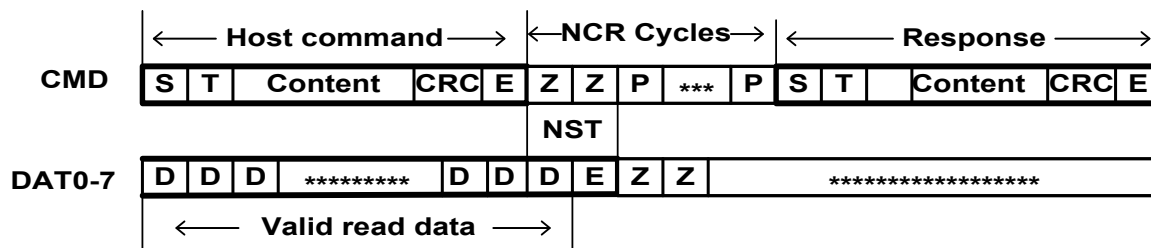


Single block read timing

14.2.2 Multiple block read



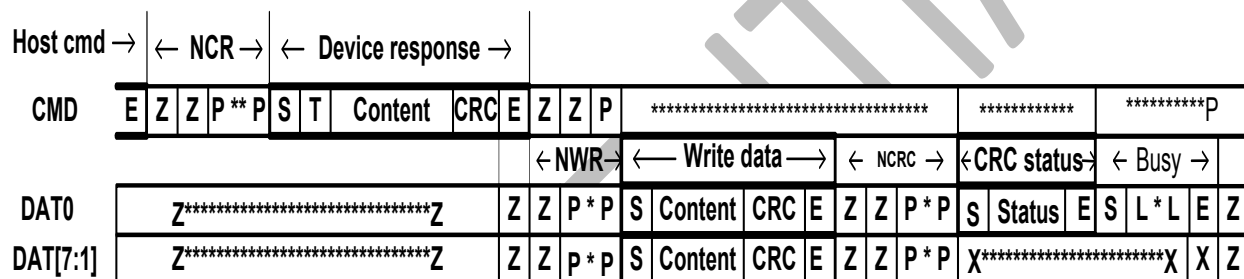
Multiple block read timing



Stop command timing (CMD12 data transfer mode)

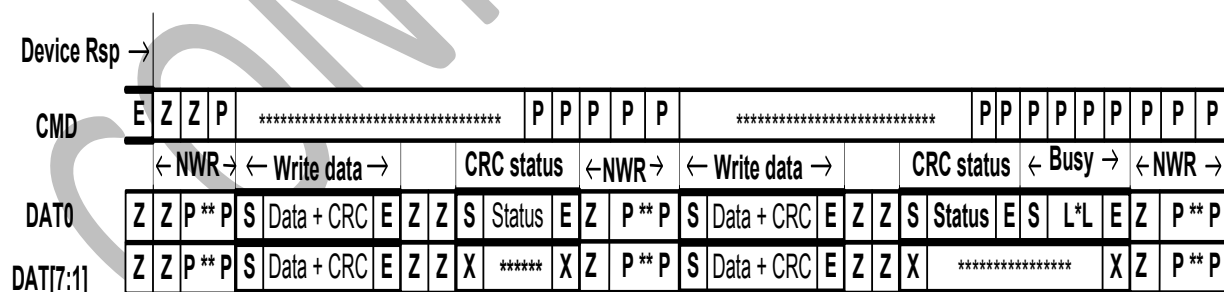
14.3 Data Write

14.3.1 Single block write



Block write command timing

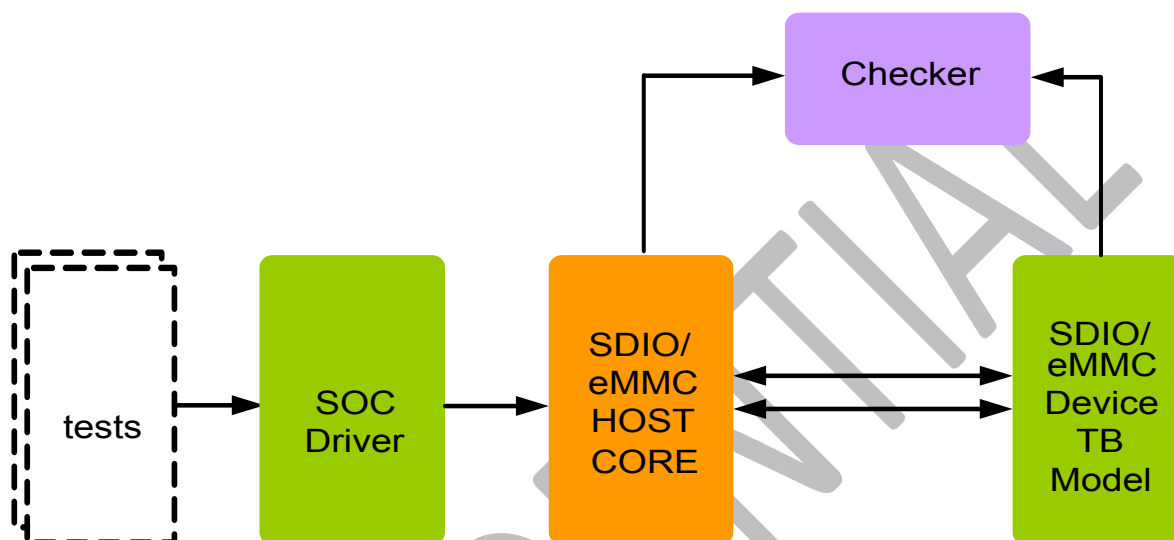
14.3.2 Multiple block write



Multiple block write timing

15 Test Environment

A basic test bench is provided with the IP to validate the functionality to help integrate with your SOC or ASIC.



15.1 Test plan

15.1.1 eMMC Test plan

Testcase	Description
test_basic	This tests all features of DUT
test_emmc_check_diff_sys_blk_addr_for_1bit_multiple_block_wr_rd	This tests case shows 1bit Single block write/read operation
test_emmc_check_diff_sys_blk_addr_for_1bit_single_block_wr_rd	This tests case shows 1bit Single block write/read operation
test_emmc_check_diff_sys_blk_addr_for_4bit_multiple_block_wr_rd	This tests case shows 4bit Single block write/read operation
test_emmc_check_diff_sys_blk_addr_for_4bit_single_block_wr_rd	This tests case shows 4bit Single block write/read operation
test_emmc_check_diff_sys_blk_addr_for_8bit_multiple_block_wr_rd	This tests case shows 8bit Single block write/read operation
test_emmc_check_diff_sys_blk_addr_for_8bit_single_block_wr_rd	This tests case shows 8bit Single block write/read operation
test_emmc_host_basic_cmd_negedge	This tests all features of DUT with command driving at negedge
test_emmc_host_basic_cmd_posedge	This tests all features of DUT with

Testcase	Description
	command driving at posedge
test_emmc_host_basic_data_negedge	This tests all features of DUT with data driving at negedge
test_emmc_host_basic_data_posedge	This tests all features of DUT with data driving at posedge
test_emmc_host_basic_read_data_sampling_negedge	This tests all features of DUT with read data sampling at negedge
test_emmc_host_basic_read_data_sampling_posedge	This tests all features of DUT with read data sampling at posedge
test_emmc_host_basic_resp_start_sampling_negedge	This tests all features of DUT with response sampling at negedge
test_emmc_host_basic_resp_start_sampling_posedge	This tests all features of DUT with response sampling at posedge
test_emmc_host_clk_update	This tests all features of DUT
test_emmc_host_ver_3_0_backward_speed_1bit_adma_cmd25_write_with_64bit_descriptor	This tests all features of DUT
test_emmc_host_ver_4_51_1bit_alt_boot_init	This testcase, test the eMMC Card alternate boot operation in 1bit bus width
test_emmc_host_ver_4_51_1bit_alt_boot_init_ack_not_en	This testcase, test the eMMC Card alternate boot operation in 1bit bus width
test_emmc_host_ver_4_51_1bit_normal_boot_init	This testcase, test the eMMC Card normal boot operation in 1bit bus width
test_emmc_host_ver_4_51_1bit_normal_boot_init_ack_not_en	This testcase, test the eMMC Card alternate boot operation in 1bit bus width
test_emmc_host_ver_4_51_4bit_alt_boot_init	This testcase, test the eMMC Card alternate boot operation in 4bit bus width
test_emmc_host_ver_4_51_4bit_alt_boot_init_ack_not_en	This testcase, test the eMMC Card alternate boot operation in 4bit bus width
test_emmc_host_ver_4_51_4bit_normal_boot_init	This testcase, test the eMMC Card normal boot operation in 4bit bus width
test_emmc_host_ver_4_51_4bit_normal_boot_init_ack_not_en	This testcase, test the eMMC Card normal boot operation in 4bit bus width
test_emmc_host_ver_4_51_8bit_alt_boot_init	This testcase, test the eMMC Card alternate boot operation in 8bit bus width
test_emmc_host_ver_4_51_8bit_alt_boot_init_ack_not_en	This testcase, test the eMMC Card alternate boot operation in 8bit bus width
test_emmc_host_ver_4_51_8bit_normal_boot_init	This testcase, test the eMMC Card normal boot operation in 8bit bus width
test_emmc_host_ver_4_51_8bit_normal_boot_init_ack_not_en	This testcase, test the eMMC Card normal boot operation in 8bit bus width
test_emmc_host_ver_4_51_adma2_1bit_multi_blk_wr_rd	This testcase, test the eMMC Card adma single block mode in 1bit bus width
test_emmc_host_ver_4_51_adma2_1bit_single_blk_wr	This testcase, test the eMMC Card adma2 single block mode in 1bit bus

Testcase	Description
	width
test_emmc_host_ver_4_51_adma2_1bit_single_blk_wr_rd	This testcase, test the eMMC Card adma2 single block mode in 1bit bus width
test_emmc_host_ver_4_51_adma2_4bit_multi_blk_wr_rd	This testcase, test the eMMC Card adma2 single block mode in 4bit bus width
test_emmc_host_ver_4_51_adma2_4bit_single_blk_wr	This testcase, test the eMMC Card adma2 single block mode in 4bit bus width
test_emmc_host_ver_4_51_adma2_4bit_single_blk_wr_rd	This testcase, test the eMMC Card adma2 single block mode in 1bit bus width
test_emmc_host_ver_4_51_adma2_8bit_multi_blk_wr_rd	This testcase, test the eMMC Card adma2 single block mode in 4bit bus width
test_emmc_host_ver_4_51_adma2_8bit_single_blk_wr	This testcase, test the eMMC Card adma2 single block mode in 8bit bus width
test_emmc_host_ver_4_51_adma2_8bit_single_blk_wr_rd	This testcase, test the eMMC Card adma2 single block mode in 1bit bus width
test_emmc_host_ver_4_51_backward_speed_1bit_cmd17_read	This testcase, test the eMMC Card CMD17 read operation in 1bit mode
test_emmc_host_ver_4_51_backward_speed_1bit_cmd18_read	This testcase, test the eMMC Card CMD18 read operation in 1bit mode
test_emmc_host_ver_4_51_backward_speed_1bit_cmd24_write	This testcase, test the eMMC Card write operation in 1bit mode
test_emmc_host_ver_4_51_backward_speed_1bit_cmd25_write	This testcase, test the eMMC Card write operation in 1bit mode
test_emmc_host_ver_4_51_backward_speed_4bit_cmd17_read	This testcase, test the eMMC Card CMD17 read operation in 4bit mode
test_emmc_host_ver_4_51_backward_speed_4bit_cmd18_read	This testcase, test the eMMC Card CMD18 read operation in 4bit mode
test_emmc_host_ver_4_51_backward_speed_4bit_cmd24_write	This testcase, test the eMMC Card write operation in 4bit mode
test_emmc_host_ver_4_51_backward_speed_4bit_cmd25_write	This testcase, test the eMMC Card write operation in 4bit mode
test_emmc_host_ver_4_51_backward_speed_8bit_cmd17_read	This testcase, test the eMMC Card CMD17 read operation in 8bit mode
test_emmc_host_ver_4_51_backward_speed_8bit_cmd18_read	This testcase, test the eMMC Card CMD18 read operation in 8bit mode
test_emmc_host_ver_4_51_backward_speed_8bit_cmd24_write	This testcase, test the eMMC Card write operation in 8bit mode
test_emmc_host_ver_4_51_backward_speed_8bit_cmd25_write	This testcase, test the eMMC Card write operation in 8bit mode
test_emmc_host_ver_4_51_high_speed_ddr	This testcase, test the eMMC Card read

Testcase	Description
_4bit_cmd17_read	operation in High speed DDR 4bit mode
test_emmc_host_ver_4_51_high_speed_ddr_4bit_cmd18_read	This testcase, test the eMMC Card read operation in High speed DDR 4bit mode
test_emmc_host_ver_4_51_high_speed_ddr_4bit_cmd24_write	This testcase, test the eMMC Card write operation in High speed DDR 4bit mode
test_emmc_host_ver_4_51_high_speed_ddr_4bit_cmd25_write	This testcase, test the eMMC Card write operation in 4bit mode high speed DDR
test_emmc_host_ver_4_51_high_speed_ddr_8bit_cmd17_read	This testcase, test the eMMC Card read operation in High speed DDR 8bit mode
test_emmc_host_ver_4_51_high_speed_ddr_8bit_cmd18_read	This testcase, test the eMMC Card read operation in High speed DDR 8bit mode
test_emmc_host_ver_4_51_high_speed_ddr_8bit_cmd24_write	This testcase, test the eMMC Card write operation in High speed DDR 8bit mode
test_emmc_host_ver_4_51_high_speed_ddr_8bit_cmd25_write	This testcase, test the eMMC Card write operation in 8bit mode high speed DDR
test_emmc_host_ver_4_51_high_speed_sdr_4bit_cmd17_read	This testcase, test the eMMC Card read operation in High speed SDR 4bit mode
test_emmc_host_ver_4_51_high_speed_sdr_4bit_cmd18_read	This testcase, test the eMMC Card read operation in High speed SDR 4bit mode
test_emmc_host_ver_4_51_high_speed_sdr_4bit_cmd24_write	This testcase, test the eMMC Card write operation in High speed SDR 4bit mode
test_emmc_host_ver_4_51_high_speed_sdr_4bit_cmd25_write	This testcase, test the eMMC Card write operation in 4bit mode high speed SDR
test_emmc_host_ver_4_51_high_speed_sdr_8bit_cmd17_read	This testcase, test the eMMC Card read operation in High speed SDR 8bit mode
test_emmc_host_ver_4_51_high_speed_sdr_8bit_cmd18_read	This testcase, test the eMMC Card read operation in High speed SDR 8bit mode
test_emmc_host_ver_4_51_high_speed_sdr_8bit_cmd24_write	This testcase, test the eMMC Card write operation in High speed SDR 8bit mode
test_emmc_host_ver_4_51_high_speed_sdr_8bit_cmd25_write	This testcase, test the eMMC Card write operation in 8bit mode high speed SDR
test_emmc_host_ver_4_51_hs200_speed_4bit_cmd17_read	This testcase, test the eMMC Card read operation in HS200 4bit mode
test_emmc_host_ver_4_51_hs200_speed_4bit_cmd18_read	This testcase, test the eMMC Card read operation in HS200 4bit mode
test_emmc_host_ver_4_51_hs200_speed_4bit_cmd24_write	This testcase, test the eMMC Card write operation in 4bit mode
test_emmc_host_ver_4_51_hs200_speed_4bit_cmd25_write	This testcase, test the eMMC Card write operation in 4bit mode
test_emmc_host_ver_4_51_hs200_speed_8bit_cmd17_read	This testcase, test the eMMC Card read operation in HS200 Speed 8bit mode
test_emmc_host_ver_4_51_hs200_speed_8bit_cmd18_read	This testcase, test the eMMC Card read operation in HS200 8bit mode
test_emmc_host_ver_4_51_hs200_speed_8bit_cmd24_write	This testcase, test the eMMC Card write operation in 8bit mode

Testcase	Description
test_emmc_host_ver_4_51_hs200_speed_8bit_cmd25_write	This testcase, test the eMMC Card write operation in 4bit mode
test_emmc_host_ver_4_51_non_dma_1bit_multi_blk_wr_rd	This testcase, test the eMMC Card non-dma multi block mode in 1bit bus width
test_emmc_host_ver_4_51_non_dma_1bit_single_blk_wr_rd	This testcase, test the eMMC Card non-dma single block mode in 1bit bus width
test_emmc_host_ver_4_51_non_dma_4bit_multi_blk_wr_rd	This testcase, test the eMMC Card non-dma multi block mode in 4bit bus width
test_emmc_host_ver_4_51_non_dma_4bit_single_blk_wr_rd	This testcase, test the eMMC Card non-dma single block mode in 4bit bus width
test_emmc_host_ver_4_51_non_dma_8bit_multi_blk_wr_rd	This testcase, test the eMMC Card non-dma multi block mode in 8bit bus width
test_emmc_host_ver_4_51_non_dma_8bit_single_blk_wr_rd	This testcase, test the eMMC Card non-dma single block mode in 8bit bus width
test_emmc_sdma_def_speed_1bit_multi_block_wr_rd_random_block_size	This testcase, test the eMMC Card multi block mode in 1bit bus width of random block size with ahb single beat transaction
test_emmc_sdma_def_speed_1bit_multi_block_wr_rd_random_block_size_ahb_128beat	This testcase, test the eMMC Card multi block mode in 1bit bus width of random block size with ahb maximum 128 beat transaction
test_emmc_sdma_def_speed_1bit_multi_block_wr_rd_random_block_size_ahb_16beat	This testcase, test the eMMC Card multi block mode in 1bit bus width of random block size with ahb maximum 16 beat transaction
test_emmc_sdma_def_speed_1bit_multi_block_wr_rd_random_block_size_ahb_256beat	This testcase, test the eMMC Card multi block mode in 1bit bus width of random block size with ahb maximum 256 beat transaction
test_emmc_sdma_def_speed_1bit_multi_block_wr_rd_random_block_size_ahb_32beat	This testcase, test the eMMC Card multi block mode in 1bit bus width of random block size with ahb maximum 32 beat transaction
test_emmc_sdma_def_speed_1bit_multi_block_wr_rd_random_block_size_ahb_4beat	This testcase, test the eMMC Card multi block mode in 1bit bus width of random block size with ahb maximum 4 beat beat transaction
test_emmc_sdma_def_speed_1bit_multi_block_wr_rd_random_block_size_ahb_64beat	This testcase, test the eMMC Card multi block mode in 1bit bus width of random block size with ahb maximum 64 beat transaction
test_emmc_sdma_def_speed_1bit_multi_block_wr_rd_random_block_size_ahb_8beat	This testcase, test the eMMC Card multi block mode in 1bit bus width of random block size with ahb maximum 8 beat

Testcase	Description
	transaction
test_emmc_sdma_def_speed_1bit_multiple_block_wr_rd	This testcase, tests the eMMC SDMA 1bit Multiple block write/read operation with default speed
test_emmc_sdma_def_speed_1bit_single_block_wr_rd	This testcase, tests the eMMC SDMA 1bit Single block write/read operation with default speed
test_emmc_sdma_def_speed_4bit_multiple_block_wr_rd	This testcase, tests the eMMC SDMA 4bit Multiple block write/read operation with default speed
test_emmc_sdma_def_speed_4bit_single_block_wr_rd	This testcase, tests the eMMC SDMA 4bit Single block write/read operation with default speed
test_emmc_sdma_def_speed_8bit_multiple_block_wr_rd	This testcase, tests the eMMC SDMA 8bit Multiple block write/read operation with default speed
test_emmc_sdma_def_speed_8bit_single_block_wr_rd	This testcase, tests the eMMC SDMA 8bit Single block write/read operation with default speed
test_emmc_ver_4_51_1bit_alt_boot_ack_err	This testcase, test the eMMC Card alternate boot ACK error operation
test_emmc_ver_4_51_1bit_alt_boot_end_bit_err	This testcase, test the eMMC Card alternate boot End bit error operation
test_emmc_ver_4_51_1bit_cmd17_data_crc_err	This tests case shows 1bit Single block write/read operation
test_emmc_ver_4_51_1bit_cmd6_crc_err	This testcase, test the 1bit CMD6 CRC error
test_emmc_ver_4_51_1bit_normal_boot_init_ack_end_bit_err	This testcase, test the eMMC Card normal boot operation in 1bit bus width
test_emmc_ver_4_51_backward_speed_1bit_cmd18_read_with_cmd23	This tests case shows 1bit Multiple block write/read operation
test_emmc_ver_4_51_backward_speed_1bit_cmd25_write_with_cmd23	This tests case shows 1bit Single block write/read operation
test_emmc_ver_4_51_backward_speed_4bit_cmd18_read_with_cmd23	This tests case shows 4bit Multiple block write/read operation
test_emmc_ver_4_51_backward_speed_4bit_cmd25_write_with_cmd23	This tests case shows 4bit Single block write/read operation
test_emmc_ver_4_51_backward_speed_8bit_cmd18_read_with_cmd23	This tests case shows 8bit Multiple block write/read operation
test_emmc_ver_4_51_backward_speed_8bit_cmd25_write_with_cmd23	This tests case shows 8bit Single block write/read operation
test_emmc_ver_4_51_cmd18_read_data_timeout_error	This tests case shows 1bit Single block write/read operation
test_emmc_ver_4_51_cmd18_read_end_bit_error	This tests case shows 1bit Single block write/read operation

Testcase	Description
test_emmc_ver_4_51_cmd6_command_index_error	This testcase, test the 1bit CMD6 Index error
test_emmc_ver_4_51_cmd6_command_timeout_error	This testcase, test the 1bit CMD6 Index error
test_emmc_ver_4_51_cmd6_resp_end_error	This testcase, test the 1bit CMD6 Index error
test_emmc_ver_4_51_high_speed_4bit_cmd25_write_with_cmd23	This tests case shows high speed 4bit Multiple block write operation with CMD23
test_emmc_ver_4_51_high_speed_8bit_cmd25_write_with_cmd23	This tests case shows high speed 8bit Multiple block write operation with CMD23
test_emmc_ver_4_51_high_speed_sdr_4bit_cmd18_read_with_cmd23	This tests case shows 4bit Multiple block write/read operation with CMD23
test_emmc_ver_4_51_high_speed_sdr_8bit_cmd18_read_with_cmd23	This tests case shows high speed sdr 8bit Multiple block write/read operation with CMD 23
test_emmc_ver_4_51_hs200_speed_4bit_cmd18_read_with_cmd23	This tests case shows hs 200 4bit Multiple block write/read operation with CMD23
test_emmc_ver_4_51_hs200_speed_4bit_cmd24_write_with_cmd21_tuning_block_emmc	This testcase, test the eMMC Card write operation in 4bit mode
test_emmc_ver_4_51_hs200_speed_4bit_cmd25_write_with_cmd23	This tests case shows hs200 4bit Multiple block write operation with CMD23
test_emmc_ver_4_51_hs200_speed_8bit_cmd18_read_with_cmd23	This tests case shows hs 200 8bit Multiple block write/read operation with CMD23
test_emmc_ver_4_51_hs200_speed_8bit_cmd24_write_with_cmd21_tuning_block_emmc	This testcase, test the eMMC Card write operation in 8bit mode
test_emmc_ver_4_51_hs200_speed_8bit_cmd25_write_with_cmd23	This tests case shows high speed 8bit Multiple block write operation with CMD23

15.1.2 SDIO Test plan

Testcase	Description
test_basic_sdio	This tests all features of DUT
test_basic_sdmem	This tests all features of SDMEM card

Testcase	Description
test_sdio_1bit_cmd53_wr_rd_incr_addr_intr	This test case is used to test SDIO card interrupt during interrupt period for 1 bit
test_sdio_1bit_cmd53_wr_rd_incr_addr_read_wait	This test case is used to test SDIO card UHS-0 Default speed 1 bit, command 53 block mode Read/Write operation with address increment operation with read wait control
test_sdio_4bit_cmd53_rd_incr_addr_intr	This test case is used to test SDIO card interrupt during interrupt period with CMD 53 read
test_sdio_4bit_cmd53_wr_incr_addr_intr	This test case is used to test SDIO card interrupt during interrupt period with CMD 53 write
test_sdio_4bit_cmd53_wr_rd_incr_addr_async_intr	This test case is used to test SDIO card Async interrupt during interrupt period
test_sdio_4bit_cmd53_wr_rd_incr_addr_read_wait	This test case is used to test SDIO card UHS-0 Default speed 4 bit, command 53 block mode Read/Write operation with address increment operation with read wait control
test_sdio_cmd53_wr_rd_incr_addr_intr	This test case is used to test SDIO card interrupt during interrupt period
test_sdio_csr_write_read_random	This testcase, perform csr register write read operation with random values for coverage purpose
test_sdio_sdma_def_speed_1bit_multiple_block_wr_rd	This testcase, tests the SDIO SDMA 1bit Multiple block write/read operation with default speed
test_sdio_sdma_def_speed_1bit_single_block_wr_rd	This testcase, tests the SDIO SDMA 1bit Single block write/read operation with default speed
test_sdio_sdma_def_speed_4bit_multiple_block_wr_rd	This testcase, tests the SDIO SDMA 4bit Multiple block write/read operation with default speed
test_sdio_sdma_def_speed_4bit_single_block_wr_rd	This testcase, tests the SDIO SDMA 4bit Single block write/read operation with default speed
test_sdio_uhs0_4bit_cmd53_data_crc_err	This test case is used to test SDIO card UHS-0, 4 bit, command 53 block mode Read/Write operation with address increment operation with DATA CRC error

Testcase	Description
test_sdio_uhs0_def_speed_1bit_cmd53_blk_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-0 Default speed 1 bit, command 53 block mode Read/Write operation with address increment operation
test_sdio_uhs0_def_speed_1bit_cmd53_byte_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-0 Default speed 1 bit, command 53 byte mode Read/Write operation with address increment operation
test_sdio_uhs0_def_speed_1bit_cmd53_infinite_blk_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-0 Default speed 1 bit, command 53 infinite block mode Read/Write operation with address increment operation
test_sdio_uhs0_def_speed_1bit_csa_cmd53_blk_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-0 Default speed 1 bit, CSA command 53 multi block mode Read/Write operation with address increment operation
test_sdio_uhs0_def_speed_1bit_csa_cmd53_byte_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-0 Default speed 1 bit, CSA command 53 byte mode Read/Write operation with address increment operation
test_sdio_uhs0_def_speed_4bit_cmd53_blk_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-0 Default speed 4 bit, command 53 block mode Read/Write operation with address increment operation
test_sdio_uhs0_def_speed_4bit_cmd53_blk_mode_wr_with_auto_cmd12_with_data_crc_error	This test case is used to test SDIO card UHS-0 Default speed 4 bit, command 53 block mode Write operation with address increment operation with auto cmd 12 enable with data crc error
test_sdio_uhs0_def_speed_4bit_cmd53_blk_mode_wr_with_data_crc_error	This test case is used to test SDIO card UHS-0 Default speed 4 bit, command 53 block mode Write operation with address increment operation with data crc error
test_sdio_uhs0_def_speed_4bit_cmd53_blk_mode_wr_with_data_timeout_error	This test case is used to test SDIO card UHS-0 Default speed 4 bit, command 53 block mode Write

Testcase	Description
	operation with address increment operation with data crc error
test_sdio_uhs0_def_speed_4bit_cmd53_blk_mode_wr_with_stop_at_blockgap_with_data_crc_error	This test case is used to test SDIO card UHS-0 Default speed 4 bit, command 53 block mode Write operation with address increment operation with stop at block gap with data crc error
test_sdio_uhs0_def_speed_4bit_cmd53_byte_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-0 Default speed 4 bit, command 53 byte mode Read/Write operation with address increment operation
test_sdio_uhs0_def_speed_4bit_cmd53_infinite_blk_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-0 Default speed 4 bit, command 53 infinite block mode Read/Write operation with address increment operation
test_sdio_uhs0_def_speed_4bit_csa_cmd53_blk_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-0 Default speed 4 bit, CSA command 53 multi block mode Read/Write operation with address increment operation
test_sdio_uhs0_def_speed_4bit_csa_cmd53_byte_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-0 Default speed 4 bit, CSA command 53 byte mode Read/Write operation with address increment operation
test_sdio_uhs0_def_speed_4bit_func1_rd_suspend_func2_rd_func1_rd_cycle_resume_func1	This test case is used to test SDIO card UHS-0 Default speed 4 bit, command 53 block mode Read operation with suspend/resume operation, function 1 read suspend function 2 read during function 1 read cycle resume function 1 read operation
test_sdio_uhs0_def_speed_4bit_rd_suspend_resume	This test case is used to test SDIO card UHS-0 Default speed 4 bit, command 53 block mode Read operation with suspend/resume operation
test_sdio_uhs0_def_speed_4bit_wr_suspend_during_busy_resume	This test case is used to test SDIO card UHS-0 Default speed 4 bit, command 53 block mode write

Testcase	Description
	operation with suspend/resume operation
test_sdio_uhs0_def_speed_cmd3_crc_err	This test case is used to test SDIO card UHS-0, 4 bit, SEND_RELATIVE_ADDR(CMD3) CRC error
test_sdio_uhs0_def_speed_cmd53_data_timeout_err	This test case is used to test SDIO card UHS-0, 4 bit, command 53 block mode Read/Write operation with address increment operation with DATA timeout error
test_sdio_uhs0_def_speed_cmd7_command_timeout_err	This test case is used to test SDIO card UHS-0, 4 bit, SELECT_DESELECT_CARD(CMD7) C
test_sdio_uhs0_def_speed_cmd8_crc_err	This test case is used to test SDIO card UHS-0, 4 bit, SEND_IF_COND(CMD8) CRC error
test_sdio_uhs0_def_speed_non_dma_1bit_block_mode_wr_rd	This test case is used to test SDIO card UHS-0 Default speed 1 bit, command 53 block mode Read/Write operation non DMA
test_sdio_uhs0_def_speed_non_dma_4bit_block_mode_wr_rd	This test case is used to test SDIO card UHS-0 Default speed 4 bit, command 53 block mode Read/Write operation non DMA
test_sdio_uhs0_high_speed_1bit_cmd53_blk_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-0 High speed 1 bit, command 53 block mode Read/Write operation with address increment operation
test_sdio_uhs0_high_speed_1bit_cmd53_byte_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-0 High speed 1 bit, command 53 byte mode Read/Write operation with address increment operation
test_sdio_uhs0_high_speed_4bit_cmd53_blk_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-0 High speed 4 bit, command 53 block mode Read/Write operation with address increment operation
test_sdio_uhs0_high_speed_4bit_cmd53_byte_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-0 High speed 4 bit, command 53 byte mode Read/Write

Testcase	Description
	operation with address increment operation
test_sdio_uhs1_ddr50_cmd53_blk_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-1 DDR 50, 4 bit, command 53 block mode Read/Write operation with address increment operation
test_sdio_uhs1_ddr50_cmd53_byte_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-1 DDR 50, 4 bit, command 53 byte mode Read/Write operation with address increment operation
test_sdio_uhs1_sdr104_cmd53_blk_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-1 SDR 104, 4 bit, command 53 block mode Read/Write operation with address increment operation
test_sdio_uhs1_sdr104_cmd53_byte_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-1 SDR 104, 4 bit, command 53 byte mode Read/Write operation with address increment operation
test_sdio_uhs1_sdr12_4bit_cmd53_data_crc_err	This test case is used to test SDIO card UHS-1 SDR 12, 4 bit, command 53 block mode Read/Write operation with address increment operation with DATA CRC error
test_sdio_uhs1_sdr12_cmd53_blk_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-1 SDR 12, 4 bit, command 53 block mode Read/Write operation with address increment operation
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-1 SDR 12, 4 bit, command 53 byte mode Read/Write operation with address increment operation
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_0_addr_lsb_0	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 0bit and address lsb bit is zero.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_0_addr_lsb_1	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 0bit and address lsb bit is One.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr	This test case is used to test sdio card

Testcase	Description
_rd_incr_addr_byte_lsb_0_addr_lsb_2	UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 0bit and address lsb bit is zero.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_0_addr_lsb_3	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 0bit and address lsb bit is Three.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_0_addr_lsb_4	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 0bit and address lsb bit is four.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_0_addr_lsb_5	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 0bit and address lsb bit is five.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_0_addr_lsb_6	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 0bit and address lsb bit is six.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_0_addr_lsb_7	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 0bit and address lsb bit is seven.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_1_addr_lsb_0	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 1bit and address lsb bit is zero.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_1_addr_lsb_1	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with

Testcase	Description
	byte lsb is 1bit and address lsb bit is One.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_1_addr_lsb_2	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 1bit and address lsb bit is Two.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_1_addr_lsb_3	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 1bit and address lsb bit is three.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_1_addr_lsb_4	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 1bit and address lsb bit is four.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_1_addr_lsb_5	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 1bit and address lsb bit is five.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_1_addr_lsb_6	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 1bit and address lsb bit is six.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_1_addr_lsb_7	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 1bit and address lsb bit is seven.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_2_addr_lsb_0	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 2bit and address lsb bit is zero.

Testcase	Description
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_2_addr_lsb_1	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 2bit and address lsb bit is One.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_2_addr_lsb_2	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 2bit and address lsb bit is Two.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_2_addr_lsb_3	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 2bit and address lsb bit is three.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_2_addr_lsb_4	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 2bit and address lsb bit is four.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_2_addr_lsb_5	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 2bit and address lsb bit is five.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_2_addr_lsb_6	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 2bit and address lsb bit is six.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_2_addr_lsb_7	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 2bit and address lsb bit is seven.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_3_addr_lsb_0	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with

Testcase	Description
	address0 increment operation with byte lsb is 3bit and address lsb bit is zero.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_3_addr_lsb_1	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 3bit and address lsb bit is One.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_3_addr_lsb_2	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 3bit and address lsb bit is Two.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_3_addr_lsb_3	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 3bit and address lsb bit is three.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_3_addr_lsb_4	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 3bit and address lsb bit is four.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_3_addr_lsb_5	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 3bit and address lsb bit is five.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_3_addr_lsb_6	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 3bit and address lsb bit is six.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_3_addr_lsb_7	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 3bit and address lsb bit is seven.

Testcase	Description
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_4_addr_lsb_0	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 4bit and address lsb bit is zero.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_4_addr_lsb_1	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 4bit and address lsb bit is One.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_4_addr_lsb_2	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 4bit and address lsb bit is Two.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_4_addr_lsb_3	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 4bit and address lsb bit is three.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_4_addr_lsb_4	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 4bit and address lsb bit is four.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_4_addr_lsb_5	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 4bit and address lsb bit is five.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_4_addr_lsb_6	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 4bit and address lsb bit is six.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_4_addr_lsb_7	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with

Testcase	Description
	address0 increment operation with byte lsb is 4bit and address lsb bit is seven.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_5_addr_lsb_0	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 4bit and address lsb bit is zero.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_5_addr_lsb_1	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 5bit and address lsb bit is One.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_5_addr_lsb_2	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 5bit and address lsb bit is Two.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_5_addr_lsb_3	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 5bit and address lsb bit is three.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_5_addr_lsb_4	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 5bit and address lsb bit is four.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_5_addr_lsb_5	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 5bit and address lsb bit is five.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_5_addr_lsb_6	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 5bit and address lsb bit is six.

Testcase	Description
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_5_addr_lsb_7	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 5bit and address lsb bit is seven.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_6_addr_lsb_0	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 4bit and address lsb bit is zero.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_6_addr_lsb_1	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 6bit and address lsb bit is One.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_6_addr_lsb_2	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 6bit and address lsb bit is Two.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_6_addr_lsb_3	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 6bit and address lsb bit is three.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_6_addr_lsb_4	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 6bit and address lsb bit is four.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_6_addr_lsb_5	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 6bit and address lsb bit is five.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_6_addr_lsb_6	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with

Testcase	Description
	address0 increment operation with byte lsb is 6bit and address lsb bit is six.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_6_addr_lsb_7	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 7bit and address lsb bit is seven.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_7_addr_lsb_0	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 7bit and address lsb bit is zero.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_7_addr_lsb_1	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 7bit and address lsb bit is One.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_7_addr_lsb_2	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 7bit and address lsb bit is Two.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_7_addr_lsb_3	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 7bit and address lsb bit is three.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_7_addr_lsb_4	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 6bit and address lsb bit is four.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_7_addr_lsb_5	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 7bit and address lsb bit is five.

Testcase	Description
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_7_addr_lsb_6	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 7bit and address lsb bit is six.
test_sdio_uhs1_sdr12_cmd53_byte_mode_wr_rd_incr_addr_byte_lsb_7_addr_lsb_7	This test case is used to test sdio card UHS-1 SDR 12, 4 bit, command 53 byte0 mode Read/Write operation with address0 increment operation with byte lsb is 7bit and address lsb bit is seven.
test_sdio_uhs1_sdr12_def_speed_cmd3_crc_err	This test case is used to test SDIO card UHS-1 SDR 12, 4 bit, SEND_RELATIVE_ADDR(CMD3) CRC error
test_sdio_uhs1_sdr12_def_speed_cmd53_data_timeout_err	This test case is used to test SDIO card UHS-1 SDR 12, 4 bit, command 53 block mode Read/Write operation with address increment operation with DATA timeout error
test_sdio_uhs1_sdr12_def_speed_cmd7_command_timeout_err	This test case is used to test SDIO card UHS-1 SDR 12, 4 bit, SELECT_DESELECT_CARD(CMD7) C
test_sdio_uhs1_sdr12_def_speed_cmd8_crc_err	This test case is used to test SDIO card UHS-1 SDR 12, 4 bit, SEND_IF_COND(CMD8) CRC error
test_sdio_uhs1_sdr25_cmd53_blk_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-1 SDR 25, 4 bit, command 53 block mode Read/Write operation with address increment operation
test_sdio_uhs1_sdr25_cmd53_byte_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-1 SDR 25, 4 bit, command 53 byte mode Read/Write operation with address increment operation
test_sdio_uhs1_sdr50_cmd53_blk_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-1 SDR 50, 4 bit, command 53 block mode Read/Write operation with address increment operation
test_sdio_uhs1_sdr50_cmd53_byte_mode_wr_rd_incr_addr	This test case is used to test SDIO card UHS-1 SDR 50, 4 bit, command 53 byte mode Read/Write operation with address increment operation

Testcase	Description
test_sdio_v2_0_1bit_multi_blk_wr_rd_with_read_abort	This test case is used to test SDIO card with abort in multi-block read function
test_sdio_v2_0_adma2_1bit_multi_blk_wr	This testcase, tests the SDIO ADMA2 1bit multi block write operation
test_sdio_v2_0_adma2_1bit_multi_blk_wr_rd	This testcase, tests the SDIO ADMA 2 ,1bit Single block write/read operation with single blk wr
test_sdio_v2_0_adma2_1bit_single_blk_wr	This testcase, tests the SDIO ADMA 2 ,1bit Single block write/read operation with single blk wr
test_sdio_v2_0_adma2_1bit_single_blk_wr_rd	This testcase, tests the SDIO ADMA 2 ,1bit Single block write/read operation with single blk wr
test_sdio_v2_0_adma2_4bit_multi_blk_wr	This testcase, tests the SDIO ADMA2 4bit multi block write operation
test_sdio_v2_0_adma2_4bit_multi_blk_wr_rd	This testcase, tests the SDIO ADMA 2 ,1bit Single block write/read operation with single blk wr
test_sdio_v2_0_adma2_4bit_single_blk_wr	This testcase, tests the SDIO ADMA 4bit Single block write/read operation with single blk wr
test_sdio_v2_0_adma2_4bit_single_blk_wr_rd	This testcase, tests the SDIO ADMA 2 ,1bit Single block write/read operation with single blk wr
test_sdio_v2_0_adma2_8bit_multi_blk_wr	This testcase, tests the SDIO ADMA2 4bit multi block write operation
test_sdio_v2_0_adma2_8bit_multi_blk_wr_rd	This testcase, tests the SDIO ADMA 2 ,1bit Single block write/read operation with single blk wr
test_sdio_v2_0_adma2_8bit_single_blk_wr	This testcase, tests the SDIO ADMA 8bit Single block write/read operation with single blk wr
test_sdio_v2_0_adma2_8bit_single_blk_wr_rd	This testcase, tests the SDIO ADMA 2 ,1bit Single block write/read operation with single blk wr
test_sdio_v2_0_non_dma_1bit_multi_blk_wr_rd	This test case,tests SDIO card 1bit Multiple block writeoperation
test_sdio_v2_0_non_dma_1bit_single_blk_wr_rd	This testcase, tests case for the Non DMA mode
test_sdio_v2_0_non_dma_4bit_multi_blk_wr_rd	This test case,tests SDIO card 4bit Multiple block writeoperation
test_sdio_v2_0_non_dma_4bit_single_blk_wr_rd	This testcase, tests case for the Non DMA mode

Testcase	Description
test_sdio_v2_0_non_dma_8bit_multi_blk_wr_rd	This test case,tests SDIO card 4bit Multiple block writeoperation
test_sdio_v2_0_non_dma_8bit_single_blk_wr_rd	This testcase, tests case for the Non DMA mode
test_sdio_v2_spi_blk_mode_wr_incr_addr	This test case,for DMA mode spi mode multi blk write increment address
test_sdio_v2_spi_blk_mode_wr_incr_addr_cmd12	This test case,for DMA mode spi mode multi blk write increment address with cmd12
test_sdio_v2_spi_blk_mode_wr_incr_addr_cmd23	This test case,for DMA mode spi mode multi blk write increment address with cmd23
test_sdio_v2_spi_byte_mode_wr_incr_addr	This test case,for DMA mode spi mode byte mode write increment address
test_sdio_v_2_spi_pio_multi_blk_wr_rd	This test case,for PIO mode , multi block SPI scenario write and read
test_sdio_v_2_spi_pio_multi_blk_wr_rd_cmd12	This test case,for PIO mode , multi block SPI scenario write and read with cmd23 operation
test_sdio_v_2_spi_pio_multi_blk_wr_rd_cmd23	This test case,for PIO mode , multi block SPI scenario write and read with cmd23 operation
test_sdio_v_2_spi_pio_single_blk_wr	This test case,for PIO mode , single block write transaction
test_sdio_v_2_spi_pio_single_blk_wr_rd	This test case,for PIO mode , SPI scenario write and read
test_sdmem_uhs0_def_speed_1bit_multiple_block_wr_rd	This test case, tests SDMEM card 1bit Multiple block write/read operation with default speed
test_sdmem_uhs0_def_speed_1bit_single_block_wr_rd	This test case, tests SDMEM card 1bit Single block write/read operation with default speed
test_sdmem_uhs0_def_speed_4bit_multiple_block_wr_rd	This test case, tests SDMEM card 4bit Multiple block write/read operation with default speed
test_sdmem_uhs0_def_speed_4bit_single_block_wr_rd	This test case, tests SDMEM card 4bit Single block write/read operation with default speed
test_sdmem_uhs0_def_speed_non_dma_1bit_multi_block_wr_rd	This test case, tests SDMEM card 1bit Multiple block write/read operation with default speed in non DMA mode
test_sdmem_uhs0_def_speed_non_dma_1bit_	This test case, tests SDMEM card 1bit

Testcase	Description
single_block_wr_rd	Single block write/read operation with default speed of non DMA mode
test_sdmem_uhs0_def_speed_non_dma_4bit_multi_block_wr_rd	This test case, tests SDMEM card 4bit Multiple block write/read operation with default speed in non DMA mode
test_sdmem_uhs0_def_speed_non_dma_4bit_single_block_wr_rd	This test case, tests SDMEM card 4bit Single block write/read operation with default speed of non DMA mode
test_sdmem_uhs0_high_speed_1bit_multiple_block_wr_rd	This test case, tests SDMEM card 1bit Multiple block write/read operation with high speed
test_sdmem_uhs0_high_speed_1bit_single_block_wr_rd	This test case, tests SDMEM card 1bit Single block write/read operation with high speed
test_sdmem_uhs0_high_speed_4bit_multiple_block_wr_rd	This test case, tests SDMEM card 4bit Multiple block write/read operation with high speed
test_sdmem_uhs0_high_speed_4bit_single_block_wr_rd	This test case, tests SDMEM card 4bit Single block write/read operation with high speed
test_sdmem_uhs1_ddr50_multiple_block_wr_rd	This test case, tests SDMEM card 4bit Multiple block write/read operation with DDR50
test_sdmem_uhs1_ddr50_single_block_wr_rd	This test case, tests SDMEM card 4bit Single block write/read operation with DDR50
test_sdmem_uhs1_sdr104_multiple_block_wr_rd	This test case, tests SDMEM card 4bit Multiple block write/read operation with SDR104
test_sdmem_uhs1_sdr104_single_block_wr_rd	This test case, tests SDMEM card 4bit Single block write/read operation with SDR104
test_sdmem_uhs1_sdr12_multiple_block_wr_rd	This test case, tests SDMEM card 4bit Multiple block write/read operation with SDR12
test_sdmem_uhs1_sdr12_single_block_wr_rd	This test case, tests SDMEM card 4bit Single block write/read operation with SDR12
test_sdmem_uhs1_sdr25_multiple_block_wr_rd	This test case, tests SDMEM card 4bit Multiple block write/read operation with SDR25
test_sdmem_uhs1_sdr25_single_block_wr_rd	This test case, tests SDMEM card 4bit Single block write/read operation with SDR25

Testcase	Description
test_sdmem_uhs1_sdr50_multiple_block_wr_rd	This test case, tests SDMEM card 4bit Multiple block write/read operation with SDR50
test_sdmem_uhs1_sdr50_single_block_wr_rd	This test case, tests SDMEM card 4bit Single block write/read operation with SDR50
test_sdmem_uhs1_sdr50_tuning_command	This test case, tests SDMEM card tuning cmd with SDR50
test_sdmem_uhs1_sdr50_tuning_command_data_crc_error	This test case, tests SDMEM card tuning cmd with data crc error with SDR50
test_sdmem_uhs1_sdr50_tuning_command_error	This test case, tests SDMEM card tuning cmd error with SDR50
test_sdmem_uhs1_sdr50_tuning_command_timeout_error	This test case, tests SDMEM card tuning cmd with command timeout error with SDR50
test_sdmem_uhs1_sdr50_tuning_command_with_1_5x_disable	This test case, tests SDMEM card tuning cmd with SDR50 with 1_5 times tuning command disable
test_sdmem_v2_0_adma2_1bit_multi_blk_wr_rd	This test case, tests SDMEM card 1bit Multiple block write
test_sdmem_v2_0_adma2_1bit_single_blk_wr_rd	This test case, tests SDMEM card 1bit single block write operation
test_sdmem_v2_0_adma2_4bit_multi_blk_wr_rd	This test case, tests SDMEM card 1bit Multiple block write
test_sdmem_v2_0_adma2_4bit_single_blk_wr_rd	This test case, tests SDMEM card 4bit single block write operation
test_sdmem_v2_0_adma2_8bit_multi_blk_wr_rd	This test case, tests SDMEM card 8bit Multiple block write
test_sdmem_v2_0_adma2_8bit_single_blk_wr_rd	This test case, tests SDMEM card 8bit single block write operation
test_sdmem_v2_0_non_dma_1bit_multi_blk_wr_rd	This testcase, tests case for the Non DMA mode multiple block write read function for 1 bit width
test_sdmem_v2_0_non_dma_1bit_single_blk_wr_rd	This testcase, tests case for the Non DMA mode for 1 bit single block write read function
test_sdmem_v2_0_non_dma_4bit_multi_blk_wr_rd	This testcase, tests case for the Non DMA mode multiple block write read function for 4 bit width
test_sdmem_v2_0_non_dma_4bit_single_blk_wr_rd	This testcase, tests case for the Non DMA mode 1 bit single block write read function
test_sdmem_v2_0_non_dma_8bit_multi_blk_wr_rd	This testcase, tests case for the Non

Testcase	Description
r_rd	DMA mode multiple block write read function for 8 bit width
test_sdmem_v2_0_non_dma_8bit_single_blk_wr_rd	This testcase, tests case for the Non DMA mode 8 bit single block write read function
test_xml_validation	This testcase is used to test the XML value and RTL value compare

15.1.3 MMC Test plan

Testcase	Description
test_mmc_v4_2_adma2_1bit_multi_blk_wr_rd	This testcase, tests the MMC ADMA 1bit multi block write
test_mmc_v4_2_adma2_1bit_single_blk_wr_rd	This testcase, tests the MMC ADMA 1bit single block write
test_mmc_v4_2_adma2_4bit_multi_blk_wr_rd	This testcase, tests the MMC ADMA 1bit multi block write
test_mmc_v4_2_adma2_4bit_single_blk_wr_rd	This testcase, tests the MMC ADMA 1bit single block write
test_mmc_v4_2_adma2_8bit_multi_blk_wr_rd	This testcase, tests the MMC ADMA 8bit multi block write
test_mmc_v4_2_adma2_8bit_single_blk_wr_rd	This testcase, tests the MMC ADMA 1bit single block write
test_mmc_v4_2_adma2_def_speed_1bit_multi_block_wr_rd	This testcase, tests the MMC ADMA2 8bit single block write
test_mmc_v4_2_adma2_def_speed_1bit_single_block_wr_rd	This testcase, tests the MMC ADMA2 1bit single block write
test_mmc_v4_2_adma2_def_speed_4bit_multi_block_wr_rd	This testcase, tests the MMC ADMA2 8bit single block write
test_mmc_v4_2_adma2_def_speed_4bit_single_block_wr_rd	This testcase, tests the MMC ADMA2 4bit single block write
test_mmc_v4_2_adma2_def_speed_8bit_multi_block_wr_rd	This testcase, tests the MMC ADMA2 8bit single block write
test_mmc_v4_2_adma2_def_speed_8bit_single_block_wr_rd	This testcase, tests the MMC ADMA2 8bit single block write
test_mmc_v4_2_basic	This tests all features of MMC v4.2 operations with initialization, write & read
test_mmc_v4_2_non_dma_1bit_multi_block_wr	This testcase, tests the MMC v4.2 non DMA 1bit single block write
test_mmc_v4_2_non_dma_1bit_multi_block_w	This testcase, tests the MMC v4.2

Testcase	Description
r_rd	non DMA 1bit multi block write and read
test_mmc_v4_2_non_dma_1bit_single_block_wr	This testcase, tests the MMC v4.2 non DMA 1bit single block write
test_mmc_v4_2_non_dma_1bit_single_block_wr_rd	This testcase, tests the MMC v4.2 non DMA 1bit single block write/read
test_mmc_v4_2_non_dma_4bit_multi_block_wr_rd	This testcase, tests the MMC v4.2 non DMA 4bit multi block write and read
test_mmc_v4_2_non_dma_4bit_single_block_wr	This testcase, tests the MMC v4.2 non DMA 4bit single block write
test_mmc_v4_2_non_dma_4bit_single_block_wr_rd	This testcase, tests the MMC v4.2 non DMA 4bit single block write/read
test_mmc_v4_2_non_dma_8bit_single_block_wr	This testcase, tests the MMC v4.2 non DMA 8bit single block write
test_mmc_v4_2_non_dma_def_speed_1bit_multi_block_wr_rd	This testcase, tests the MMC v4.2 non DMA 1bit multi block write and read
test_mmc_v4_2_non_dma_def_speed_1bit_single_block_wr_rd	This testcase, tests the MMC v4.2 non DMA 1bit single block write/read
test_mmc_v4_2_non_dma_def_speed_4bit_multi_block_wr_rd	This testcase, tests the MMC v4.2 non DMA 1bit multi block write and read
test_mmc_v4_2_non_dma_def_speed_4bit_single_block_wr_rd	This testcase, tests the MMC v4.2 non DMA 4bit single block write/read
test_mmc_v4_2_non_dma_def_speed_8bit_multi_block_wr_rd	This testcase, tests the MMC v4.2 non DMA 1bit multi block write and read
test_mmc_v4_2_non_dma_def_speed_8bit_single_block_wr_rd	This testcase, tests the MMC v4.2 non DMA 8bit single block write/read
test_mmc_v4_2_non_dma_high_speed_sdr_4bit_multi_blk_wr_rd	This testcase, tests the MMC SDMA 8bit multi block write
test_mmc_v4_2_non_dma_high_speed_sdr_4bit_single_blk_wr_rd	This testcase, tests the MMC SDMA 4bit multi block write
test_mmc_v4_2_non_dma_high_speed_sdr_8bit_multi_blk_wr_rd	This testcase, tests the MMC SDMA 8bit multi block write
test_mmc_v4_2_non_dma_high_speed_sdr_8bit_single_blk_wr_rd	This testcase, tests the MMC SDMA 8bit multi block write
test_mmc_v4_2_sdma_def_speed_1bit_multi_	This testcase, tests the MMC

Testcase	Description
block_wr_rd	SDMA 1bit multi block write
test_mmc_v4_2_sdma_def_speed_1bit_single_block_wr_rd	This testcase, tests the MMC SDMA 1bit multi block write
test_mmc_v4_2_sdma_def_speed_4bit_multi_block_wr_rd	This testcase, tests the MMC SDMA 1bit multi block write
test_mmc_v4_2_sdma_def_speed_4bit_single_block_wr_rd	This testcase, tests the MMC SDMA 1bit multi block write
test_mmc_v4_2_sdma_def_speed_8bit_multi_block_wr_rd	This testcase, tests the MMC SDMA 8bit multi block write
test_mmc_v4_2_sdma_def_speed_8bit_single_block_wr_rd	This testcase, tests the MMC SDMA 8bit multi block write
test_mmc_v4_2_sdma_high_speed_sdr_4bit_cmd24_write	This testcase, tests the MMC SDMA 4bit multi block write
test_mmc_v4_2_sdma_high_speed_sdr_4bit_cmd25_write	This testcase, tests the MMC SDMA 4bit multi block write
test_mmc_v4_2_sdma_high_speed_sdr_4bit_multi_blk_wr_rd	This testcase, tests the MMC SDMA 8bit multi block write
test_mmc_v4_2_sdma_high_speed_sdr_4bit_single_blk_wr_rd	This testcase, tests the MMC SDMA 4bit single block write
test_mmc_v4_2_sdma_high_speed_sdr_8bit_cmd24_write	This testcase, tests the MMC SDMA 8bit multi block write
test_mmc_v4_2_sdma_high_speed_sdr_8bit_cmd25_write	This testcase, tests the MMC SDMA 8bit multi block write
test_mmc_v4_2_sdma_high_speed_sdr_8bit_multi_blk_wr_rd	This testcase, tests the MMC SDMA 8bit multi block write
test_mmc_v4_2_sdma_high_speed_sdr_8bit_single_blk_wr_rd	This testcase, tests the MMC SDMA 8bit multi block write

16 Speed and Size

Core is certified to works up to 200Mhz as required by the SDIO/eMMC specification. Size of the SDIO/eMMC HOST IP depends on the configuration. Please send us your requirement and we will let you know the gate count.

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17 Reporting Problems

Please generate the Log file and FSDB file or VCD file with debug level set to VERBOSE (please use `bzip2 -9 filename.vcd` to compress the file) and put into Smartdv's FTP server. And send detailed mail explaining what you expect to see and what you are seeing.

1. Generate log files with debug level set to VERBOSE.
2. Generate the FSDB or VCD file.
3. Compress log file and FSDB/VCD file with `bzip2 -9 <filename>`
4. Ftp the FSDB/VCD and log file to smart-dv's FTP server.
5. Send mail to support about the problem you are seeing with time stamp where you are seeing the problem.
6. If you need help with Synthesis, send synthesis reports and script you are using to us to help close the issue.

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18 Known Limitations

There are no limitations in the SDIO/eMMC HOST IIP

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